Senior High School

DepED
DEPARTMENT OF EDUCATION

Lesson Exemplar in General Mathematics

Quarter 4

Unit

11

Lesson Exemplar for General Mathematics
Quarter 4: Unit 11

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Republic of the Philippines

Department of Education

BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR					
Learning Area:	**GENERAL MATHEMATICS**			Grade Level:	**11**
Semester:	**SECOND**	Quarter:	**FOURTH**	Unit:	**11**

I. LEARNING GOALS	
Content Standard	The learners demonstrate knowledge and understanding of basic concepts of (a) statistical hypothesis testing, (b) parametric test of difference, (c) test of relationships, and (d) simple linear regression. **The learner demonstrates understanding of the fundamental concepts and procedures of inferential statistics, including statistical hypothesis testing, tests of difference, tests of relationships, and simple linear regression, to analyze sample data and make evidence-based conclusions.**
Performance Standard	*By the end of the quarter, the learners are able to* create a financial proposal using comparisons of different investment plans (e.g., time deposits, mutual funds, pension plans) to recommend and explain the best financial strategy. Learners make inferences and explain their reasoning for real-life problems in different disciplines (e.g., market trends, election results, sports statistics) given specific tests of difference and association. Learners evaluate arguments using logic in real-world contexts (e.g., critically analyzing misleading arguments or statements in advertisements, politics, or social media). **The learners are able to make informed decisions and draw meaningful conclusions from sample data by applying appropriate inferential statistical tools, including hypothesis testing and correlation analysis highlighting implications for real-life situations.**
Learning Competencies	The learners… 1. illustrate (a) null hypothesis, (b) alternative hypothesis, (c) level of significance, and (d) types of error; 2. formulate the null and alternative hypothesis of a given problem; 3. determine the (a) level of significance and (b) types of error appropriate to the given hypothesis; 4. test hypothesis for one and two population means using technology employing the following statistical tools: a. *z*-test for known standard deviation b. *z*-test for unknown standard deviation c. one-sample t-test d. independent sample t-test e. paired sample t-test

	5. construct a scatter plot and estimate the strength of association between variables; 6. draw the line of best fit for a scatter plot; 7. determine the degree of relationship between two variables using Pearson's r correlation; 8. conclude from the results of hypothesis testing on a. one and two population means using t-test and z-test b. relationship using Pearson r correlations
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	Materials ○ Calculators ○ Visual aids (3D shapes) or posters of 3D shapes/PowerPoint presentation ○ Worksheets (printed or digital) ○ LCD Projector or TV (if using video lessons)	
III. CONTENT	**Lesson 11.1: Null and Alternative Hypothesis** *Recommended time allotment: 2 hours*	
IV. OBJECTIVES	At the end of the lesson, learners should be able: 1. to explain the fundamental concepts of the null hypothesis (H_0) and the alternative hypothesis (H_1/Ha) as the basis of statistical decision-making. 2. to demonstrate how to formulate appropriate null and alternative hypotheses in relation to different research questions or real-life situations. 3. to develop critical thinking skills in interpreting statistical test results by comparing evidence against the null and alternative hypotheses.	
V. PROCEDURES	**ACTIVITIES** (Instructions for Learners)	**ANNOTATION** (Instructions for Teachers)

<table>
<tr><td rowspan="2">A. Activating Prior Knowledge</td><td colspan="2">A.1 Eliciting Prior Knowledge</td></tr>
<tr><td>
Activity A.1: Leveling Learner Readiness

Option 1. Quick Recall
Instructions:
1. Think (Individual Work)
Read each statement carefully.
Ask yourself: What is the speaker claiming or assuming?
Think quietly about your answer.
Prompt: What assumption or claim is being made in each case?
A teacher says that studying at night is better than studying in the morning.
A company claims their bottled water boosts energy.
A classmate says that listening to music while studying improves test scores.
A friend argues that online classes are less effective than face-to-face classes.
A sports coach insists that eating bananas before games improves performance.
2. Pair (Discussion with a Partner)
Share your ideas with a partner.
Discuss whether you both identified the same claim.
Clarify or revise your answer if needed.
3. Share (Class Discussion)
Be ready to share your answer with the class if called.
Listen to how others interpret the same statement.
4. Focus on the Claim
Do not judge if the statement is true or false.
Only identify what is being claimed.
</td><td>
Activity A.1: Leveling Learner Readiness

Option 1. Quick Recall

This activity helps learners recall and relate everyday assumptions or claims to statistical thinking. By discussing in pairs, they build confidence before sharing in class. It encourages critical reflection on how hypotheses mirror real-life statements.

In addition, learners will engage in a Think-Pair-Share exercise to critically examine everyday claims. Each claim contains an implicit assumption or belief that can be explored and questioned. Learners will first think individually about each claim, then pair up to discuss their interpretations, and finally share their insights with the class. This activity serves as a foundation for understanding how assumptions lead to the formulation of null and alternative hypotheses in research.

To facilitate the activity, follow these steps.
Think Phase (Individual Reflection):
What is the main claim being made in each statement?
What assumption underlies this claim?
Is the claim based on personal experience, belief, or evidence?
Pair Phase (Discussion with a Partner):
Do you and your partner agree on the assumption behind the claim?
Can you think of counterexamples or alternative explanations?
How could this claim be tested scientifically?
Share Phase (Class Discussion):
Which claims seem most testable? Why?
How would you formulate a null hypothesis and an alternative hypothesis for one of these claims?
What kind of data or experiment would help validate or refute the claim?
Examples (recall-based prompts):
Studying at night vs. morning effectiveness.
</td></tr>
</table>

Option 2.
Real-Life Scenarios (Yes/No Hypotheses)

Instructions:
1. Read each scenario carefully and focus on what is being claimed or compared.

Scenarios:
- A new headache medicine claims faster relief than the leading brand.
- A new teaching method claims to improve math scores.
- A restaurant claims their low-fat recipe tastes the same as the original.
- A learner says boys are taller than girls in Grade 10.
- A phone company claims their battery lasts longer than competitors.

2. Identify the Null Hypothesis (H_0)
   - State that there is no difference, no change, or no effect.
   - Keep the statement neutral and simple.

3. Identify the Alternative Hypothesis (H_1)
   - State that there is a difference or improvement.
   - This should directly match the claim in the scenario.

4. Write Hypotheses as Statements
   - Do not use questions.
   - Avoid explanations or justifications.

- Bottled water boosts energy.
- Music improves test scores.
- Online vs. face-to-face classes
- Bananas improve sports performance.

Case 2.
Real-Life Scenarios (Yes/No Hypotheses)

In this activity, learners will analyze real-life claims and practice identifying the null hypothesis (H_0) and alternative hypothesis (H_1) for each scenario. These claims reflect common assumptions found in advertising, education, and daily conversations. By translating these into formal hypotheses, learners will develop foundational skills in scientific thinking and hypothesis testing.

Learners will work individually or in pairs to examine each scenario, determine the underlying claim, and express it in terms of H_0 and H_1. This activity emphasizes the binary nature of hypothesis testing whether there is a significant effect or not and prepares learners for designing experiments and interpreting data.

To facilitate the activity, follow these steps.
Individual or Pair Work:
1. What is the main claim or assumption in the scenario?
2. What would be the **null hypothesis (H_0)** if we were to test this claim?
3. What would be the **alternative hypothesis (H_1)**?
4. Is the alternative hypothesis **directional** (e.g., “better,” “longer,” “taller”) or **non-directional** (e.g., “different”)?

Class Discussion:
1. Which scenarios suggest a **one-tailed test** vs. a **two-tailed test**?
2. How could each hypothesis be tested in a real experiment?
3. What kind of data would you need to accept or reject the null hypothesis?

Scenarios anchor abstract statistical concepts in familiar contexts. By identifying null and alternative hypotheses, learners learn to formalize claims as testable statements. This builds the foundation for hypothesis formulation.

Examples (scenario with hypotheses):

1. New headache medicine → H_0: no difference; H_1: faster relief.
2. New teaching method → H_0: no improvement; H_1: improves scores.
3. Low-fat recipe → H_0: no taste difference; H_1: taste is different.
4. Boys vs. girls height → H_0: no difference; H_1: boys taller.
5. Phone battery → H_0: no difference; H_1: new battery lasts longer.

Option 3. Prediction Activity
Instructions:

1. Read Each Situation Carefully
   - State what you expect if the null is true
   - Identify what H_0 is assuming (fairness, no difference, no effect).

Scenarios:

- **Coin Toss:** Toss a coin 50 times → Expect ~25 heads (H_0: fair coin).
- **Dice Roll:** Roll a die 60 times → Expect ~10 for each face (H_0: fair die).
- **Classroom Survey:** Ask if learners prefer online vs. face-to-face → Expect ~equal if no preference.
- **Plant Growth:** Two plants given same sunlight → Expect equal growth (H_0: no difference).
- **Exam Scores:** Compare two classes with same teacher → Expect no difference in average scores (H_0: teaching doesn't matter).

2. State the Expected Outcome Under H_0
   - Predict what should happen if the null hypothesis is true.
   - Use approximate or equal values when appropriate.

3. Do Not Analyze Actual Results Yet
   - This activity focuses only on expectations, not conclusions.

Case 3. Prediction Activity

In this activity, learners will explore the concept of expected outcomes under the null hypothesis (H_0). Each scenario presents a situation where a claim is being tested, and learners are asked to predict what results they would expect if the null hypothesis is true. This helps reinforce the idea that the null hypothesis represents a baseline or default assumption—typically, no effect or no difference.

Learners will work individually or in small groups to analyze each scenario, identify the null hypothesis, and state the expected outcome. This activity builds foundational skills in statistical reasoning and prepares learners for interpreting experimental results.

To facilitate the activity, follow these steps.
Individual or Group Work:

1. What is the null hypothesis in each scenario?
2. What outcome would you expect if the null hypothesis is true?
3. Would small differences in results still support the null hypothesis?

Class Discussion:

1. How do random variation and sample size affect your expectations?

	4. Think About Deviations • Ask yourself: What would it mean if results were very different from what I expected? **Option 4. Concept Mapping** **Instructions:** 1. Work Individually or in Small Groups • Follow your teacher's instructions for grouping. 2. Read the Prompt Words • Each line shows a sequence of related concepts. • Think about how one idea leads to the next. 3. Create a Concept Map • Rewrite the words using arrows, diagrams, or bubbles. • Add short phrases explaining each connection (if instructed). 4. Discuss Your Map • Share ideas with your group or the class. • Compare how different learners made connections. 5. Listen to the Teacher's Explanation The teacher will connect your ideas to: • H_0 (Null Hypothesis) • H_1 (Alternative Hypothesis) • Significance Level • Type I and Type II Errors **Task:** Brainstorm and connect prior knowledge words. **Prompt words per sample:** • Assumption → Evidence → Decision → Proof • Claim → Experiment → Result → Error • Chance → Probability → Risk → Testing • Statement → Comparison → Data → Conclusion • Belief → Research → Significance → Truth	2. What kind of results would lead you to reject the null hypothesis? 3. How can you tell if an observed difference is statistically significant? **Option 4. Concept Mapping** In this activity, learners will create concept maps using sets of related words that reflect key ideas in research, logic, and decision-making. Each group of words represents a thematic progression—from assumptions and beliefs to evidence and conclusions. Learners will brainstorm definitions, relationships, and examples for each word, then visually organize them into a concept map that shows how the ideas connect. This activity encourages learners to activate prior knowledge, deepen understanding of research vocabulary, and see how concepts interrelate in the process of inquiry and hypothesis testing. To facilitate the activity, follow these steps. **Brainstorming Phase:** 1. What does each word mean in the context of research or reasoning? 2. Can you give a real-life or classroom example for each word? 3. Which words are causes, processes, or outcomes? **Mapping Phase:** 1. How are the words connected logically or sequentially? 2. Which word is the starting point of the concept map? 3. Are there feedback loops or multiple pathways between concepts? **Sharing Phase:** 1. How does your concept map reflect the process of scientific thinking? 2. Did you discover any surprising connections between the words? 3. How could this map help someone understand hypothesis testing or decision-making?

	A.2. Establishing the Purpose of the Lesson	
	Activity A. 2: Appreciating Lesson Relevance **Option 1. Real-Life Claim Debate** **Instructions:** 1. Read the Claim Carefully • Focus on what is being asserted or believed. 2. Choose a Side • Decide whether you Agree or Disagree with the claim. • Be ready to explain your reasoning briefly. 3. Participate in the Class Discussion • Share your opinion when called. • Listen respectfully to opposing viewpoints. 4. Connect the Claim to Hypotheses The teacher will show how the claim can be tested by forming: • A null hypothesis (H_0) • An alternative hypothesis (H_1) 5. Record the Hypotheses • Write down the correct H_0 and H_1 for each claim discussed.	**Activity A. 2: Appreciating Lesson Relevance** **Option 1. Real-Life Claim Debate** These debate-style activities encourage learners to critically examine everyday claims. By framing each as a null (no effect) and alternative (effect exists) hypothesis, learners realize that scientific testing is needed to resolve arguments. This develops critical thinking and logical reasoning**.** **Instructions** 1. Start with a provocative claim learners can relate to (e.g., “Energy drinks improve focus”). 2. Divide the class into Agree/Disagree groups → let each side give 1–2 arguments. 3. Write both H_0 and H_1 on the board to show how debates are structured scientifically. 4. Emphasize: *“In statistics, we don’t stop at debate—we test these claims with data.”* **Examples and Answer Key:** Claim: “Energy drinks improve concentration during exams.” $\mathbf{H_0}$: Energy drinks have no effect on concentration. $\mathbf{H_1}$: Energy drinks improve concentration. Claim: “Using gadgets while studying makes learning easier.” $\mathbf{H_0}$: Gadgets do not improve learning. $\mathbf{H_1}$: Gadgets improve learning. Claim: “Organic food is healthier than non-organic food.” $\mathbf{H_0}$: No difference in health effects. $\mathbf{H_1}$: Organic food is healthier. Claim: “Learners who play sports perform better academically.”

	Option 2. True or False Predictions **Instructions:** 1. Read the Statement Carefully • Each item presents a real-life claim (example: “Eating breakfast improves test scores.”). 2. Make a Prediction • Decide whether you think the statement is True or False. • This is only a guess based on your intuition or prior knowledge. 3. Identify the Null Hypothesis (H_0) • Write a statement that assumes no change or no difference. • The null hypothesis is usually neutral and conservative. 4. Identify the Alternative Hypothesis (H_1) • Write a statement that supports the original claim. • This hypothesis suggests that the stated effect or difference exists. 5. Use Clear and Simple Language • Do not include explanations, reasons, or data. • Just clearly state each hypothesis.	$\mathbf{H_0}$: Sports participation does not affect academics. $\mathbf{H_1}$: Sports participation improves academics. Claim: “Listening to classical music improves memory.” $\mathbf{H_0}$: Music does not affect memory. $\mathbf{H_1}$: Classical music improves memory. **Option 2. True or False Predictions** Prediction activities help learners tap into their prior knowledge and assumptions. When these are tested against hypotheses, they see that personal beliefs are not enough—statistical evidence is necessary. This activity fosters awareness of bias and builds readiness for hypothesis testing. **Instructions:** 1. Present statements one by one and let learners vote “True” or “False” (raise hands or use a digital poll). 2. Record the class results on the board (e.g., 70% True, 30% False). 3. Show how each statement can be framed as H_0 vs. H_1. 4. Reinforce: *“Our predictions may vary, but hypothesis testing gives objective answers.”* **Examples & Answer Key:** • Statement: “Boys are better at math than girls.” ○ $\mathbf{H_0}$**:** No difference in math performance. ○ $\mathbf{H_1}$**:** Boys perform better in math. • Statement: “Drinking milk makes bones stronger.” ○ $\mathbf{H_0}$**:** Milk has no effect on bone strength. ○ $\mathbf{H_1}$**:** Milk strengthens bones. • Statement: “Learners with part-time jobs get lower grades.” ○ $\mathbf{H_0}$**:** No difference in grades. ○ $\mathbf{H_1}$**:** Part-time jobs lower grades. • Statement: “Taller people run faster.” ○ $\mathbf{H_0}$**:** Height does not affect running speed. ○ $\mathbf{H_1}$**:** Taller people run faster. • Statement: “Eating breakfast improves test scores.” ○ $\mathbf{H_0}$**:** Breakfast has no effect on test scores. ○ $\mathbf{H_1}$**:** Breakfast improves test scores.

	Option 3. Guess the Experiment **Instructions:** 1. Read the Question Carefully • Each item asks whether one factor affects another (example: “Does a new teaching method improve reading comprehension?”). 2. Suggest How to Test the Question Briefly describe: • Who or what will be studied • What is being changed or compared • What will be measured Keep your description simple (1–2 sentences). 3. Write the Null Hypothesis (H_0) • State that the factor being tested has no effect. • Use clear and neutral language. 4. Write the Alternative Hypothesis (H_1) • State that the factor does have an effect, matching the question. • This hypothesis should support the idea being tested. 5. Do Not Include Results • You are designing the experiment, not conducting it.	**Option 3. Guess the Experiment** This activity will train learners to transform research questions into testable hypotheses. By identifying H_0 and H_1, they practice formulating hypotheses and thinking like researchers. This activity builds problem-solving and application skills**,** preparing learners for real statistical investigations. **Instructions:** 1. Pose a research question (e.g., “Does exercising daily reduce stress?”). 2. Ask learners in small groups to suggest how they would test it experimentally. 3. Guide them to phrase H_0 and H_1 correctly. 4. Use graphic organizers (T-chart: Null vs. Alternative) for clarity. 5. Reinforce: *“Every research question must begin with clear hypotheses.”* **Examples & Answer Key:** 1. Q: Does a new teaching method improve reading comprehension? **H_0:** New method has no effect. **H_1:** New method improves comprehension. 2. Q: Does exercising 30 minutes daily reduce stress? **H_0:** Exercise has no effect on stress. **H_1:** Exercise reduces stress. 3. Q: Does a vitamin supplement increase energy levels? **H_0:** Supplement has no effect on energy. **H_1:** Supplement increases energy. 4. Q: Do learners who review notes daily score higher in exams? **H_0:** Reviewing notes has no effect. **H_1:** Reviewing notes improves exam scores. 5. Q: Does playing video games improve problem-solving skills? **H_0:** Video games have no effect. **H_1:** Video games improve problem-solving skills **Option 4. Quick poll and Reflection**

Option 4. Quick poll and Reflection

Instructions:

1. Participate in the Class Poll
    - Your teacher will present a topic (example: Online learning vs. face-to-face learning).
    - Answer honestly based on your experience or opinion.

2. Observe the Poll Results
    - Take note of how many learners chose each option.
    - Look for visible differences or trends in the responses.

3. Discuss as a Class
    - Share observations about which option seems more common.
    - Discuss possible reasons for the differences.
    - Remember: discussion is informal—no conclusions yet.

4. Write the Null Hypothesis (H_0)
    - State that there is no difference or no effect between the two options.
    - Use neutral wording.

5. Write the Alternative Hypothesis (H_1)
    - State that one option is better, stronger, or more effective.
    - This should reflect the trend seen in the poll or the class discussion.

Option 5. Case Study

Instructions:

1. Read the Case Carefully
    - Each case describes a situation where something new is being tested (example: a product, method, or treatment).

Quick polls engage learners and reveal diverse opinions within the classroom. By translating opinions into hypotheses, learners see why data-driven evidence is essential for decision-making. This activity encourages participation and shows the relevance of statistics in resolving debates.

Strategy for Teachers:

1. Begin with a **class poll** (e.g., "Who prefers online learning over face-to-face?").
2. Display poll results on the board.
3. Ask: *"Can we conclude from this poll alone?"* (Expected: No, because it's just opinions.)
4. Translate the poll question into **H_0 and H_1**.
5. Highlight: *"This is why we rely on hypothesis testing—to move from opinion to evidence."*

Examples & Answer Key:

1. Online learning vs. face-to-face:
   H_0: No difference in effectiveness.
   H_1: Online is more effective.
2. Sleeping early improves test performance:
   H_0: Sleep time has no effect.
   H_1: Sleeping early improves performance.
3. Group study vs. solo study:
   H_0: No difference in results.
   H_1: Group study improves results.
4. Left-handed people are more creative:
   H_0: No difference in creativity.
   H_1: Left-handed are more creative.
5. Social media reduces attention span:
   H_0: Social media has no effect.
   H_1: Social media reduces attention span.

Option 5. Case Study

Stories make abstract concepts concrete and relatable**.** By identifying null and alternative hypotheses within real-world cases, learners understand the practical applications of hypothesis testing in fields like medicine, education, agriculture, and business. This activity connects learning to everyday decision-making and professional contexts**.**

<table>
<tr>
<td></td>
<td>2. Identify the Claim or Change
• Determine what is being introduced, improved, or compared.
• Think about what outcome is expected to change.

3. Write the Null Hypothesis (H_0)
• State that the new change has no effect or there is no difference compared to before.
• Use neutral and general language.

4. Write the Alternative Hypothesis (H_1)
• State that the new change does improve, increase, reduce, or affect the outcome.
• This hypothesis should directly reflect the claim in the case.

5. Keep Statements Clear and Simple
• Do not explain or justify.
• Write short and direct hypotheses.</td>
<td>Instructions:
1. Tell a short story or case (e.g., a company testing a new battery).
2. Ask guiding questions: “What’s the claim? What’s the starting assumption? How can we test it?”
3. Write both hypotheses (H_0 vs. H_1) on the board.
4. Let learners discuss real-world consequences of making the wrong decision (Type I & II errors).
5. Reinforce: “Hypotheses guide real-world decisions in science, medicine, and business.”

Examples & Answer Key:
1. The phone company claims new battery lasts 15 hrs (old: 12 hrs).
H_0: No difference in battery life.
H_1: New battery lasts longer.
2. Farmer tests new fertilizer.
H_0: No effect on yield.
H_1: New fertilizer increases yield.
3. The uses visual aids in lessons.
H_0: Visual aids have no effect on grades.
H_1: Visual aids improve grades.
4. The clothing brand claims new fabric is more durable.
H_0: No difference in durability.
H_1: New fabric is more durable.
5. The hospital tests new drug for recovery time.
H_0: No difference in recovery time.
H_1: New drug reduces recovery time.</td>
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<tr>
<td rowspan="2">B. Instituting New Knowledge</td>
<td colspan="2">B.1. Presenting Examples</td>
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<tr>
<td>Activity B.1 Exploring Key Concepts
Instructions:
1. The teacher will present the real-life scenario.
2. Restate these scenarios in H_0 and H_1 in simple sentences.
3. Write responses on the board in T-chart format for clarity.
4. Identify: “Which is the null hypothesis? Which is the alternative? Why?”

Example 1 – Medicine Effect
Scenario: A pharmaceutical company claims their new pain reliever works faster.</td>
<td>Activity B.1 Exploring Key Concepts

Presenting examples is crucial because it bridges abstract statistical concepts with concrete, real-life situations. By showing learners how to formulate both null and alternative hypotheses in contexts like medicine, education, technology, and agriculture, the teacher helps learners understand that hypothesis testing is not just theoretical but has practical applications in decision-making.
This activity develops:</td>
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</table>

H_0: The new pain reliever has no difference in effect compared to the old one.
H_1: The new pain reliever works faster than the old one.

Example 2 – Teaching Strategy
Scenario: A teacher introduces a new reading strategy and believes it improves comprehension.
H_0: The new reading strategy does not improve comprehension.
H_1: The new reading strategy improves comprehension.

Example 3 – Crop Fertilizer
Scenario: A farmer tries a new fertilizer and claims it increases crop yield.
H_0: The new fertilizer does not affect crop yield.
H_1: The new fertilizer increases crop yield.

Example 4 – Battery Life
Scenario: A cellphone brand says their new battery lasts longer than the standard one.
H_0: The new battery has the same lifespan as the standard one.
H_1: The new battery lasts longer than the standard one.

Example 5 – Study Habits
Scenario: Learners who review their notes daily perform better in exams.
H_0: Reviewing notes daily has no effect on exam performance.
H_1: Reviewing notes daily improves exam performance.

1. **Conceptual understanding** → Learners clearly distinguish H_0 (no effect/difference) from H_1 (effect/difference).
2. **Application skills** → Learners practice applying the concept to diverse scenarios.
3. **Critical thinking** → Learners see how assumptions and claims are tested scientifically.
4. **Engagement** → Real-world examples make the lesson relatable and meaningful.

By presenting guided examples, the teacher scaffolds new knowledge, providing models that learners can later use to formulate their own hypotheses during independent or group activities.

To facilitate this activity, apply of the suggested strategies.
1. Conceptual Understanding
1. Use a T-chart on the board: one side labeled “Null Hypothesis (H_0),” the other “Alternative Hypothesis (H_1).”
2. As each example is presented, ask the class: *“Which is the starting assumption? Which is the claim?”* and fill in the chart together.

2: Application Skills
1. After 2–3 teacher-led examples, switch to guided practice: provide new scenarios and let learners write H_0 and H_1 in pairs.
2. Use **think-pair-share** so all learners get involved before answers are discussed.

3. Critical Thinking
b. For each example, ask a reflective question:
   a. *“If we assume H_0, what would it mean in real life?”*
   b. *“If H_1 is true, what changes or benefits might we expect?”*
c. Encourage discussion of consequences of rejecting or failing to reject H_0 **(**connect later to errors in hypothesis testing).

4. Engagement

1. Use learner-centered examples (school performance, study habits, technology use).
2. Add short debates: *"Who believes the new method works? Who doesn't?"* before writing H_0 and H_1.
3. Incorporate multimedia or polling apps (Kahoot, Mentimeter) for quick learner votes before presenting the formal hypotheses.

B.2. Discussing the Concept

Activity B.2: Deepening Conceptual Understanding

Instructions:
Participate actively in the discussion and give extra attention to the following:
1. Understanding the roles of the null hypothesis (H_0) and the alternative hypothesis (H_1).
2. Clearly differentiating between H_0 and H_1.
3. Explaining why both hypotheses are necessary in statistical testing.

a. Null Hypothesis (H_0):

It is a statement of no difference between sample means or proportions or no difference between a sample mean or proportion and a population mean or proportion. In other words, the difference equals 0.

- ✔ A statement of **no effect, no difference,** or **no relationship**.
- ✔ It is the **starting assumption** we test against.

Example: *"There is no difference in scores between learners who study in the morning and those who study at night."*

b. Alternative Hypothesis (H_1 / Ha):

It is a claim about the population that is contradictory to H_0 and what we conclude when we reject H_0.

- ✔ A statement that **contradicts the null hypothesis**.

Activity B.2: Deepening Conceptual Understanding

This stage of the lesson is designed to provide learners with a clear and structured understanding of the new concept. By formally defining the Null Hypothesis (H_0) and Alternative Hypothesis (H_1) and discussing their features, the teacher establishes the foundation for hypothesis testing**.**
The discussion helps learners:

a. **Differentiate** between H_0 (the default assumption of no effect) and H_1 (the research claim).
b. **Understand purpose**: Hypothesis testing is not about proving H_0 true, but about evaluating whether there is enough evidence to reject it.
c. **Recognize structure**: Hypotheses must be written as statements, not questions, and always come in pairs (H_0 vs. H_1).
d. **Engage critically**: Guiding questions encourage learners to think about *why* we need hypotheses and how they are used in scientific testing.

By grounding the discussion in both formal definitions and guided questioning, learners gain not only conceptual clarity but also the ability to apply the concept to real-life claims later in the lesson.

To facilitate discussion, employ any of the following strategies.

Strategies for Discussing the Concept
1. Direct Instruction with Visuals

a. Use a T-chart on the board or projector labeled: *Null Hypothesis (H_0)* vs. *Alternative Hypothesis (H_1).*
b. Fill in the chart with simple, real-life examples (medicine, teaching method, battery life).
c. Highlight keywords like *"no effect/difference"* for H_0 and *"there is an effect/difference"* for H_1.

- ✔ Represents the **claim or effect the researcher wants to test**.

Example: *"Learners who study at night perform better than those who study in the morning."*

Mathematical Symbols Used in H_0 and H_1 or H_a:

H_0	H_1 or H_a
equal (=)	not equal (≠) or greater than (>) or less than (<)
greater than or equal to (≥)	less than (<)
less than or equal to (≤)	more than (>)

Note: *H_0 always has a symbol with an equal in it. H_1 or H_a never has a symbol with an equal in it. The choice of symbol depends on the wording of the hypothesis test. However, be aware that many researchers (including one of the co-authors in research work) use = in the null hypothesis, even with > or < as the symbol in the alternative hypothesis. This practice is acceptable because we only make the decision to reject or not reject the null hypothesis.*

3. Important Features

a. Hypotheses are **statements**, not questions.
b. H_0 is assumed true **until proven otherwise**.
c. The purpose of hypothesis testing is to see if we have **enough evidence to reject H_0** in favor of H_1.

4. Guided Class Discussion Questions

a. *"Which statement comes first, H_0 or H_1?"* (Answer: H_0, the starting assumption.)
b. *"Why can't we just start with H_1 right away?"* (Because we need a baseline to compare against.)
c. *"Can we ever prove H_0 true?"* (No, we can only fail to reject it.)
d. *"Why do we need both H_0 and H_1?"* (To cover both possibilities and test claims objectively.)

2. Guided Questioning

a. Ask learners probing questions to clarify understanding:
   - ✔ *"Which one is the default assumption, H_0 or H_1?"*
   - ✔ *"Why do we need a starting point in testing claims?"*
   - ✔ *"Can we ever 100% prove the null hypothesis true?"*
b. Use wait time to let learners think before responding.

3. Use Analogies

a. Compare hypotheses to a court trial:
   - ✔ H_0 = presumption of innocence.
   - ✔ H_1 = prosecution's claim.
   - ✔ Data = the evidence that may or may not reject H_0.
b. This makes the abstract idea relatable and easier to grasp.

4. Think-Pair-Share Activity

a. Pose a scenario: *"A learner believes studying at night improves scores. What is H_0? What is H_1?"*
b. Let learners think individually, discuss with a partner, then share with the class.
c. Reinforce correct phrasing and structure.

5. Summarize and Emphasize Key Takeaways

a. End the discussion by summarizing:
   - ✔ H_0 = no effect/difference.
   - ✔ H_1 = effect/difference (claim).
   - ✔ Hypothesis testing = checking if evidence is strong enough to reject H_0.
b. Write this as a **"Key Rule Box"** on the board or in a handout.

5. Key Takeaways (for the board or handout)

a. H_0 = no effect, no difference (default assumption).
b. H_1 = effect, difference, or claim being tested.
c. Hypothesis testing is about **evidence**, not assumptions.

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Part 1: Let's Try Together.

1. Read each scenario carefully.
2. Identify the **Null Hypothesis (H_0)** – statement of no effect/difference.
3. Identify the **Alternative Hypothesis (H_1)** – statement that there is an effect/difference.

Scenarios:

1. A company claims its new fertilizer increases corn yield.
2. A teacher says her new math technique improves problem-solving skills.
3. A hospital tests whether a new patient scheduling system reduces wait times.
4. A gym claims its new workout program improves stamina in 4 weeks.

Part 2: State Your Hypotheses

Instructions:

Write H_0 and H_1 for each scenario.

1. A doctor says a new drug lowers blood pressure better than the current one.
2. A coffee shop claims its coffee keeps people awake longer.
3. A company introduces a battery it claims lasts 20% longer.
4. A school believes athletes have higher GPAs than non-athletes.
5. A clothing brand claims its new fabric is more durable.

Activity B.3: Practicing Learned Skills

This activity helps learners move from understanding the definitions of null (H_0) and alternative (H_1) hypotheses to formulating them independently in real-life contexts.

It emphasizes:
Practice and repetition – Reinforce correct phrasing of hypotheses
Peer interaction – Encourage discussion and error correction
Critical thinking – Recognize the implications of H_0 and H_1 in research

Part 1 – Teacher-Guided Practice

- Read each scenario aloud with learners.
- Model how to write H_0 and H_1 for the first two scenarios.
- Ask learners: *"Which statement assumes no effect? Which statement is the claim being tested?"*
- Encourage learners to justify their choices verbally.

2. Part 2 – Independent Practice

- Learners write H_0 and H_1 for each scenario individually.
- Pair learners to compare answers and discuss reasoning.
- Teacher circulates to check accuracy and provide feedback.

3. Extension / Reflection

- Discuss tricky scenarios or common mistakes.
- Ask reflective questions:

6. A university claims its online learning platform improves learner retention.
7. A tech company tests whether its new app improves user satisfaction.
8. A city claims that installing more streetlights reduces crime rates.
9. A nutritionist says eating almonds daily improves concentration.
10. A bookstore claims its loyalty program increases monthly sales.

   - *"How do we know if a statement should be H_0 or H_1?"*
   - *"What happens if H_0 is rejected but was actually true?"*

4. Assessment / Follow-Up

- Collect worksheets or check answers during peer review.
- Provide targeted practice for learners who struggle with identifying H_0 and H_1.

Answer Key:
Part I.

Scenario	H_0 (Null Hypothesis)	H_1 (Alternative Hypothesis)
A company claims its new fertilizer increases corn yield	The fertilizer does not increase corn yield	The fertilizer increases corn yield
A teacher says her new math technique improves problem-solving skills	The new math technique does not improve problem-solving skills	The new math technique improves problem-solving skills
A hospital tests whether a new patient scheduling system reduces wait times	The scheduling system does not reduce wait times	The scheduling system reduces wait times
A gym claims its new workout program improves stamina in 4 weeks	The workout program does not improve stamina in 4 weeks	The workout program improves stamina in 4 weeks

Part 2.

Scenario	H_0 (Null Hypothesis)	H_1 (Alternative Hypothesis)
1. A doctor says a new drug lowers blood pressure better than the current one	The new drug does not lower blood pressure better than the current one	The new drug lowers blood pressure better than the current one
2. A coffee shop claims its coffee keeps people awake longer	Coffee does not keep people awake longer	Coffee keeps people awake longer

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<tr><td>3. A company introduces a battery it claims lasts 20% longer</td><td>The battery does not last 20% longer</td><td>The battery lasts 20% longer</td></tr>
<tr><td>4. A school believes athletes have higher GPAs than non-athletes</td><td>Athletes do not have higher GPAs than non-athletes</td><td>Athletes have higher GPAs than non-athletes</td></tr>
<tr><td>5. A clothing brand claims its new fabric is more durable</td><td>The new fabric is not more durable</td><td>The new fabric is more durable</td></tr>
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<td rowspan="2">C. Demonstrating Knowledge and Skills</td>
<td colspan="2">C.1. Finding Practical Application</td>
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<td>Activity C.1: Making Real-World Connections
Option 1: School-Based Research Idea
Instructions:
1. Think of a School-Based Issue
• Choose a topic related to school life, learning, or learner behavior.
• The question should be measurable and realistic.</td>
<td>Activity C.1: Making Real-World Connections
This phase ensures learners demonstrate mastery by applying knowledge in authentic contexts. Moving from controlled classroom examples to open-ended, real-life situations strengthens:
1. Critical thinking → seeing connections between theory and practice.
2. Problem-solving → identifying testable claims in real-world settings.</td>
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2. Write Your Research Question
   - Phrase it as a clear, testable question.
   - Focus on how one factor may affect another.

3. Write the Null Hypothesis (H_0)

   - State that the factor you are testing has no effect on the outcome.
   - Use neutral wording.

4. Write the Alternative Hypothesis (H_1)
   - State that the factor does affect, improve, or reduce the outcome.
   - This should align with your research question.

5. Keep Statements Clear and Simple
   - Hypotheses must be written as statements, not questions.
   - Do not include explanations or results.

Task:
Learners propose a researchable school problem and state H_0 and H_1.
Example: *"Does using background music while studying affect test scores?"*

H_0: Background music has no effect on test scores.
H_1: Background music improves test scores.

3. **Communication skills** → presenting hypotheses clearly in oral or written form.
4. **Research readiness** → preparing learners for actual scientific investigations.

Teacher Strategies
1. **Contextualization:** Use local issues (school, community, environment) to make activities relevant.
2. **Collaborative Learning:** Encourage teamwork (group research ideas, poster-making).
3. **Inquiry-Based Approach:** Let learners generate their own testable problems.
4. **Feedback-Oriented:** Give immediate feedback on correctness of H_0 and H_1.
5. **Integration:** Link to upcoming lessons (level of significance, testing, types of error).

The following activities can be conducted to apply hypothesis testing in real-life situations.

Activity 1: School-Based Research Idea
Description:
In this activity, learners brainstorm a school-related problem or question that can be investigated through research. They then write a clear and testable hypothesis pair (H_0 and H_1) based on their proposed idea. This activity promotes creativity, relevance, and critical thinking.
Instructions:
1. **Introduce the Task:**
   a. Explain that learners will think of a realistic and researchable problem within the school setting.
   b. Examples include academic performance, classroom strategies, learner behavior, or school facilities.
2. **Provide a Sample:**
   Example Research Idea:
   "Does using background music while studying affect test scores?"
   a. **H_0:** Background music has no effect on test scores.
   b. **H_1:** Background music improves test scores.

	Option 2: Community Application **Instructions:** 1. Form a Group • Work with your assigned group members. 2. Identify a Community Issue	3. **Learner Task:** a. Learners write their own school-based research question. b. Then, they formulate: ● **H_0 (Null Hypothesis):** Assumes no effect or no difference. ● **H_1 (Alternative Hypothesis):** Reflects the expected effect or difference. 4. **Encourage Creativity and Relevance:** a. Prompt learners to think about issues they care about or have observed in school. b. Examples to inspire them: ● "Does peer tutoring improve math scores?" ● "Do learners who eat breakfast perform better in morning classes?" ● "Does classroom temperature affect concentration?" 5. **Review and Share:** a. Have learners share their ideas in pairs or small groups. b. Provide feedback to ensure hypotheses are clear, testable, and aligned with the research question. **Tips for Teachers:** ● Remind learners that H_0 is the default assumption and should be neutral. ● Encourage them to avoid vague or overly broad questions. ● Use this activity to prepare learners for future research or project-based learning. **Option 2: Community Application** **Description:** In this group activity, learners identify a relevant issue in their community and create a pair of hypotheses (H_0 and H_1) based on a researchable question. This encourages collaboration, critical thinking, and real-world relevance. **Instructions:** 1. **Introduce the Task:**

	• Choose a problem related to:
	  a. Health
	  b. Environment
	  c. Education
	  d. Business
	  e. Public safety
	• The issue should be local, realistic, and measurable.
	
	3. Create a Research Question
	• Write a question that investigates whether an action or program affects a community outcome.
	• The question should be testable using data.
	
	4. Write the Null Hypothesis (H_0)
	• State that the action or program has no effect on the issue.
	• Use clear and neutral wording.
	
	5. Write the Alternative Hypothesis (H_1)
	• State that the action or program does affect or improve the issue.
	• This should directly match your research question.
	
	6. Prepare to Share
	• Be ready to explain your question and hypotheses to the class.
	
	Task: Groups identify a community issue (e.g., health, environment, business) and create hypotheses. **Example**: *"Does recycling awareness reduce waste in the barangay?"*
	H_0: Recycling awareness has no effect on waste reduction.
	H_1: Recycling awareness reduces waste.
		a. Explain that learners will work in groups to choose a community issue they care about or observe.
		b. The issue should be specific, measurable, and researchable.
		2. **Provide a Sample:**
		Example Research Idea:
		"Does recycling awareness reduce waste in the barangay?"
		  a. **H_0:** Recycling awareness has no effect on waste reduction.
		  b. **H_1:** Recycling awareness reduces waste.
		3. **Learner Task:**
		  a. In groups, learners:
		    ● Identify a community issue (e.g., public health, local business, environmental concern).
		    ● Write a clear research question.
		    ● Formulate:
		      ■ **H_0 (Null Hypothesis):** Assumes no effect or no difference.
		      ■ **H_1 (Alternative Hypothesis):** Reflects the expected effect or difference.
		4. **Encourage Exploration:**
		  a. Sample topics to inspire learners:
		    ● "Does a local fitness program reduce obesity rates?"
		    ● "Do weekend markets increase local business revenue?"
		    ● "Does planting trees reduce heat in urban areas?"
		5. **Present and Discuss:**
		  a. Have each group present their research idea and hypotheses.
		  b. Facilitate a class discussion to provide feedback and refine ideas.
		
		Tips for Teachers:
		● Encourage learners to choose issues they are passionate about or that affect their daily lives.
		● Remind them that hypotheses should be testable and specific.

	Option 3: Mini-Case Studies **Instructions:** 1. Read the Mini-Case Carefully • Identify what is being introduced, tested, or compared. • Determine the outcome being measured. 2. Identify the Variables • What is the factor or treatment? • What is the result or outcome? 3. Write the Null Hypothesis (H_0) • State that the factor being tested has no effect on the outcome. • Use clear, neutral wording. 4. Write the Alternative Hypothesis (H_1) • State that the factor does affect or improve the outcome. • Match the claim implied in the case. 5. Write Hypotheses as Statements • Do not use questions. • Keep answers short and direct. **Task:** Teacher provides short case studies → learners identify hypotheses. **Example**: A farmer applies organic fertilizer on half of his rice fields to test yield difference. **H_0**: Organic fertilizer has no effect on rice yield. **H_1**: Organic fertilizer increases rice yield.	• Use this activity to prepare learners for project-based learning or community research initiatives. **Option 3: Mini-Case Studies** **Instructions:** 1. **Present Case Studies:** Share short, simple scenarios like: **Example Case Study:** A farmer applies organic fertilizer on half of his rice fields to test yield difference. a. **H_0:** Organic fertilizer has no effect on rice yield. b. **H_1:** Organic fertilizer increases rice yield. 2. **Learner Task:** a. Learners read each case. b. Identify and write H_0 and H_1. c. Optionally, explain their reasoning. 3. **Discussion:** a. Review answers as a class. b. Clarify any misconceptions. **Tips for Teachers:** • Use diverse scenarios (education, agriculture, health, etc.). • Encourage learners to justify their hypotheses.
	C.2. Making Generalization	
	Activity C.2: Wrapping up the Lesson **Part 1: Summarization Ladder** Instructions: 1. Write **5 key words** on the lesson. 2. Turn them into **3 short phrases**.	**Activity C.2: Wrapping up the Lesson** This stage ensures learners don't just know how to write hypotheses, but also why they matter and how they connect to the bigger picture of research.

	3. Then write **1 summary sentence** about null and alternative hypotheses. **Part 2: Exit Reflection** **Instructions:** In one or two sentences, explain why hypothesis testing is important in research.	The for the *generalization phase* facilitate the provided activities.
	C.3. Evaluating Learning	
	Activity C.3: Assessing Learning Outcomes **Instructions**: Do as indicated. A. Choose the Correct Answer 1. The null hypothesis (H_0) usually states: a. There is an effect or difference b. There is no effect or difference 2. Which of the following is an example of an alternative hypothesis (H_1)? a. "The new drug has no effect on headaches." b. "The new drug reduces headaches better than the old one." 3. Which statement best describes the role of H_1? a. It represents what researchers want to disprove b. It represents the claim or effect being tested 4. In hypothesis testing, researchers usually try to: a. Prove H_0 is true b. Find evidence against H_0 5. Which pair of hypotheses is correctly stated? a. a) H_0: There is a difference; H_1: There is no difference b. b) H_0: There is no difference; H_1: There is a difference B. Identify Whether the Statement is H_0 or H_1 1. "The new teaching method has no effect on learner performance." 2. "Athletes have higher GPAs than non-athletes." 3. "There is no significant difference in test scores between online and face-to-face classes." 4. "Learners who sleep at least 8 hours perform better on exams."	**Activity C.3: Assessing Learning Outcomes** Description This activity checks learners' understanding of null and alternative hypotheses (H_0 and H_1) through recognition, identification, and creation tasks. It also surfaces common misconceptions to prepare learners for formal hypothesis testing. Instructions: 1. Review Key Concepts (Before Activity) • Remind learners: ○ H_0 = no effect/difference ○ H_1 = effect/difference exists • Highlight common misconceptions (H_0 is "always wrong," H_1 must be positive). 2. Guide Recognition (Part A: Multiple Choice) • Let learners answer individually. • Ask: *"Why is this correct? Why are the others wrong?"* • Focus on keywords: no effect, difference, better, higher. 3. Support Identification (Part B: H_0 or H_1) • Use Think–Pair–Share before discussing as a class. • Ask: *"Does this statement claim no change or a difference?"* • Clarify direction and phrasing as needed. 4. Scaffold Formulation (Part C: Constructing Hypotheses) • Model one example, then let learners try independently. • Prompt: *"What would happen if nothing changes?"* → *H_0, "What is being claimed?"* → *H_1* • Check for logical pairing and clarity. 5. Facilitate Reflection & Misconceptions (Part D: Synthesis Questions)

	5. "The new phone battery does not last longer than the previous model." C. Constructing Hypotheses 1. Create one research question of your own related to school, health, or daily life. 2. Write the corresponding: ○ H_0 (null hypothesis) ○ H_1 (alternative hypothesis)	• Discuss: 1. Why H_0 is not always false 2. Common wording challenges 3. Importance of careful hypothesis writing • Use responses to decide readiness for significance testing. **Reflection Questions** Answer in 2–3 sentences each. 1. Many learners think H_0 is always false. Why is this incorrect, and what is the real purpose of H_0? 2. What difficulties did you encounter when deciding whether a statement belongs to H_0 or H_1? 3. Why is it important to write hypotheses carefully before conducting any statistical test? Answer Key A. 1. b 2. b 3. b 4. b 5. b B. 1. H_0 2. H_1 3. H_0 4. H_1 5. H_0
	C.4. Additional Activities	
	Activity C.4: Extending and Reinforcing Learning **Remediation** **A. Scenario-Based Hypothesis Formulation** **Instructions:** a. Read each research scenario carefully. b. Write the null hypothesis (H_0) – statement of no effect/difference. c. Write the alternative hypothesis (H_1) – statement that there is an effect/difference. d. Be prepared to explain your reasoning. **Scenarios** 1. A company wants to test if a new training program improves employee productivity.	**Activity C.4: Extending and Reinforcing Learning** This activity helps learners apply, reinforce, and extend their understanding of null (H_0) and alternative (H_1) hypotheses. • Remediation activities support learners who need extra guidance in identifying and writing hypotheses correctly. • Enhancement activities challenge learners who are ready for higher-order thinking, including correcting errors and creating mini research proposals. The goal is for learners to develop confidence, accuracy, and independence in formulating hypotheses in real-life contexts. **Instructions:**

	2. Researchers are studying whether a new fertilizer increases crop yield. 3. A school investigates if longer recess periods improve learners' focus in class. 4. Scientists want to determine if a new drug lowers blood pressure. 5. A tech company wants to test whether a new app improves user satisfaction. **B. Hypothesis Matching Game** **Instructions**: a. You are given a list of research questions and a separate list of hypotheses. b. Match each research question to the correct **H_0 and H_1**. c. Discuss your choices with a partner before confirming. **Scenarios**: Does caffeine affect reaction time? 1. Does listening to classical music improve memory recall? 2. Does a new teaching method improve reading comprehension? 3. Does regular meditation reduce anxiety levels? 4. Does using blue light filters improve sleep quality? **Enhancement** **B. Hypothesis Correction Challenge** **Instructions:** a. You are provided with incorrectly written hypotheses. b. Identify the errors in the hypotheses. c. Rewrite them correctly as H_0 and H_1. d. Explain why the original hypothesis was incorrect. **Scenarios:** 1. H_0: Learners who study score higher. H_1: Learners who study score differently. 2. H_0: Exercise improves health. H_1: Exercise has no effect on health.	1. Introduce the Activity (5–7 minutes) • Review H_0 and H_1 definitions. • Explain the difference between the two and why both are important. • Briefly describe the two tracks: Remediation for guided practice, Enhancement for deeper application. 2. Conduct Remediation Activities (20–25 minutes) Scenario-Based Hypothesis Formulation • Have learners work individually first. • Pair learners to explain reasoning to each other. • Teacher circulates and provides feedback, highlighting correct phrasing and misconceptions. Hypothesis Matching Game • Learners match research questions to hypotheses in pairs. • Facilitate a whole-class discussion to confirm answers. • Address errors in matching and explain why certain statements are H_0 or H_1. 3. Conduct Enhancement Activities (25–30 minutes) Hypothesis Correction Challenge • Learners work individually or in pairs to identify errors and rewrite hypotheses. • Encourage explanation of why the original statement was incorrect. • Teacher reviews common mistakes with the class. Mini Research Proposal • Learners select a topic and formulate H_0 and H_1. • Ask them to identify variables and a simple testing method. • Optionally, have learners present proposals to peers for feedback. 4. Reflection and Synthesis (10 minutes) • Ask learners to answer reflection questions: 1. What was the most challenging part of writing H_0 and H_1? 2. How did peer review or correction help improve your understanding? 3. Why is it important to clearly state hypotheses before testing?

	3. H_0: Music makes people happy. H_1: Music doesn't affect happiness. 4. H_0: New drug cures disease. H_1: Drug has some effect. 5. H_0: Reading improves IQ. H_1: Reading sometimes improves IQ. **B. Mini Research Proposal** **Instructions:** a. Choose a research topic of interest. b. Formulate the **H_0 and H_1** for your study. c. Identify independent and dependent variables. d. Suggest a simple method to test your hypothesis. **Research Topics:** 1. Topic: Effect of caffeine on study efficiency. 2. Topic: Influence of social media usage on sleep quality. 3. Topic: Impact of background music on concentration. 4. Topic: Effect of morning exercise on mood. 5. Topic: Influence of screen time on attention span in children.	• Discuss misconceptions observed during activities and clarify. 5. Assessment/Follow-Up • Check learner worksheets for accuracy. • Formatively assess readiness for significance testing. • Provide additional practice for learners struggling with H_0/H_1 formulation **Answer Keys:** **Remediation** **A. Scenario-Based Hypothesis Formulation** 1. H_0: The new training program does not improve employee productivity. H_1: The new training program improves employee productivity. 2. H_0: The new fertilizer has no effect on crop yield. H_1: The new fertilizer increases crop yield. 3. H_0: Longer recess periods do not improve learners' focus. H_1: Longer recess periods improve learners' focus. 4. H_0: The new drug has no effect on blood pressure. H_1: The new drug lowers blood pressure. 5. H_0: The new app does not improve user satisfaction. H_1: The new app improves user satisfaction. **B. Hypothesis Matching Game** 1. H_0: Caffeine has no effect on reaction time. H_1: Caffeine affects reaction time. 2. H_0: Listening to classical music does not affect memory recall. H_1: Listening to classical music improves memory recall. 3. H_0: The new teaching method does not improve reading comprehension. H_1: The new teaching method improves reading comprehension. 4. H_0: Regular meditation does not reduce anxiety levels. H_1: Regular meditation reduces anxiety levels. 5. H_0: Using blue light filters does not improve sleep quality. H_1: Using blue light filters improves sleep quality.

		Enhancement **A. Hypothesis Correction Challenge** 1. H_0: Learners who study do not score differently. H_1: Learners who study score differently. 2. H_0: Exercise has no effect on health. H_1: Exercise improves health. 3. H_0: Music does not affect happiness. H_1: Music affects happiness. 4. H_0: New drug has no effect on disease. H_1: New drug cures the disease. 5. H_0: Reading does not improve IQ. H_1: Reading improves IQ. **B. Mini Research Proposal** 1. H_0: Caffeine has no effect on study efficiency. H_1: Caffeine improves study efficiency. 2. H_0: Social media usage does not affect sleep quality. H_1: Social media usage reduces sleep quality. 3. H_0: Background music does not affect concentration. H_1: Background music improves concentration. 4. H_0: Morning exercise does not affect mood. H_1: Morning exercise improves mood. 5. H_0: Screen time does not affect attention span. H_1: Increased screen time reduces attention span.
III. CONTENT	**Lesson 1.2: Type I and Type II Error** *Recommended time allotment: 2 hours*	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. define the level of significance and explain its role in hypothesis testing. 2. identify and differentiate between Type I and Type II errors. 3. understand the importance of balancing errors to make informed decisions in statistical analysis.	
V. PROCEDURES	**ACTIVITIES**	**ANNOTATION**
A. Activating Prior Knowledge	***A.1 Eliciting Prior Knowledge***	
	Activity A.1: Leveling Learner Readiness **Instructions** 1. 1Answer all questions to the best of your ability. 2. Work individually first, then discuss with a partner. 3. Be ready to explain your answers in class.	**Activity A.1: Leveling Learner Readiness** This pre-assessment checks learners' prior understanding of key concepts before introducing formal hypothesis testing procedures. It uses multiple formats such as multiple choice,

	1. Which of the following best defines the level of significance (α) in hypothesis testing? a. The probability of accepting a true null hypothesis. b. The probability of rejecting a false null hypothesis c. The probability of rejecting a true null hypothesis d. The probability of accepting a false alternative hypothesis 2. A medical researcher tests a new drug. H_0: The drug has no effect. The researcher rejects H_0 even though the drug actually has no effect. What type of error occurred? a. Type I b. Type II **3.** A school tests a new teaching method. H_0: The new method has no effect. The study fails to reject H_0 even though the method actually improves learning. What type of error occurred? a) Type I b) Type II 4. Match the error type with the correct description. Error Type — Description 1. Type I — a. Failing to detect an effect that actually exists 2. Type II — b. Detecting an effect that does not exist *4.* Why is it important to balance Type I and Type II errors when making decisions in research? *(Expect 1–2 sentences.)*	scenario-based, and matching aim to assess knowledge, reasoning, and application. • Assess learners' prior knowledge of statistical errors and significance. • Identify misconceptions before formal instruction. **Strategies** **1. Think-Pair-Share** • Have learners answer individually, then discuss in pairs, then share with the class. • Encourages reasoning and justification of answers. **2. Scenario-Based Discussion** • Present real-life examples (medical testing, product testing, education) and ask learners to identify errors. • Helps learners connect abstract concepts to practical situations. **3. Concept Mapping (Optional)** • Ask learners to draw a simple map showing H_0, H_1, Type I, Type II, and α. • Supports visual understanding of relationships between concepts. **Reflection Questions** 1. Which scenario was most confusing when identifying Type I or Type II errors? Why? 2. How does the level of significance influence the likelihood of making errors? 3. Why is it important to consider both types of errors when making decisions in research?
	A.2 Establishing the Purpose	
	Activity A. 2: Appreciating Lesson Relevance **Instructions**: 1. Read each question or scenario carefully. 2. Answer using your own reasoning — there are no strict right or wrong answers yet. 3. Discuss your answers with a partner or group. 4. Be ready to share your thoughts with the class.	**Activity A. 2: Appreciating Lesson Relevance** This activity helps learners connect real-life decision-making with statistical concepts like confidence and the level of significance (α). Learners will explore how much certainty is required before making decisions, recognize different levels of risk-taking, and reflect on the consequences of mistakes.

Part 1:
Estimating Confidence and Level of Significance

Directions:

1. Imagine each situation.
2. Ask yourself: "How confident would I want to be before making this decision?"
3. Write your estimated confidence as a percentage.
4. Calculate the level of significance (α) as: α = 100% – confidence.

Situations:

- A doctor is deciding whether to tell a patient they have cancer based on a test result.
- You're checking the weather forecast to decide whether to carry an umbrella.
- A company is testing whether a new drug is safe for public use.
- A learner checks if their essay is plagiarized using an online checker.
- A smart security system decides whether to alert the police about an intruder.

Part 2: **Strict or Lenient?**
Directions:

1. For each situation, decide whether the decision-maker is being strict (very cautious) or lenient (more relaxed).
2. Briefly explain why.

Situations:

- A scientist only accepts results that are 99.9% certain.
- A teacher accepts a learner's answer if they're 80% confident it's correct.
- A weather app sends alerts only if there's at least a 95% chance of a storm.
- A spam filter marks an email as junk even if there's only a 10% chance it's spam.
- A doctor treats a patient only if the diagnosis is correct with 99% certainty.

Part 3: Thinking About Mistakes

Instructions:
1. Set the Context

- Begin with a short discussion:
  "In real life, do we ever wait until we are 100% sure before making decisions?"
- Allow 2–3 learner responses.

Key Teacher Points to Emphasize

- Real-life decisions are made under uncertainty.
- Statistics helps us make decisions by controlling how much risk we accept.
- Today's activity is about thinking, not getting correct answers yet.

Important:
Do not introduce formulas, hypothesis testing steps, or α formally yet. Let intuition come first.

B. During the Activity
Part 1:
Estimating Confidence and Level of Significance
Teacher Role

1. Read the first scenario aloud (doctor diagnosing cancer).
2. Ask guiding questions:
   - "What could happen if the decision is wrong?"
   - "Would you want to be very sure or just somewhat sure?"
3. Model one example:
   - If a learner says 95% confidence, write:
     - Confidence = 95%
     - α = 100% – 95% = 5% or 0.05
4. Let learners work individually, then compare in pairs.

Key Facilitation Prompts

- "Why did you choose a higher confidence here than in the weather example?"
- "Which situations need more caution? Why?"

What to Listen For

- Learners naturally assigning lower α to high-risk situations (medicine, security)

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Directions:

1. Reflect on each question and write a short response or discuss with a partner.
2. There are no right or wrong answers — this is about connecting your experiences to statistical decision-making.

Questions:

- Think of a time when you made a decision and later realized you were wrong. What happened?
- Have you ever done something “just to be safe,” even if you weren’t completely sure? Describe the situation.
- Can you think of a real-life situation where being wrong could lead to serious consequences?
- Why don’t people always wait until they’re 100% sure before making a decision?
- How sure do you usually want to be before making an important decision?

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- Higher α for low-risk decisions (umbrella, spam filter).

Reinforce that α reflects risk tolerance, not correctness.

Part 2:
Strict or Lenient Decisions
Teacher Role

1. Explain:
    - Strict = very cautious, low tolerance for error
    - Lenient = more relaxed, higher tolerance for error
2. Let learners classify each situation individually, then discuss in pairs.
3. Facilitate a whole-class check by asking:
    - “Who labeled this as strict?”
    - “Who labeled it as lenient?”
    - “Convince us why.”

Guiding Questions

- **“**What kind of mistake is the decision-maker trying to avoid?”
- “Is it worse to act when you shouldn’t, or not act when you should?”

Teacher Insight to Highlight

- Strict decisions reduce false positives (Type I errors).
- Lenient decisions reduce false negatives (Type II errors).
  (Do not yet formalize—just plant the idea.)

Part 3: Thinking About Mistakes
Teacher Role

1. Ask learners to write brief responses or discuss with a partner.
2. Choose 2–3 questions for class sharing**.**

Facilitation Prompts

- “What was the consequence of being wrong?”
- “Would you rather make that mistake again—or a different one?”

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Purpose of This Part

- Normalize mistakes as part of decision-making.
- Prepare learners emotionally and cognitively for Type I and Type II errors.

Emphasize:
"Statistics doesn't eliminate mistakes—it helps us manage them."

C. After the Activity (Synthesis & Transition)
Guide learners to articulate these ideas:

- We rarely wait for 100% certainty.
- Different decisions require different confidence levels.
- The level of significance (α) sets a boundary for acceptable risk.
-

Key Connecting Question

- "If we set α too low, what kind of mistake might we avoid?
  If we set it too high, what kind of mistake might increase?"
- Let learners answer intuitively.

D. Addressing Common Misconceptions

Misconception	Teacher Response
α is the chance of being wrong overall	"α controls one specific kind of mistake."
Higher confidence is always better	"Higher confidence can delay action or miss real effects."
Statistics gives certainty	"Statistics manages uncertainty—it doesn't remove it."

E. Transition to Next Lesson
Close with:

- "Today, you decided how much risk you were willing to take.
  In our next lesson, we'll name the two kinds of mistakes we can make—and see how α controls them."
- This smoothly leads into Type I and Type II errors and formal hypothesis testing.

B. Instituting New Knowledge	***B.1. Presenting Examples***	
	Activity B.1 Exploring Key Concepts **Instructions**: i. Read each situation carefully. ii. Identify what decision was made. iii. Decide whether the situation shows a Type I error or Type II error. iv. Be ready to explain why. 1. A medical test shows that a patient has a disease, but further testing reveals the patient is actually healthy. 2. A medical test fails to detect a disease in a patient who is actually sick. 3. A fire alarm goes off in a building, but there is no actual fire. 4. A fire alarm does not go off even though there is a real fire. 5. A spam filter allows a dangerous phishing email into the inbox. 6. A spam filter marks an important email as spam even though it is legitimate. 7. A court convicts a person of a crime, but later evidence proves the person is innocent. 8. A guilty person is found not guilty due to lack of evidence. 9. A teacher concludes that a new teaching strategy improved test scores, but the improvement happened only by chance. 10. A researcher concludes that a new fertilizer has no effect, even though it truly increases crop yield.	**Activity B.1 Exploring Key Concepts** This activity is designed to help learners clearly differentiate between Type I and Type II errors through real-life, relatable situations. Learners analyze two sets of scenarios: one illustrating Type I errors (false positives) and the other illustrating Type II errors (false negatives). By examining the decision made versus what is true in reality, learners develop a deeper understanding of how errors occur in hypothesis testing. The activity emphasizes critical thinking, discussion, and justification of answers, allowing learners to recognize that statistical decisions often involve trade-offs. Through guided reflection, learners connect these statistical concepts to everyday decision-making and real-world consequences. It aims to differentiate Type I error from Type II error using real-life situations, explain why each error happens in decision-making and recognize that avoiding one error may increase the other To facilitate the activity, you may apply any of the provided strategies: **Strategy 1: Contrast and Compare** • Present Set A first, then Set B. • Ask: "What do all the situations in Set A have in common?" "How are the situations in Set B different?" • Guide learners to articulate: Set A = Something is detected but not real Set B = Something is real but not detected **Strategy 2: Decision Mapping** On the board, draw a simple table:

Reality	Decision	Error Type
No effect	Says there is an effect	Type I
Effect exists	Says there is no effect	Type II

Refer back to this table for every scenario.

Strategy 3: Think-Pair-Explain

- Learners first classify the error individually.
- Pair learners and ask them to justify their choice.
- Call on pairs to explain using this sentence frame: "This is a Type ___ error because the decision says ___, but in reality ___."

Reflection Questions

1. Which type of error is more dangerous in medical testing? Why?
2. Which type of error might be more acceptable in daily life?
3. Why can't we eliminate both Type I and Type II errors completely?
4. How does being very strict reduce one error but increase the other?

Answer Key:
1.Type I
2. Type II
3. Type I
4. Type II
5. Type II
6. Type I
7. Type I
8. Type II
9. Type I
10. Type II

B.2. Discussing the Concept

Activity B.2:
Deepening Conceptual Understanding

Activity B.2:
Deepening Conceptual Understanding

Topic: Level of Significance (α) and Types of Errors

	Instructions: To better understand the level of significance of types of error, participate actively to the discussion. Topic**:** **Level of Significance (α) and Types of Errors** **Step 1:** Concept Discussion **What is the Level of Significance (α)?** a. In hypothesis testing, we never know with 100% certainty whether we're right or wrong. b. The level of significance (α) is the threshold of risk we are willing to take to reject the null hypothesis. c. Common values are: ○ **α = 0.05** → 5% risk of making a **Type I error** ○ **α = 0.01** → 1% risk ○ **α = 0.10** → 10% risk **Think of α as your personal "risk limit"** before you say, "Yes, I think there is an effect." What are Type I and Type II Errors?	This activity deepens learners' conceptual understanding of hypothesis testing by focusing on the level of significance (α) and Type I and Type II errors. Through structured discussion, real-world analogies, guided examples, and collaborative learner activities, learners explore how statistical decisions involve balancing risk and consequences. The activity emphasizes that hypothesis testing is not about certainty, but about managing uncertainty responsibly. By analyzing authentic scenarios and making justified choices of α, learners develop critical thinking skills and connect abstract statistical concepts to real-life decision-making in health, safety, education, and technology. To facilitate the discussion, apply any of the following strategies: Strategy 1: **Explicit Teaching + Socratic Questioning** How to Facilitate: 1. Begin Step 1 by explicitly explaining α, Type I error, and Type II error using simple language. 2. Use analogies (courtroom, fire alarm, medical testing) to ground abstract ideas. 3. Ask guiding questions such as: • "What happens if we are wrong in this situation?" • "Which mistake is more dangerous here?" 4. Emphasize that α is a choice, not a fixed rule, and that different situations demand different risk levels. Tip: Pause frequently to check understanding using thumbs-up/thumbs-down or quick verbal responses. Strategy 2: **Think–Pair–Share and Small-Group Reasoning** **How to Facilitate:** 1. During Activity 1 and Activity 2, have learners: 2. Think individually, 3. Discuss with a partner or small group, 4. Share reasoning with the class.

Term	What it means	Example
Type I Error	Rejecting a true null hypothesis (a **false positive**)	Alarm rings, but there's no fire
Type II Error	Failing to reject a false null hypothesis (a **false negative**)	No alarm, but there is a fire

Real-World Analogy: Courtroom
- **H_0 (null)**: The defendant is innocent
- **H_1 (alternative)**: The defendant is guilty
- **Type I Error**: Convicting an innocent person
- **Type II Error**: Letting a guilty person go free
- Courts try to avoid **Type I errors** because punishing an innocent person is serious.

When you perform a hypothesis test, there are four possible outcomes depending on the actual truth (or falseness) of the null hypothesis H_0 and the decision to reject or not. The outcomes are summarized in the following table:

Action	H_0 is Actually	…
	True	*False*
Do not reject H_0	*Correct Outcomes*	*Type II Error*
Reject H_0	*Type I Error*	*Correct Outcomes*

The four possible outcomes in the table are:

1. The decision is not to reject H0 when H0 is true (correct decision).
2. The decision is to reject H0 when H0 is true (incorrect decision known as a Type I error).
3. The decision is not to reject H0 when, in fact, H0 is false (incorrect decision known as a Type II error).
4. The decision is to reject H0 when H0 is false (correct decision whose probability is called the Power of the Test).

Each of the errors occurs with a particular probability. The Greek letters α and β represent the probabilities.

- ✔ α = probability of a Type I error = P(Type I error) = probability of rejecting the null hypothesis when the null hypothesis is true.
- ✔ β = probability of a Type II error = P(Type II error) = probability of not rejecting the null hypothesis when the null hypothesis is false.

α and β should be as small as possible because they are probabilities of errors. They are rarely zero.

The Power of the Test is 1 – β. Ideally, we want a high power that is as close to one as possible. Increasing the sample size can increase the Power of the Test.
The following are examples of Type I and Type II errors.

Example 1. Suppose the null hypothesis, H_0, is: Frank's rock climbing equipment is safe.

- ✔ Type I error: Frank thinks that his rock-climbing equipment may not be safe when, in fact, it really is safe. Type II error: Frank thinks that his rock-climbing equipment may be safe when, in fact, it is not safe.

5. Require learners to justify their answers, not just label them.
6. When incorrect answers arise, treat them as learning opportunities:

- "That's an interesting choice—what assumption led you there?"

Tip: Write contrasting answers on the board and ask learners to compare reasoning, not just correctness.

Strategy 3:
Error Comparison and Consequence Mapping

How to Facilitate:

1. For Activities 1 and 3, ask learners to complete this sentence:
- "If we make this mistake, the consequence is…"
2. Draw a simple two-column table on the board:
- Column 1: Type I Error – "False Alarm"
- Column 2: Type II Error – "Missed Detection"
3. Guide learners to recognize that:
- Lower α reduces Type I errors but may increase Type II errors.
- Decisions depend on which error is more costly.

Tip: Revisit earlier examples (fire alarm, medical testing) during the wrap-up to reinforce transfer of learning.

Wrap-Up Facilitation

1. Revisit the discussion prompt:
- "When is it okay to take more risk? When should we be more cautious?"
2. Encourage learners to share new real-life examples, reinforcing ownership of learning.
3. Summarize with a key message:
- "Statistics helps us make better decisions—not perfect ones."

Answer Key:
Activity 1

1. **Type I Error**
2. **Type II Error**
3. **Correct Decision**
4. **Type I Error**

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- ✔ α = probability that Frank thinks his rock-climbing equipment may not be safe when, in fact, it really is safe. β = probability that Frank thinks his rock-climbing equipment may be safe when, in fact, it is not safe.

Notice that, in this case, the error with the greater consequence is the Type II error. (If Frank thinks his rock-climbing equipment is safe, he will go ahead and use it.)

Example 2. Suppose the null hypothesis, H0, is: The victim of an automobile accident is alive when he arrives at the emergency room of a hospital.

- ✔ Type I error: The emergency crew thinks that the victim is dead when, in fact, the victim is alive. Type II error: The emergency crew does not know if the victim is alive when, in fact, the victim is dead.
- ✔ α = probability that the emergency crew thinks the victim is dead when, in fact, he is really alive = P (Type I error). β = probability that the emergency crew does not know if the victim is alive when, in fact, the victim is dead = P (Type II error).

The error with the greater consequence is the Type I error. (If the emergency crew thinks the victim is dead, they will not treat him.)

Step 2: Learner Activities
Activity 1: "Name That Error!"
Instructions to Learners: Read each scenario below. Decide whether it describes a Type I error, a Type II error, or neither. Be prepared to explain your reasoning.

Scenarios:

1. A pregnancy test says a woman is pregnant, but she is not.
2. A smoke detector doesn't go off, even though there's smoke in the room.
3. A COVID-19 test correctly detects that someone is infected.

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5. **Type II Error**

Activity 2
Suggested Answers:

1. **0.01** – Need high certainty; avoid false positives
2. **0.10** – False positives are acceptable to catch dishonest applicants
3. **0.05** – Balance between safety and false alarms
4. **0.01** – Health consequences demand caution
5. **0.10** – Rain is minor; better to over-alert than under-alert

Activity 3
Learner response might look like:

- ✔ H_0: The new method is **not more effective** than the current one.
- ✔ H_1: The new method **is more effective**.
- ✔ Type I Error: They switch to the new method, but it's actually not better.
- ✔ Type II Error: They keep the old method, but the new one is actually better.

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4. A security system sends police even though there is no intruder.
5. A learner is cheating, but the teacher fails to notice

Activity 2: “Pick Your Risk Level (α)”
Instructions to Learners: For each situation, choose what level of significance (α) would be most appropriate: 0.01, 0.05, or 0.10. Then justify your choice.
Situations:
1. A doctor testing a new drug for a serious disease.
2. A hiring manager scanning resumes for fake qualifications.
3. A school setting off a fire alarm system.
4. A scientist testing a new vaccine.
5. A weather app deciding whether to send a rain alert.

Activity 3: “Error Detective”
Instructions to Learners: For each situation below:
1. Identify the null hypothesis (H_0) and alternative hypothesis (H_1).
2. Describe what a Type I and Type II error would be.
Situation Example:
"A school is testing a new method to improve math scores. The current method is believed to be effective."

You can create 2–3 more similar situations and have learners complete them in groups.

Step 3: Wrap-Up Discussion Prompt
The learners will reflect:
“Now that you’ve explored these ideas, when do you think it's okay to take more risk (higher α)? When do we need to be extra cautious (lower α)? Can you think of any new examples from real life?
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<tr><td></td><td colspan="2">B.3. Developing Mastery</td></tr>
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Activity B.3: Practicing Learned Skills

Instructions:
Do as indicated.
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Activity B.3: Practicing Learned Skills

This activity helps learners apply their understanding of Type I and Type II errors and the level of significance (α) in real-life contexts. Through a combination of quizzes,
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	Part 1: Error Analysis Quiz **Directions:** Read each question carefully and choose the correct answer. For each, explain why your choice is correct. 1. In a clinical trial, the researchers reject the null hypothesis when the new drug actually has no effect. This is an example of: a. Type I Error b. Type II Error c. Correct Decision 2. A smoke detector fails to sound during a fire. What type of error is this? a. Type I Error b. Type II Error c. Correct Decision 3. A test with $\alpha = 0.05$ means: a. There is a 5% chance of making a Type II error b. There is a 5% chance of making a Type I error c. There is a 95% chance of a Type I error 4. What happens if the significance level α is set very low (e.g., 0.01)? a. Increased chance of Type I error b. Increased chance of Type II error c. No change in errors 5. When the consequences of a false positive are serious, you should: a. Increase α to 0.10 b. Decrease α to 0.01 c. Ignore α **Part 2: Scenario-Based Decision Making** **Directions:** For each scenario below: 1. Identify the null and alternative hypotheses. 2. Determine which type of error is more serious. 3. Recommend an appropriate α level and explain your choice. Scenarios:	scenario analysis, and debate, learners practice identifying errors, evaluating risk, and making informed decisions about significance levels. The activity strengthens conceptual understanding, encourages critical thinking, and promotes collaborative learning. Learners also reflect on how different contexts change the consequences of errors and the choice of α. **Facilitation Guide and Strategies** **Strategy 1:** **Guided Error Analysis** **How to Facilitate:** 1. Present Activity 1: Error Analysis Quiz. 2. Have learners answer individually, then pair up to discuss their reasoning. 3. Encourage justification: *"Why is this Type I or Type II?"* 4. Circulate and provide clarifications, highlighting common misconceptions: ○ Confusing α with β ○ Thinking Type I or II errors are always "bad" in all contexts **Tip:** Use a real-world example (medical test, fire alarm) to connect abstract errors to practical outcomes. **Strategy 2:** **Scenario-Based Decision Making** **How to Facilitate:** 1. Divide learners into small groups. 2. Provide Part 2: Scenario-Based Decision Making. 3. For each scenario, ask groups to: ○ Identify H_0 and H_1 ○ Decide which error has more serious consequences ○ Recommend an α level and justify it **Prompts to guide discussion:** • "If we set α too high, what could go wrong?" • "Which error is more costly here, Type I or Type II?" • "How would this differ in another context?" **Tip:** Encourage groups to share their reasoning with the class to compare decisions and α choices. Strategy 3: **Structured Debate** (Whole Class)

	1. Testing if a new airplane model is safer than the current model. 2. Screening job applicants for criminal records. 3. Detecting a disease outbreak in a community. **Part 3: Group Debate** **Directions:** Divide the class into two groups. One group argues that Type I errors are more serious, the other that Type II errors are more serious. Each group must: ● Provide real-world examples. ● Explain consequences. ● Discuss how significance level affects their position. After the debate, conduct a reflection discussing that seriousness depends on context.	**How to Facilitate:** 1. Divide the class into two groups: o Group A: Type I errors are more serious o Group B: Type II errors are more serious 2. Give 5–10 minutes for preparation. Each group must: o Provide examples o Explain consequences of their assigned error o Discuss how α affects the likelihood of their error 3. Conduct the debate, allowing rebuttals and discussion. 4. Conclude by emphasizing context: the seriousness of Type I vs Type II errors depends on the situation. **Tip:** Use a **timer** and ensure equal participation to keep the debate focused and productive. **Reflection Questions** 1. In which situations is a Type I error more serious? Give real-life examples. 2. In which situations is a Type II error more serious? Give real-life examples. 3. How does the level of significance (α) influence the balance between Type I and Type II errors? 4. After completing these activities, how confident are you in identifying errors in a hypothesis test? 5. How can understanding these errors help you make better decisions in daily life or in scientific research? **Answer Key:** 1.a. Type I Error — Rejecting a true null hypothesis. 2.b. Type II Error — Failing to reject a false null hypothesis. 3. b. 5% chance of Type I error (false positive). 4. b. Increased chance of Type II error (more cautious, less false positives, but more false negatives). 5. b. Decrease α to 0.01 to minimize false positives. **Sample Answer for #1:** ● H_0: New airplane is **not** safer. ● H_1: New airplane **is** safer. ● More serious error: Type I (declaring safer when it's not). ● Recommended α: 0.01 to minimize risk of false positives (avoiding unsafe planes).

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td rowspan="2">C. Demonstrating Knowledge and Skills</td><td colspan="2">C.1. Finding Practical Application</td></tr>
<tr><td>

Activity C.1: Making Real-World Connections

Instructions:
Step 1: Understand Your Variable

1. Review your assigned variable from the performance task (e.g., hours of study, sleep hours, practice tests).
2. Think about how this variable might affect academic performance.

Step 2: Formulate Hypotheses

1. Write a null hypothesis (H_0): Assume the variable has no effect on academic performance.
2. Write an alternative hypothesis (H_1): Assume the variable does have an effect.

Step 3: Identify Possible Errors

1. Describe a Type I error for your variable: What would happen if you wrongly conclude there is an effect when there isn't?
2. Describe a Type II error for your variable: What would happen if you fail to detect an effect when there actually is one?

Step 4: Decide on Level of Significance (α)

1. Consider the consequences of each error.
2. Choose an α level (0.01, 0.05, or 0.10) that balances caution and risk.
3. Justify your choice in writing.

Step 5: Organize Your Ideas

1. Create a simple table to summarize your variable, hypotheses, errors, and α choice. Example:

<table>
<tr><th>Variable</th><th>H_0</th><th>H_1</th><th>Type I Error</th><th>Type II Error</th><th>α</th><th>Reason</th></tr>
<tr><td>Study Hours</td><td>No effect on grades</td><td>Has effect</td><td>Conclude effect when none exists</td><td>Fail to detect effect</td><td>0.05</td><td>Moderate risk; false positives may mislead strategies</td></tr>
</table>

Step 6: Reflection

1. Discuss with your partner or group:

</td><td>

Activity C.1: Making Real-World Connections

In this activity, learners will explore how the concepts of level of significance (α) and Type I and Type II errors apply to real-life decisions. Using your assigned variable (e.g., study hours, sleep, practice tests), you will analyze possible outcomes and consider the risks of making mistakes when interpreting data. This prepares you for your upcoming performance task, where you will investigate the relationship between study habits and academic performance.

By completing this activity, they will:

- Understand the potential consequences of errors in data analysis.
- Decide on an appropriate level of significance (α) for your investigation.
- Connect abstract statistical concepts to real-life decision-making in education.

Instructions:
Step 1: Formulating the Context

- Before performing calculations, learners should frame the research question:

"Is there a significant relationship between study hours and mathematics performance?"

- Discuss what it means to accept or reject the null hypothesis (H_0) in this context:
  - H_0: There is no relationship between study habits and math performance.
  - H_1: There is a relationship between study habits and math performance.

Teacher Tip: Emphasize that α is a "risk threshold", which reflects how willing they are to make a Type I error (e.g., claiming study habits affect performance when they actually do not).

Step 2: Identifying Possible Errors
Ask learners to think critically about the consequences of errors in this real-life scenario:

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</table>

- Which error is more serious for your variable?
- How does your α choice influence the risk of making errors?

2. Be prepared to share your reasoning with the class.

Error Type	Practical Example in Task	Consequences
Type I	Concluding that study hours significantly affect math performance when in reality there is no effect	Learners or educators might change study strategies unnecessarily or invest time in ineffective methods
Type II	Concluding that study hours do not affect math performance when in reality there is an effect	Learners might miss strategies that could improve performance; opportunities for intervention are lost

Facilitation Tip: Use realistic consequences to help learners weigh which error is more serious in their context.

Step 3: Applying α in Decision-Making

- Introduce α (level of significance) as a decision threshold:
  - High α (e.g., 0.10): More willing to risk Type I error → more likely to detect an effect even if it may not exist.
  - Low α (e.g., 0.01): More cautious → reduces Type I but increases Type II error risk.
- Learners should justify their choice of α in the study context:
  - Example: Since educational interventions affect learners' time and effort, a lower α (0.05 or 0.01) may be appropriate to avoid false positives.

Strategy: Let learners rank the seriousness of errors for their assigned variable (e.g., hours of study, sleep, distractions), linking it to α choice.

Step 4: Preparing for Data Analysis Conceptually

Even if formal t-tests or correlation are not yet taught, learners can:

1. Organize data (hours of study, test scores) into tables or charts.
2. Visualize patterns with scatter plots, highlighting trends.

		3. Predict possible outcomes: Will rejecting H_0 be likely? What errors could occur? 4. Discuss implications: How could Type I or II errors affect learners' recommendations? **Teacher Tip**: Emphasize conceptual reasoning over calculations at this stage. **Step 5: Reflection and Group Discussion** Have learners answer guiding questions to connect lesson concepts to the performance task: 1. Which variable(s) might produce a serious Type I error if we make a wrong conclusion? Why? 2. Which variable(s) might produce a serious Type II error? Why? 3. If you had to choose α for your study, would you prefer a lower or higher value? Explain. 4. How does considering errors and risk change how you interpret trends in your data? 5. How could understanding errors influence future recommendations for learners or educators? **Facilitation Strategies** 1. **Scenario-Based Group Discussion** o Assign each group one or two variables from the task (e.g., study hours, sleep, practice tests). o Ask them to identify potential Type I/II errors and discuss which error would have more serious consequences. 2. **Think-Pair-Share** o Learners first reflect individually on α and error implications for their variable. o Then pair up to justify their reasoning. o Share with class to generate a range of perspectives. 3. **Visual Mapping** o Have learners create a table linking variables, potential errors, α levels, and consequences. o Helps them conceptualize risk management before formal statistical testing. **Example Table (for Learners)**

Variable	H_0	H_1	Type I Error	Type II Error	α	Reason
Study hours per week	No effect on grades	Effect exists	Conclude effect when none exists	Miss a real effect	0.05	Moderate risk; false positives can mislead study strategies
Sleep hours	No effect on grades	Effect exists	Conclude effect when none exists	Miss a real effect	0.01	False positives could waste time; missing effect could affect health

C.2. Making Generalization

Activity C.2: Wrapping Up the Lesson

Option 1: Case Comparison to Generalization
Instructions:

1. Review 3–4 real-life scenarios (e.g., drug approval, airport security, hiring decisions).
2. For each case, identify:
   - H_0 and H_1
   - Type I and Type II errors
   - Chosen or recommended significance level (α)
   - Which error is more serious and why
3. Reflect on the question:

"What general conclusions can you draw about when to use a low or high α level? In what situations is avoiding Type I errors more important? What about Type II errors?"

Expected Generalizations:

- Serious consequences of a false positive → use **low α** (e.g., 0.01)
- Serious consequences of missing an effect → may accept **higher α**

Option 2:
"Always, Sometimes, Never: Generalization Challenge"

Activity C.2: Wrapping Up the Lesson

This set of optional activities allows you to apply and deepen your understanding of hypothesis testing, significance levels (α), and Type I/II errors. By examining real-life scenarios, comparing cases, and reflecting on trade-offs, you can draw generalizations about when different α levels are appropriate and which errors are more serious. You may complete all activities or select the ones that help you most in consolidating your learning.

Instructions:

1. Review the statements below and decide if each is **Always, Sometimes, or Never True**. Explain your reasoning.

Statements:

1. A smaller α always decreases Type II error.
2. Type I errors are more serious than Type II errors.
3. A lower α level leads to fewer Type I errors.
4. We should always choose $\alpha = 0.05$.
5. If $p < \alpha$, we always accept H_1.
6. Write a generalization based on patterns in your answers.

Example:
"Lowering α reduces the chance of Type I error but increases Type II error — there is a trade-off."

Option 3: Group Sort and Generalize
Materials: Cards or slides with scenarios, significance levels, and error implications.
Instructions:

1. In small groups, sort cases by:
   - High or low α levels
   - Type I or Type II error being more serious
2. Write 2–3 generalizations from your sorted groups.

Example Generalizations:

- In medical or legal settings, avoiding Type I error is prioritized.
- In security or screening systems, more Type I errors may be tolerated to prevent Type II errors.

Option 4:
Reflection – From Examples to Rules
Prompt:
"After examining several examples, what general rules can you make about choosing α? How do real-life contexts influence whether Type I or Type II errors are more serious?"
Sentence Frames:

1. "In situations involving ____, it's better to avoid Type I error, so α should be ____."

2. "When the risk of missing something important is high, we should use a ____ α level to reduce Type II error."

Option 5:
Assessment: Generalization Skills Checklist

Criteria	✓ Yes	~ Partial	✗ Not Yet
Identifies patterns across multiple cases			
Makes logical, clearly stated generalizations			
Supports generalizations with examples			
Recognizes exceptions or trade-offs (α vs. error types)			

Sample Generalizations & Examples:

Generalization	Supporting Example
Lower α reduces Type I errors	Medical testing α = 0.01
Higher α increases Type I errors but reduces Type II errors	Airport security α = 0.10
Type I errors are worse when false positives are harmful	Legal trials, drug approvals
Type II errors are worse when missing real effects is dangerous	Disease detection, safety systems

C.3. Evaluating Learnings

Activity C.3: Assessing Learning Outcomes
Instructions:
Perform the indicated tasks.

Part 1. Multiple-Choice
Directions:
Choose the correct answer.
1. A significance level (α) of 0.01 means:
   a, There is a 1% chance of making a Type II error

Activity C.3: Assessing Learning Outcomes
This activity evaluates learners' understanding of level of significance (α), Type I and Type II errors, and decision-making in hypothesis testing. It combines multiple-choice questions, scenario-based short answers, and matching exercises to check both conceptual knowledge and application skills.

The activity is designed to:

b. There is a 1% chance of making a Type I error
c. There is a 99% chance of making a Type I error

2. Rejecting H_0 when it is actually true is called:
   a. Type I Error
   b. Type II Error
   c. Correct Decision
3. Failing to reject H_0 when it is actually false is called:
   a. Type I Error
   b. Type II Error
   c. Correct Decision
4. Which scenario requires a very low α level?
   a. Choosing a flavor for ice cream
   b. Approving a new life-saving drug
   c. Choosing a class activity
5. Increasing α from 0.01 to 0.10 generally:
   a. Increases Type I error and decreases Type II error
   b. Decreases both Type I and Type II errors
   c. Increases Type II error and decreases Type I error

Part 2. Scenario-Based Short Answer
Directions:
For each scenario:

1. Identify H_0 and H_1
2. State which error (Type I or II) is more serious
3. Suggest an appropriate α level and justify

Scenarios:

1. A smoke detector in a school detects fire.
2. A company tests a new software for security vulnerabilities.
3. A hospital tests a new vaccine for safety.
4. Airport security screens luggage for explosives.
5. A plagiarism detection tool checks learner essays.

Part 3. Matching Activity
Directions:
Match the scenarios to the correct error type.
Scenarios:
A. A pregnancy test says pregnant, but the woman is not.
B. A fire alarm does not go off during a real fire.

1. Assess if learners can identify α and its role in hypothesis testing.
2. Determine if learners can differentiate between Type I and Type II errors.
3. Evaluate learners' ability to apply concepts to real-life scenarios and justify decisions.

This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

Reflection Questions
After completing all parts, ask learners to reflect:

1. "How does choosing a higher or lower α affect the likelihood of Type I and Type II errors?"
2. "Can you think of a real-life situation where a Type I error would be more serious than a Type II error, and vice versa?"
3. "Based on these activities, what general rules can you draw about deciding on an appropriate significance level in different contexts?"

Expected Generalizations:

- Lower α → reduces Type I error, increases Type II risk.
- Higher α → reduces Type II error, increases Type I risk.
- The seriousness of Type I vs. Type II errors depends on **context and consequences**.

Answer Key
Part 1 – Multiple Choice

1. b. There is a 1% chance of making a Type I error
2. a. Type I Error
3. b. Type II Error
4. b. Approving a new life-saving drug
5. a. Increases Type I error and decreases Type II error

Part 2 – Scenario-Based Short Answer (Sample Answers)

1. Smoke detector:
   - H_0: No fire

C. A COVID-19 test correctly identifies infection.
D. Security alerts police with no intruder.
E. A learner cheats but the teacher does not notice.

Error Types:

1. Type I Error
2. Type II Error
3. Correct Decision

- H_1: Fire exists
- More serious error: Type II (missed detection)
- α: 0.01 (high certainty)

2. Company software:
   - H_0: Software is secure
   - H_1: Software has vulnerabilities
   - More serious error: Type II (undetected vulnerabilities)
   - α: 0.05
3. Hospital vaccine:
   - H_0: Vaccine is safe
   - H_1: Vaccine is harmful
   - More serious error: Type I (approving harmful vaccine)
   - α: 0.01
4. Airport security:
   - H_0: No explosives
   - H_1: Explosives present
   - More serious error: Type II (missed explosives)
   - α: 0.01
5. Plagiarism tool:
   - H_0: Essay is original
   - H_1: Essay is plagiarized
   - More serious error: Type II (undetected plagiarism)
   - α: 0.05

Part 3 – Matching Activity

- A → Type I
- B → Type II
- C → Correct Decision
- D → Type I
- E → Type II

C.4. Additional Activities

Activity C.4:
Extending and Reinforcing Learning

Remediation Activities
Activity R.1: "Error Sorting Practice"
Instructions:

1. Read each scenario carefully.
2. Decide if it represents Type I error, Type II error, or correct decision.

Activity C.4:
Extending and Reinforcing Learning

This activity is designed to help learners deepen their understanding of hypothesis testing concepts, particularly the level of significance (α), Type I errors, and Type II errors. It includes both remediation and enhancement components:

	3. Explain your choice in 1–2 sentences.
	Scenarios:
	• A smoke alarm goes off when there's no fire.
	• A fire alarm does not ring during a real fire.
	• A pregnancy test correctly shows negative.
	• Security calls the police even though no one broke in.
	
	Activity R.2: "Significance Level Matching"
	Instructions:
	1. Match each scenario with a suitable α level (0.01, 0.05, or 0.10).
	2. Explain why this α is appropriate.
	Scenarios:
	• Approving a life-saving drug.
	• Choosing the flavor of ice cream for a party.
	• Detecting explosives at an airport.
	
	Enhancement Activities
	Activity E.1: "Design Your Own Experiment"
	Instructions:
	1. Think of a research scenario (e.g., testing a new study technique).
	2. Write the null (H_0) and alternative (H_1) hypotheses.
	3. Identify what would be a Type I error and Type II error in your scenario.
	4. Choose an appropriate α level and justify your choice.
	
	Activity E.2: "Error Consequence Debate"
	Instructions:
	1. In pairs or small groups, select one scenario (medical testing, airport security, school grading).
	2. Decide whether Type I or Type II error is more serious and why.
	3. Present your reasoning to the class.
		Remediation activities target learners who may need additional support in recognizing errors and understanding the consequences of different α levels.
		
		Enhancement activities challenge advanced learners to apply concepts creatively, justify decisions, and critically evaluate trade-offs between errors in real-life scenarios.
		
		Through these activities, learners will strengthen conceptual understanding, practice decision-making in realistic contexts, and make informed judgments about the seriousness of errors.
		
		Instructions:
		Before the Activity:
		1. Review the scenarios and answers for both remediation and enhancement activities.
		2. Prepare materials: worksheets, slides, or printed scenario cards.
		3. Ensure learners understand key concepts: H_0, H_1, Type I error, Type II error, and significance level α.
		
		During the Activity:
		• **Remediation Activities:**
		  ○ Guide learners through R.1 "Error Sorting Practice" and R.2 "Significance Level Matching".
		  ○ Encourage think-pair-share: learners decide individually, then discuss with a partner.
		  ○ Provide immediate feedback and clarify misconceptions.
		  ○ Use real-world analogies (fire alarms, medical tests, security alerts) to reinforce understanding.
		
		• **Enhancement Activities:**
		  ○ For E.1 "Design Your Own Experiment", encourage learners to choose realistic or personally meaningful scenarios.

		○ Promote peer review: have learners exchange scenarios and evaluate each other's identification of H_0, H_1, and error types. ○ For E.2 "Error Consequence Debate", organize small group debates. Each group argues why Type I or Type II error is more serious in a given context, using evidence and examples. ○ Facilitate discussion on trade-offs between α levels and error risks. **After the Activity:** • **Summarize key takeaways:** 1. Type I errors are false positives; Type II errors are false negatives. 2. The choice of α depends on the consequences of each error. 3. Balancing Type I and II errors is crucial in real-life decision-making. **Answer Key**: R.1 1 → Type I (false positive) 2 → Type II (false negative) 3 → Correct Decision 4 → Type I (false positive) R.2 1 → $\alpha = 0.01$ (high certainty needed, avoid Type I error) 2 → $\alpha = 0.10$ (low risk, less serious) 3 → $\alpha = 0.01$ (high certainty, avoid missing danger Sample Answer: E.1. Scenario: Testing if a new study app improves math scores H_0: The app has no effect on scores H_1: The app improves scores Type I Error: Concluding the app works when it doesn't Type II Error: Concluding the app doesn't work when it does

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E.2
Medical testing: Type I error (approving harmful drug) more serious
Airport security: Type II error (missing a threat) more serious
School grading: Type I error (marking a learner incorrectly) less serious</td></tr>
<tr><td>III. CONTENT</td><td colspan="2">Lesson 11.3: Test of Mean for One Population
Recommended time allotment: 2 hours</td></tr>
<tr><td>IV. OBJECTIVES</td><td colspan="2">By the end of the lesson, the learners…
1. perform appropriate statistical tests involving hypothesis testing about the population mean
2. use appropriate technology (e.g. MS Excel, Jamovi) in testing hypothesis for the mean of one population involving real-life situations
3. analyze and interpret the results of hypothesis testing to draw meaningful conclusions about the population</td></tr>
<tr><td>V. PROCEDURES</td><td>ACTIVITIES</td><td>ANNOTATION</td></tr>
<tr><td rowspan="2">A. Activating Prior Knowledge</td><td colspan="2">A.1. Eliciting Prior Knowledge</td></tr>
<tr><td>Activity A.1: Leveling Learner Readiness

Instructions:
1. Read each statement carefully.
1. The average height of Filipino teenagers is 5'4".
2. Most learners prefer online learning.
3. The mean score in the National Achievement Test is 70.
4. Learners in our barangay walk 2 kilometers to school daily.
5. The average baon of Grade 10 learners is ₱100.
6. Many learners feel stressed before exams.
7. Our barangay’s average electricity bill last month was ₱1,200.
8. Girls tend to have higher average grades than boys.
9. The mean age of teachers in our school is 42.
10. The mean rainfall in a certain municipality during October is 300 mm.</td><td>Activity A.1: Leveling Learner Readiness

This activity is designed to help learners distinguish between claims (statements that require evidence or verification) and realities (statements that are measurable, documented, or verifiable). It encourages critical thinking by prompting learners to question information, consider the type of evidence needed, and reflect on why some statements are harder to classify.

For this activity you may choose any of the following strategies.

Strategy 1. Think-Pair-Share

This will guide learners in moving from personal reflection to collaborative verification, helping them distinguish between unverified claims and evidence-based realities through respectful dialogue, critical questioning, and shared inquiry.

THINK (Individual Reflection)</td></tr>
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2. Decide: Is this a claim or a reality?

3. Be ready to explain your answer.

- ✔ What kind of evidence would help prove or disprove it?
- ✔ Discuss why some statements are harder to classify.
- ✔ What kind of evidence would help?

Possible Answers:

a. The average height of Filipino teenagers is 5'4".
- Claim (needs official data to verify, may vary by age group and region)

b. Most learners prefer online learning.
- Claim (preference is subjective; requires survey data to confirm)

c. The mean score in the National Achievement Test is 70.
- Reality (if based on official DepEd reports)

d. Learners in our barangay walk 2 kilometers to school daily.
- Claim (local observation; not verified with actual data collection)

e. The average baon of Grade 10 learners is ₱100.
- Claim (need survey data; varies by school and/or family income)

f. Many learners feel stressed before exams.
- Claim (emotional states are subjective; requires survey data to confirm)

g. Our barangay's average electricity bill last month was ₱1,200.
- Reality (if based on actual billing records which are measurable and verifiable)

h. Girls tend to have higher average grades than boys.

1. Give the following statements
   a. The average height of Filipino teenagers is 5'4".
   b. Most learners prefer online learning.
   c. The mean score in the National Achievement Test is 70.
   d. Learners in our barangay walk 2 kilometers to school daily.
   e. The average baon of Grade 10 learners is ₱100.
   f. Many learners feel stressed before exams.
   g. Our barangay's average electricity bill last month was ₱1,200.
   h. Girls tend to have higher average grades than boys.
   i. The mean age of teachers in our school is 42.
   j. The mean rainfall in a certain municipality during October is 300 mm.

2. Each learner will decide whether the statement is a claim or a reality. What kind of data would prove or disprove it? What assumptions might be hidden in this statement?

PAIR (Partner Discussion)
1. Each learner will share their thoughts with a partner.
2. Learners will compare their answers.
3. Ask: Did you both classify it the same way?
4. Each pair will discuss the following:
   - ✔ What evidence would you need to verify this?
   - ✔ Could this average be misleading? Why or why not?
   - ✔ How might this claim affect learners, teachers, or parents?
   - ✔

SHARE (Whole-Class Dialogue)
Ask the following prompts:

	- *Claim (needs school data to confirm)* i. The mean age of teachers in our school is 42. - *Reality (if based on HR data where the ages of teachers are documented and can be averaged)* j. The mean rainfall in a certain municipality during October is 300 mm. - *Reality (if based on weather bureau data; rainfall is measurable and recorded)* ***Some factors that make certain statements difficult to classify*** - *Subjective.* Statements may rely on perceptions, emotions, or preferences. - *Scope and representativeness.* Statements may be true in one area but not generalizable. - *Data availability.* Some realities are regularly documented, and others require data collection making them harder to prove immediately.	▪ Who classified it as a claim? Why? ▪ Who believes it's a reality? What data supports that? ▪ What other claims about learner life could we investigate using data? ▪ How can we use statistics to clarify not confuse realities in our community? Strategy 2. **Claim Carousel Strategy** This strategy will help learners distinguish between claims and realities, ask critical questions, and identify credible data sources while engaging in movement-based, peer-supported learning. 1. Prepare the given statements related to your lesson theme. Mix obvious realities with common but unverified claims. 2. Write each claim on a large sheet of paper and post around the room. Leave space below each for learners to write: - Classification: Claim or Reality - One verification question - Suggested data source (e.g., PSA, classroom survey, barangay records) 3. Divide learners into small groups (3–5 members). Assign each group a starting station. After 3–5 minutes, signal them to rotate clockwise to the next claim. Continue until all groups have visited each station. 4. Gather the class and review each claim. Highlight interesting disagreements or creative verification questions. Ask the following questions: ✔ Which claims were hardest to classify? ✔ What kinds of data would help us decide? ✔ How do our values or assumptions affect what we believe?
	A.2. Establishing the Purpose of the Lesson	

	Activity A. 2: Appreciating Lesson Relevance **Instructions**: 1. Read the following challenge: *You are now part of the Barangay Data Squad. Your mission is to decide which claims can be tested using statistics specifically, a test of mean for one population.* 2. Look at the following challenge cards. In pairs or small groups, you will classify each statement if testable using one-population mean test or not testable. a. The average height of Filipino teenagers is 5'4". b. Most learners prefer online learning. c. The mean score in the National Achievement Test is 70. d. Learners in our barangay walk 2 kilometers to school daily. e. The average baon of Grade 10 learners is ₱100. f. Many learners feel stressed before exams. g. Our barangay's average electricity bill last month was ₱1,200. h. Girls tend to have higher average grades than boys. i. The mean age of teachers in our school is 42. j. The mean rainfall in a certain municipality during October is 300 mm. 3. Justify your answer using the following guide questions: ▪ Is the variable numerical? ▪ Is the population clearly defined? ▪ Is the claim about a mean? ***Possible Answers:*** a. The average height of Filipino teenagers is 5'4". - Is the variable numerical? *Yes, height in feet/inches* - Is the population clearly defined? *Yes, Filipino teenagers* - Is the claim about a mean? *Yes, average height* b. Most learners prefer online learning.	**Activity A. 2: Appreciating Lesson Relevance** This activity will help learners identify which real-life claims can be tested using a one-population mean test and explain why. For this activity you may choose any of the following strategies. Strategy 1. **Claim Sorting Relay** 1. Prepare 10 challenge cards with real-life statements. a. The average height of Filipino teenagers is 5'4". b. Most learners prefer online learning. c. The mean score in the National Achievement Test is 70. d. Learners in our barangay walk 2 kilometers to school daily. e. The average baon of Grade 10 learners is ₱100. f. Many learners feel stressed before exams. g. Our barangay's average electricity bill last month was ₱1,200. h. Girls tend to have higher average grades than boys. i. The mean age of teachers in our school is 42. j. The mean rainfall in a certain municipality during October is 300 mm. 2. Divide the class into Barangay Data Squads (small groups). 3. Each squad races to sort the cards into: ✔ *Testable using one-population mean test* ✔ *Not testable using this method*

<table>
<tr>
<td></td>
<td>- Is the variable numerical?
No (preference is categorical)
- Is the population clearly defined?
Not clearly stated
- Is the claim about a mean?
No, proportion

c. The mean score in the National Achievement Test is 70.
- Is the variable numerical?
Yes, test scores
- Is the population clearly defined?
Yes, learners who took National Achievement Test
- Is the claim about a mean?
Yes, mean score

d. Learners in our barangay walk 2 kilometers to school daily.
- Is the variable numerical?
Yes, distance in kilometers
- Is the population clearly defined?
Yes, learners in the barangay
- Is the claim about a mean?
No, statement of typical distance

e. The average baon of Grade 10 learners is ₱100.
- Is the variable numerical?
Yes, amount in pesos
- Is the population clearly defined?
Yes, Grade 10 learners
- Is the claim about a mean?
Yes, average baon

f. Many learners feel stressed before exams.
- Is the variable numerical?
No, stress is qualitative
- Is the population clearly defined?
No, not clearly stated
- Is the claim about a mean?
No, about frequency / emotions or feelings

g. Our barangay’s average electricity bill last month was ₱1,200.
- Is the variable numerical?</td>
<td>4. For each card, squads must justify their choice using the guide questions:
○ Is the variable numerical?
○ Is the population clearly defined?
○ Is the claim about a mean?

Encourage learners to think aloud and explain their reasoning.

5. Debrief with a class discussion. Highlight common misconceptions (e.g., confusing proportions with means).

Strategy 2. Sabi-sabi o Sukat? Debate Circle

1. Present a mix of real-life claims.

2. Divide the class into two sides:
○ Team Sabi-sabi: Argues that the claim is not testable using a one-population mean test.
○ Team Sukat: Argues that the claim is testable.

3. Each team must use the guide questions to justify their stance:
✔ Is the variable numerical?
✔ Is the population clearly defined?
✔ Is the claim about a mean?

4. Rotate roles and reflect on how statistical reasoning can clarify community beliefs.

5. Debrief with a reflection question prompt: “How can statistics help us move from hearsay to truth?”</td>
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<tr>
<td></td>
<td>Yes, amount in pesos
- Is the population clearly defined?
Yes, households in barangay
- Is the claim about a mean?
Yes, average bill

h. Girls tend to have higher average grades than boys.
- Is the variable numerical?
Yes, grades
- Is the population clearly defined?
Yes, boys and girls in school
- Is the claim about a mean?
Yes, comparison of average grades

i. The mean age of teachers in our school is 42.
- Is the variable numerical?
Yes, age in years
- Is the population clearly defined?
Yes, teachers
- Is the claim about a mean?
Yes, mean age

j. The mean rainfall in a certain municipality during October is 300 mm.
- Is the variable numerical?
Yes, rainfall in mm
- Is the population clearly defined?
Yes, specific municipality (during the month of October)
- Is the claim about a mean?
Yes, average rainfall</td>
<td></td>
</tr>
<tr>
<td rowspan="2">B. Instituting New Knowledge</td>
<td colspan="2">B.1. Presenting Examples</td>
</tr>
<tr>
<td>Activity B.1 Exploring Key Concepts

Your task is to investigate whether certain community claims are supported by actual data. You'll work with sample information and decide: Is the claim believable or does it need to be tested more formally?

1. Read the claim cards carefully. Each card contains a real-life statement, such as:
- The average baon of Grade 10 learners is ₱100.
- Learners sleep 6 hours per night.</td>
<td>Activity B.1 Exploring Key Concepts

This activity engages learners in investigating real-life claims to determine whether they are supported by actual data or require further verification. Learners will work with sample information, calculate averages, and critically compare their results to the stated claims. The activity promotes analytical thinking, understanding of numerical data, and practical application of statistical concepts.
Whole-Class Investigation (Traditional Format)</td>
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<td>- Varsity players make 8 free throws out of 10.

2. Discuss each claim and examine the sample data provided. You may also collect your own data from classmates if time allows.

3. Use the data to find the average value for each claim. You can use calculators or do it manually.

4. Compare the sample mean to the claimed mean. Ask yourselves:
- Is the sample mean close to the claimed value?
- Does it seem believable based on the data?

5. Use the following guide questions to justify your answer.
- Is the variable numerical?
(e.g., baon, sleep hours, scores)
- Is the population clearly defined?
(e.g., Grade 10 learners, varsity players)
- Is the claim about a mean?
(not a preference or category)

If you answer “YES” to all three, the claim can be tested using a one-population mean test.</td>
<td>(This strategy will help learners investigate real-life claims using sample data, compute averages, and decide whether the claims are believable or need formal testing.)

1. Choose 3–5 real-life numerical claims relevant to learner life.
Examples:
- The average baon of Grade 10 learners is ₱100.
- Learners sleep 6 hours per night.
- Varsity players make 8 free throws out of 10.

2. Write each claim on an index card, slide, or printed sheet.

3. Provide sample data for each claim. You may:
- Use pre-collected class data.
- Share fictional but realistic data sets.
- Display data on the board, handouts, or slides.

4. Divide the class into groups and assign one claim per group. Each group will:
- Compute the sample mean.
- Compare it to the claimed value.
- Discuss whether the claim seems believable.

5. Groups present their findings using manila paper or slides. Encourage them to answer these guide questions:
✔ Is the variable numerical?
✔ Is the population clearly defined?
✔ Is the claim about a mean?
✔ Is the sample mean close to the claimed value?
✔ Is the claim believable based on the data?</td>
</tr>
<tr>
<td></td>
<td colspan="2">B.2. Discussing the Concept</td>
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</table>

	Activity B.2: **Deepening Conceptual Understanding** **Hypothesis Testing** *Hypothesis testing* is a statistical method used to make decisions or draw conclusions about a population by analyzing data from a sample. It begins when we observe a pattern and want to test whether that pattern reflects a real effect or is simply due to chance. This process, also known as *significance testing*, involves forming two competing claims: the null hypothesis (which assumes no effect or difference) and the alternative hypothesis (which suggests there is an effect or difference). Moreover, hypothesis testing helps us move from assumptions to evidence-based conclusions, guiding decisions in fields like education, health, and social research. No matter what measure is applied during the decision-making process, the outcome ultimately leads to a conclusion about whether the null hypothesis should be rejected or retained. The procedure for hypothesis testing follows a series of structured steps that guide us in evaluating whether the observed data provides enough evidence to challenge the initial assumption. **Steps in Hypothesis Testing** *(adapted from Arcilla et al., 2019)* 1. Formulate the null hypothesis (H_0) and alternative hypothesis (H_1). 2. Decide on a level of significance (α). *(Usually, alpha level is set at 0.05)* 3. Select the appropriate test statistic. 4. Establish the critical region and decision rule. 5. Calculate the value of the test statistic. 6. Decide whether to reject or fail to reject the null hypothesis. 7. Write down the conclusion based on the decision.	**Activity B.2:** **Deepening Conceptual Understanding** To facilitate the discussion of the following topics, you may apply the following strategy: **Small Group Discussion** Part A: Understanding Testing Hypothesis About Mean of One Population Instructions: 1. Divide the class into small groups (4–5 learners per group). 2. Prepare printed or digital copies of the lesson handouts and sample dataset. 3. Distribute handout to all groups and say: "After exploring basic concepts in inferential statistics, each group will now explore the procedures for conducting tests of hypotheses about a single population. You will also look at real-world scenarios and interpret the results." 4. Instruct each group to: • Read the definitions and examples in the handout. • Highlight key concepts and discuss their insights. • Answer guiding questions like: - *What real-life situation are we investigating?* - *Why is it important to test this claim?* - *What is the population mean we are testing against?* - *Is this a one-tailed or two-tailed test? Why?* - *Do we know the population standard deviation?* - *Should we use a z-test or a t-test?* - *Is our sample size large enough to assume normality?* - *What is the computed value of the test statistic?* - *Is the test statistic in the rejection region?* - *Do we reject or fail to reject the null hypothesis?* - *What does our conclusion mean in the real world?*

Hypothesis Testing About the Population Mean μ

a. Testing Hypothesis About the Population Mean μ when Standard Deviation σ is known

Suppose a random sample of size n is drawn from a normally distributed population. In order to test the null hypothesis $H_0: \mu = \mu_0$ against the chosen alternative H_1, the following test statistic is applied:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where:

z = z-test value
$\bar{x}$ = sample mean
μ = hypothesized population mean
σ = population standard deviation
n = sample size

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	*H_1*	*Reject H_0 if*
Two-tailed test	$\mu \neq \mu_0$	$z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
One-tailed test (left-tailed)	$\mu < \mu_0$	$z < -z_{\alpha}$
One-tailed test (right-tailed)	$\mu > \mu_0$	$z > z_{\alpha}$

Otherwise, fail to reject H_0.

Example:
A senior high school in a certain province claims that its Grade 12 learners study an average of 3.5 hours per day during exam week. A learner researcher wants to test

whether this claim is accurate. She collects a random sample of 40 learners and finds that the average study time is 3.2 hours per day. From previous school records, the standard deviation of study hours is known to be 0.8 hours. Using a significance level of 0.05, can we conclude that the actual mean study time is different from the school's claim?

Solution:

Step 1. Null hypothesis (H_0): The average study time of Grade 12 learners during exam week is equal to 3.5 hours per day (i.e., $\mu = 3.5$).

Alternative hypothesis (H_1): The average study time of Grade 12 learners during exam week is different from 3.5 hours per day (i.e., $\mu \neq 3.5$).

Step 2. $\alpha = 0.05$

Step 3. Since the population standard deviation is given, the sample size is sufficiently large, and the hypothesis concerns the population mean, the appropriate test statistic to use is

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Step 4. Two-tailed test is used since the alternative hypothesis is non-directional. For a two-tailed test at $\alpha = 0.05$:

Reject H_0 if $z < -1.96$ or $z > 1.96$.
Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x} = 3.2$ $\mu = 3.5$
$\sigma = 0.8$ $n = 40$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{3.2 - 3.5}{\frac{0.8}{\sqrt{40}}} = -2.37$$

Step 6. Given that computed value of z falls in the critical region, that is, $-2.37 < -1.96$, then the null hypothesis is rejected.

Step 7. At $\alpha = 0.05$, there is sufficient evidence to conclude that the average study time of Grade 12 learners during exam week is significantly different from the school's claimed average of 3.5 hours per day.

b. Testing Hypothesis About the Population Mean μ when Standard Deviation σ is Unknown and Sample Size n ≥ 30

Suppose a large random sample of size n is drawn from a non-normal population. In order to test the null hypothesis H_0: $\mu = \mu_0$ against the chosen alternative H_1, the following test statistic is applied:

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where:

z = z-test value
$\bar{x}$ = sample mean
μ = hypothesized population mean
s = sample standard deviation
n = sample size

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	H_1	*Reject H_0 if*
Two-tailed test	$\mu \neq \mu_0$	$z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
One-tailed test (left-tailed)	$\mu < \mu_0$	$z < -z_{\alpha}$
One-tailed test (right-tailed)	$\mu > \mu_0$	$z > z_{\alpha}$

Otherwise, fail to reject H_0.

Example:
A senior high school in a certain province introduces a new volleyball training program aimed at improving the vertical jump height of a player. The school claims that after the program, players will have an average vertical jump of at least 60 cm. Mr. Malakas, a learner researcher, measures the jump heights of 100 randomly selected players after completing the program. The sample shows an average jump height of 58.7 cm, with a sample standard deviation of 5.5 cm. Using a significance level of 0.05, can we conclude that the average jump height is less than what the school claims?

Solution:

Step 1. Null hypothesis (H_0): The average vertical jump height after the training program is at least 60 cm (i.e., $\mu \geq 60$).

Alternative hypothesis (H_1): The average vertical jump height after the training program is lower than 60 cm (i.e., $\mu < 60$).

Step 2. $\alpha = 0.05$

Step 3. Since the population standard deviation σ is not known and the sample size n is large, the appropriate test statistic to use is

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Step 4. One-tailed test is used since the alternative hypothesis is directional (left-tailed). For a one-tailed test at $\alpha = 0.05$:

Reject H_0 if $z < -1.645$
Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x} = 58.7$ $\quad \mu = 60$
$s = 5.5$ $\quad n = 100$

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{58.7 - 60}{\frac{5.5}{\sqrt{100}}} = -2.36$$

Step 6. Given that computed value of z falls in the critical region, that is, -2.36 < -1.645, then the null hypothesis is rejected.

Step 7. At the 0.05 significance level, there is sufficient statistical evidence to conclude that the true average vertical jump height of players after completing the program is less than the 60 cm claimed by the school.

c. Testing Hypothesis About the Population Mean μ when Standard Deviation σ is Unknown and Sample Size n small

Suppose a large random sample of size n is drawn from a normal population with an unknown standard deviation σ. In order to test the null hypothesis $H_0: \mu = \mu_0$ against the chosen alternative H_1, the following test statistic is used with $n - 1$ degrees of freedom:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

where:

t = t-test value
$\bar{x}$ = sample mean
μ = hypothesized population mean
s = sample standard deviation
n = sample size

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	H_1	*Reject H_0 if*
Two-tailed	$\mu \neq \mu_0$	$t < -t_{\alpha/2\,(n-1)}$ or $t > t_{\alpha/2\,(n-1)}$
One-tailed (left-tailed)	$\mu < \mu_0$	$t < -t_\alpha$

One-tailed (right-tailed)	$\mu > \mu_0$	$t > t_\alpha$

Otherwise, fail to reject H_0.

Example:
To promote both learner health and environmental care, Likas National High School launched a "*Lakad Para sa Kalikasan*" campaign encouraging learners to walk or bike to school instead of using motorized transport. Ms. Mahinahon, school nurse, believes that learners who participate in the campaign get at least 30 minutes of physical activity per day. To test this claim, she randomly selects 10 learners who joined the campaign and records their walking/biking durations. The data shows a sample mean of 32.1 minutes and a sample standard deviation of 4.02 minutes. Using a significance level of 0.05, can we conclude that learners are getting less than 30 minutes of physical activity per day?

Solution:
Step 1. Null hypothesis (H_0): Learners get at least 30 minutes of physical activity per day (i.e., $\mu \geq 30$).

Alternative hypothesis (H_1): Learners get less than 30 minutes of physical activity per day (i.e., $\mu < 30$).

Step 2. $\alpha = 0.05$

Step 3. Because the population standard deviation σ is unknown and the sample size n is relatively small, the appropriate test statistic to use is

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Step 4. One-tailed test is used since the alternative hypothesis is directional (left-tailed). For a one-tailed test at $\alpha = 0.05$ with degrees of freedom $df = n - 1 = 9$

Reject H_0 if $t < -1.833$.
Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x} = 32.1$ $\mu = 30$
$s = 4.02$ $n = 10$

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{32.1 - 30}{\frac{4.02}{\sqrt{10}}} = 1.65$$

Step 6. Given that computed value of t does not fall in the critical region, that is, 1.65 > -1.833, then the null hypothesis is not rejected.

Step 7. There is not enough evidence to support the claim that learners are getting less than 30 minutes of physical activity per day. In fact, the sample suggests they may be getting more, supporting the goals of the *"Lakad Para sa Kalikasan"* campaign.

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Instructions:
Read and solve each problem.

1. To promote digital wellness, the school recommends that learners limit their recreational screen time to no more than 2 hours per day. Mr. Malinaw, a learner researcher, surveys 100 learners and finds that the average screen time is 2.3 hours, with a sample standard deviation of 0.8 hours. Using a significance level of 0.05, can we conclude that learners are spending more time than recommended?

2. A school wellness committee of Maalaga National High School recommends that learners get at least 7 hours of sleep per night, even during exam week. A learner-researcher surveys 15 classmates and finds that the average sleep duration is 6.4 hours, with a sample standard deviation of 0.8 hours. Using a significance level of 0.05, can we conclude that learners are sleeping less than the recommended amount?

3. The Mabusog-lusog National High School cafeteria claims that their budget meal contains an average of 500 calories. A nutritionist believes the meals contain more than this amount. From a previous nationwide study, the population standard deviation is known to be 25

Activity B.3: Practicing Learned Skills

This activity allows learners to apply statistical concepts and hypothesis-testing skills they have learned to real-life scenarios. Using sample data, learners will determine whether observed outcomes provide sufficient evidence to support or refute a given claim. The activity emphasizes critical thinking, numerical analysis, and decision-making based on statistical evidence.
To facilitate this activity the teacher may apply any of the following strategies.

Strategy 1. **Hypothesis Quest: Step by Step to the Truth**

(The strategy will guide learners through the essential steps of hypothesis testing using a structured, scenario-based approach that builds statistical reasoning and real-world relevance).

1. Begin with a motivating question: *"Can we use data to uncover the truth behind everyday claims?"*
2. Present the different scenario and explain that learners will become Truth Trackers, navigating steps to test a claim.
3. Divide the class into small groups (5 learners). Assign each team a unique scenario or let them choose one from

calories. The nutritionist analyzes a sample of 100 meals and finds an average of 506 calories. Using a 0.05 significance level, is there enough evidence to support the claim of the nutritionist?

4. Sama-sama National High School promotes shared meals at home, recommending 5 meals per week with parents. Suppose Ms. Dimagibagiba, a learner researcher, surveys 8 learners and finds that the average is 6.1 shared meals, with a sample standard deviation of 1.0 meal. Using a significance level of 0.05, can we conclude that learners are sharing more meals than recommended?

Answer Key:

1. H_0: The average screen time is no more than 2 hours. ($\mu \leq 2$).
   H_1: The average screen time is greater than 2 hours. ($\mu > 2$).

   Given: $\bar{x} = 2.3$ $\quad \mu = 2$
   $s = 0.8$ $\quad n = 100$

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{2.3 - 2}{\frac{0.8}{\sqrt{100}}} = 3.75$$

   For a right-tailed test at 0.05 significance level, use z-critical = 1.645

   Reject H_0 if *z*-computed > *z*-critical. Since the computed *z*-value is greater than the *z*-critical, the null hypothesis is rejected.

   At the 0.05 significance level, there is sufficient evidence to conclude that learners are spending more than 2 hours per day on recreational screen time, exceeding the school's recommendation.

2. H_0: The average sleep is at least 7 hours. ($\mu \geq 7$).
   H_1: The average sleep is less than 7 hours. ($\mu < 7$).

   Given: $\bar{x} = 6.4$ $\quad \mu = 7$
   $s = 0.8$ $\quad n = 15$

a curated list. Provide each team with a Hypothesis Quest Map (flowchart or worksheet with labeled steps).
4. Each team works through the steps in hypothesis testing.
5. The teams post their final conclusions around the room.
6. Facilitate a short discussion: *"What did you learn about testing claims with evidence?"*

Strategy 2. **Stat Squad: Mission Mean Possible**
(This strategy will help learners master each step of hypothesis testing through collaborative, role-based learning that reflects real-life scenarios).

1. Choose 1–2 real-life scenarios. Prepare a Hypothesis Quest Map or flowchart with five labeled steps. Provide calculators, t/*z*-tables, and sample data.

2. Divide the class into 5 stations or role-based teams. Each team focuses on one step of the hypothesis test:
Hypothesis Team – defines H_0 and H_1
Computation Team – calculates the test statistic
Critical Value Team – finds and interprets the critical value
Decision Team – compares and decides
Conclusion Team – writes the final statement

3. Provide sentence starters:
Step 1: "We want to test if…"
Step 2: "Using the formula…"
Step 3: "The critical value at α = 0.05 is…"
Step 4: "Since our test statistic is…"
Step 5: "There is enough evidence to conclude that…"

You may offer bilingual support if needed, especially for null hypothesis H_0 and alternative hypothesis H_1 phrasing.

4. After completing the map, ask learners: "What can we do to improve this situation?"

5. Create a conclusion gallery where learners post their final statements.

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{6.4 - 7}{\frac{0.8}{\sqrt{15}}} = -2.905$$

For a left-tailed test with degrees of freedom df = 14 at 0.05 significance level, use t-critical = - 1.761.

Reject H_0 if t-computed < t-critical. Since the computed t-value is less than the t-critical, the null hypothesis is rejected.

At the 0.05 significance level, there is sufficient evidence to conclude that the average sleep duration of learners during exam week is less than the recommended 7 hours.

3. H_0: The budget meal contains an average of 500 calories. ($\mu = 500$).
   H_1: The budget meal contains more than 500 calories. ($\mu > 500$).

   Given: $\bar{x} = 506$ $\mu = 500$
   $\sigma = 25$ $n = 100$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{506 - 500}{\frac{25}{\sqrt{100}}} = 2.40$$

For a right-tailed test at 0.05 significance level, use z-critical = 1.645.

Reject H_0 if z-computed > z-critical. Since the computed z-value is greater than the z-critical, the null hypothesis is rejected.

At the 0.05 significance level, there is sufficient evidence to support the nutritionist's claim that the budget meals contain significantly more than the advertised 500 calories.

4. H_0: The average number of shared meals is 5 meals per week. ($\mu = 5$).

	H_1: The average number of shared meals is greater than 5 meals per week ($\mu > 5$). Given: $\bar{x} = 6.1$ $\mu = 5$ $s = 1$ $n = 8$ $$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{6.1 - 5}{\frac{1}{\sqrt{8}}} = 3{,}11$$ For a right-tailed test with degrees of freedom df = 7 at 0.05 significance level, use t-critical = 1.895. Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected. At the 0.05 significance level, there is sufficient evidence to conclude that learners at Sama-sama National High School are sharing more than the recommended 5 meals per week with their parents.	
C. Demonstrating Knowledge and Skills	***C.1. Finding Practical Application***	
	Activity C.1: Making Real-World Connections **Instructions**: Read each problem and perform the indicated tasks. *Problem:* A consumer advocacy group in a certain province is concerned that residents may be paying more for electricity than the national average of ₱2,500 per month. To investigate, they randomly survey 50 households and record their monthly electricity bills. The population standard deviation is known to be ₱400. The sample yields an average monthly bill of ₱2,580. *Question:* At a 5% significance level, is there enough evidence to conclude that the average monthly electricity bill exceeds the national average? **Gaming Time: Just Right or Too Much?** *Problem:* Suppose Masunurin National High School conducts a study on learners' digital habits. Ma'am Maaro, school guidance counselor, is concerned that learners may be spending more time playing mobile	**Activity C.1: Making Real-World Connections** This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. ***Answer Key:*** **Power Check: Are We Paying More?** H_0: The average monthly electricity bill is ₱2500. ($\mu = 2500$). H_1: The average monthly electricity bill is greater than ₱2500. ($\mu > 2500$). Given: $\bar{x} = 2580$ $\mu = 2500$ $\sigma = 400$ $n = 50$

	games than the recommended maximum of 2 hours per day. She randomly selects a sample of 25 Grade 12 learners and records their average daily gaming time during a regular school week. The results show: *Sample mean: 2.4 hours* *Sample standard deviation: 0.6 hours* *Sample size: n = 25* *Question:* Assuming the population of learners has a normal distribution, test at the 5% significance level whether learners are spending more than 2 hours per day on mobile games.	$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{2580 - 2500}{\frac{400}{\sqrt{50}}} = 1.41$$ For a right-tailed test at 0.05 significance level, use z-critical = 1.645 Reject H_0 if z-computed > z-critical. Since the computed z-value is less than the z-critical, the null hypothesis is not rejected. At the 0.05 significance level, there is not enough evidence to conclude that the average monthly electricity bill exceeds the national average of ₱2,500. **Gaming Time: Just Right or Too Much?** H_0: Learners are spending 2 hours per day on mobile games. ($\mu = 2$). H_1: Learners are spending more than 2 hours per day on mobile games. ($\mu > 2$). Given: $\bar{x} = 2.4$ $\mu = 2$ $s = 0.6$ $n = 25$ $$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{2.4 - 2}{\frac{0.6}{\sqrt{25}}} = 3{,}33$$ For a right-tailed test with degrees of freedom df = 24 at 0.05 significance level, use t-critical = 1.711. Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected. At the 0.05 significance level, there is sufficient evidence to conclude that Grade 12 learners at Masunurin National High School are spending significantly more than 2 hours per day playing mobile games.

	C.2. Making Generalization	
	Activity C.2: Wrapping up the Lesson **Instructions**: To create a space where learners can reflect on what they've learned from statistical investigations whether through t-tests, z-tests, or real-life data stories and reconnect with the values and questions that matter most. 1. Choose a visible space like a classroom bulletin board, or corner wall. Add a title banner, "Data with Heart: What We Learned, What We Feel". 2. Use scaffolded prompts that invite both analytical and emotional responses. You can print these on cards, post them on the wall, or write them on a whiteboard. Suggested Prompts: *What did our data reveal?* *What surprised me about the results?* *How does this connect to my life or community?* *What values are reflected in this investigation?* *What questions do I still have?* *What can we do with this insight?* 3. Offer multiple ways for learners to respond (e.g. sticky notes or index cards, speech bubble cutouts, or heart-shaped or lightbulb-shaped templates) 4. Organize the wall or bulletin board into thematic zones. *Discoveries* – Key findings from one sample t-tests or z-tests *Surprises* – Unexpected results or shifts in thinking *Connections* – Links to real life, culture, or community *Next Steps* – Actions, advocacy, or further inquiry *Values* – Reflections on empathy, equity, and stewardship 5. Encourage others to respond to posted reflections.	**Activity C.2: Wrapping up the Lesson** This activity provides learners with an opportunity to reflect on their statistical investigations and connect their analytical findings to personal, community, and ethical perspectives. By combining data analysis with reflective prompts, learners can articulate insights, explore surprises, and consider the broader meaning of their investigations. The activity fosters critical thinking, emotional engagement, and values integration. For the summary, you may facilitate the provided activity.

C.3. Evaluating Learning

Activity C.3: Assessing Learning Outcomes

Instructions:
Solve the following problems as required. Show a complete solution to get complete points.

1. A disaster response team claims that each relief pack weighs exactly 5 kilograms. A volunteer measures a random sample of 40 relief packs and finds an average weight of 5.1 kg. The population standard deviation is known to be 0.3 kg. At a 5% significance level, is there enough evidence to conclude that the average weight of the relief packs is different from the claimed 5 kg?

2. A teacher believes learners sleep less than 8 hours during exam week. A sample of 20 STEM learners shows an average of 7.3 hours, with a sample standard deviation of 0.7 hours. At a 5% significance level, is the average sleep time significantly less than 8 hours?

3. A government agency claims citizens complete a feedback form in 10 minutes or less. A sample of 100 responses shows an average of 10.4 minutes, with a known population standard deviation of 2.5 minutes. At a 5% significance level, is the average completion time greater than 10 minutes?

Reflection Questions:
1. How can hypothesis testing help validate or challenge claims in each situation?

2. How does the choice between a z-test and a t-test affect the reliability of your conclusion in each scenario?

3. If we fail to reject the null hypothesis or find no significant difference, can we conclude that the original claim is always true? Explain your answer.

Activity C.3: Assessing Learning Outcomes

This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

Answer Key:

1. H_0: The average weight of the relief packs is 5 kg. calories. ($\mu = 5$).
   H_1: The average weight of the relief packs is different from 5 kg. ($\mu \neq 5$).

   Given: $\bar{x} = 5.1$ $\mu = 5$
   $\sigma = 0.3$ $n = 40$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{5.1 - 5}{\frac{0.3}{\sqrt{40}}} = 2.11$$

   For a two-tailed test at 0.05 significance level, use z-critical = ±1.96.

   Reject H_0 if *z*-computed > *z*-critical. Since the computed *z*-value is greater than the z-critical, the null hypothesis is rejected.

   At the 0.05 significance level, there is sufficient evidence to conclude that hat the average weight of the relief packs is significantly different from the claimed 5 kg.

2. H_0: The average sleep time among STEM learners during exam week is 8 hours. ($\mu = 8$).
   H_1: The average sleep time among STEM learners during exam week is less than 8 hours ($\mu < 8$).

   Given: $\bar{x} = 7.3$ $\mu = 8$
   $s = 0.7$ $n = 20$

		$$t=\frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}=\frac{7.3-8}{\frac{0.7}{\sqrt{20}}}=-4.47$$ For a left-tailed test with degrees of freedom df = 19 at 0.05 significance level, use t-critical = -1.729. Reject H_0 if t-computed < t-critical. Since the computed t-value is less than the t-critical, the null hypothesis is rejected. At the 0.05 significance level, there is sufficient evidence to conclude that STEM learners sleep significantly less than 8 hours during exam week. 3. H_0: The average completion time to complete the feedback form is 10 minutes or less. ($\mu \leq 10$). H_1: The average completion time to complete the feedback form is more than 10 minutes. ($\mu > 10$). Given: $\bar{x}$ = 10.4 μ = 10 σ = 2.5 n = 100 $$z=\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}}=\frac{10.4-10}{\frac{2.5}{\sqrt{100}}}=1.60$$ For a right-tailed test at 0.05 significance level, use z-critical = 1.645. Reject H_0 if z-computed > z-critical. Since the computed z-value is less than the z-critical, the null hypothesis is not rejected. At the 0.05 significance level, there is not enough statistical evidence to conclude that citizens take longer than 10 minutes, on average, to complete the feedback form.
	C.4. Additional Activities	
	Activity C.4: Extending and Reinforcing Learning	**Activity C.4: Extending and Reinforcing Learning**

	Solve the following problems as required. 1. A PE teacher believes that learner athletes have a lower resting heart rate than the general average of 72 bpm. A sample of 15 varsity players shows an average of 68 bpm, with a sample standard deviation of 5 bpm. At a 5% significance level, is the average heart rate lower than 72 bpm? 2. A public health campaign claims that people wear masks for at least 2 hours daily. A sample of 100 commuters shows an average of 1.8 hours, with a known population standard deviation of 0.5 hours. At a 5% significance level, is the average mask-wearing time less than the claimed duration?	This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. ***Answer Key:*** 1. H_0: The average heart rate of learner athletes is 72 bpm. ($\mu = 72$). H_1: The average heart rate of learner athletes is less than 72 bpm. ($\mu < 72$). Given: $\bar{x} = 68$ $\mu = 72$ $s = 5$ $n = 15$ $$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{68 - 72}{\frac{5}{\sqrt{15}}} = -3.10$$ For a left-tailed test with degrees of freedom df = 14 at 0.05 significance level, use t-critical = -1.761. Reject H_0 if t-computed < t-critical. Since the computed t-value is less than the t-critical, the null hypothesis is rejected. At the 0.05 significance level, there is sufficient statistical evidence to conclude that learner athletes have a lower average resting heart rate than 72 bpm. 2. H_0: The average mask-wearing time is at least 2 hours daily. ($\mu \geq 2$). H_1: The average mask-wearing time is less than 2 hours daily. ($\mu < 2$). Given: $\bar{x} = 1.8$ $\mu = 2$ $\sigma = 0.5$ $n = 100$ $$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{1.8 - 2}{\frac{0.5}{\sqrt{100}}} = -4$$

		For a left-tailed test at 0.05 significance level, use z-critical = -1.645. Reject H_0 if z-computed < z-critical. Since the computed z-value is less than the z-critical, the null hypothesis is rejected. At the 0.05 significance level, there is enough statistical evidence to conclude that that commuters wear masks for less than the claimed 2 hours daily.
III. CONTENT	**Lesson 11.4: Test of Mean for Two Populations (Paired and Unpaired)** *Recommended time allotment: 2 hours*	
IV. OBJECTIVES	By the end of the lesson, the learners… 1. perform appropriate statistical tests involving the mean of two populations. 2. use appropriate technology (e.g. MS Excel, Jamovi) in testing hypothesis for two population means involving real-life situations. 3. analyze and interpret the results of hypothesis tests to draw meaningful conclusions about the populations being compared.	
V. PROCEDURES	**ACTIVITIES**	**ANNOTATION**
A. Activating Prior Knowledge	***A.1. Eliciting Prior Knowledge***	
	Activity A.1: Leveling Learner Readiness **Instructions**: 1. Consider the following real-life scenarios: a. Two Grade 11 sections have different average scores in their quarterly assessments. *What could explain the difference?* b. Learners in morning classes scored higher on average than those in afternoon classes. *What might account for the difference between the two groups?* c. Learners who eat breakfast regularly scored higher on their first-period exams than those who skip it. *What might be causing this gap?* d. Learners who live near the school have higher attendance and better average scores than those who commute far. *What factors could be influencing the outcome?* e. Varsity athletes scored higher on a fitness test than non-athletes. *What might explain this?*	**Activity A.1: Leveling Learner Readiness** This activity prepares learners to think critically about differences in group averages by exploring real-life situations where outcomes vary between groups. Instead of jumping to conclusions, learners are encouraged to identify possible variables and factors that may influence results. The activity builds foundational understanding for statistical analysis by highlighting that observed differences often have multiple explanations. For this activity you may choose any of the following strategies. Strategy 1. **Think-Pair-Compare-Connect** This strategy will help learners activate their prior knowledge, deepen their reasoning through dialogue, and build shared understanding by comparing perspectives—before introducing formal concepts or procedures.

	2. Write down at least three possible explanations. 3. With your seatmate, share your ideas. Listen respectfully and add new ideas to your list. 4. Be ready to answer the following questions: *a. What variables could be influencing the averages?* *b. What measurable factors might explain the difference?* *c. Which group characteristics might be affecting the results?*	**THINK (Individual Reflection)** 1. Introduce the given real-life scenario. You may write the scenario on the board or project it visually. Ask learners to reflect silently*: "What could explain the difference?"* 2. Ask learners to write down 3–5 possible reasons. Encourage them to think about academic, personal, and environmental factors. Remind them: No need for formulas, just real-life reasoning. **PAIR (Partner Sharing)** 3. Learners pair up and share their ideas. Together, list down at least three possible reasons for the difference. **COMPARE (Small Group Discussion)** 4. Join another pair to form a group of four. Compare your lists and discuss the following: - *Which reasons are most likely?* - *Which ones could be tested or measured with data?* - *What kind of information would help us know for sure?* **CONNECT (Teacher-led Discussion)** 5. Guide learners into the next part of the lesson. You may ask: "*How can we use statistics to test whether these differences are real or just due to chance?"* Strategy 2. **Think, Sort, Decide** (This strategy will help learners identify and classify possible explanations for group differences, preparing them for variable selection and data collection in future lessons. 1. Choose a relatable prompt. Write it on the board or project it visually. Ask: "*What could be the possible reason?"* 2. Prepare 5-10 cards with possible explanations. Each card contains one idea. 3, Divide the learner to form small groups (3–5 learners). Give each group a full set of sorting cards.

		4. Ask groups to sort cards into two piles (measurable vis-à-vis non-measurable). ○ *Measurable - can be counted, observed, or recorded* ○ *Non-Measurable- based on feelings, assumptions, or hard-to-quantify factors* 5. Ask groups to re-sort the cards based on their judgment (likely vis-à-vis unlikely) ○ Likely - strong, believable explanation ○ Unlikely - less convincing or harder to prove 6. Ask groups to classify each card into: ○ Academic - related to teaching, learning, or schoolwork ○ Personal - related to habits, health, motivation ○ Environmental - related to surroundings, schedule, commute 7. Ask learners to respond individually: ○ *What measurable factors might explain the difference?* ○ *How could we test if it really matters?* ○ *How can we use statistics to test whether these differences are real or just due to chance?*
	A.2. Establishing the Purpose of the Lesson	
	Activity A. 2: Appreciating Lesson Relevance **Instructions:** Perform the indicated tasks. 1. Consider the following sample data set the previous real-life scenarios: a. Two Grade 11 sections have different average scores in their quarterly assessments.	**Activity A. 2: Appreciating Lesson Relevance** This activity will guide learners from intuitive reasoning to formal statistical inquiry by investigating whether observed differences between groups are real or due to chance using the logic and purpose of hypothesis testing. For this activity you may choose any of the following strategies.

Section A	Section B
43	34
34	35
45	23
43	43
34	45
45	23
36	34
45	35
49	29

b. Learners in morning classes scored higher on average than those in afternoon classes

Morning class	Afternoon class
23	14
24	15
25	23
23	13
24	15
30	23
30	14
25	25
29	29

c. Learners who eat breakfast regularly scored higher on their first-period exams than those who skip it.

Eating breakfast	Not eating breakfast
35	23
34	31
40	31
45	24
44	15
45	23
36	17
35	12
39	10

Strategy 1. **Scenario-Based Grouping: Hypothesis Hunter Squads**

This strategy will guide learners from intuitive reasoning to formal statistical inquiry by assigning them real-life scenarios and datasets to investigate whether observed group differences are real or due to chance.

1. Print or project the five scenarios (a–e) with their corresponding sample datasets. Create *Mission Cards* for each scenario, including:
    - ✔ Scenario title and description
    - ✔ Sample data table
    - ✔ Guiding questions (e.g., "Is this difference real or random?")
    - ✔ Space for squad name and notes

2. Introduce the activity. Explain to learners:
"Today, you will become Hypothesis Hunter Squads, learner researchers investigating real-life claims. Your mission is to uncover whether the difference between two groups is real or just chance.

3. Divide the class into 5 groups (or more if duplicating scenarios)
- Assign each group one scenario from the list:
    a. Section A vs. Section B
    b. Morning vs. Afternoon Classes
    c. Breakfast vs. No Breakfast
    d. Near vs. Far Commute
    e. Varsity Athletes vs. Non-Athletes

Let learners name their squads to build ownership and engagement.

4. For the investigation, each squad:
    a. Reads their Mission Card

d. Learners who live near the school have higher attendance and better average scores than those who commute far.

Live near the school	Commute far
80%	70%
85%	65%
90%	76%
76%	54%
78%	58%
80%	67%
82%	79%
65%	54%
75%	45%

e. Varsity athletes scored higher on a fitness test than non-athletes.

Varsity athlete	Non-athlete
94	82
87	45
89	70
90	78
93	56
87	67
89	65
93	78
76	67

2. Compute the mean for each group and observe the difference.

3. Answer the following questions:
   - *Is this enough to say one group is better? Could this be due to chance?*
   - *What does this tell us about small samples? Can we trust one small dataset to make a big claim?*

Answer Key:
a. Mean (Section A) = 41.56
   Mean (Section B) = 33.44

b. Computes the mean for each group

c. Observes the difference

d. Discuss the following:
   - ✔ *Is this enough to say one group is better?*
   - ✔ *Could this be due to chance?*
   - ✔ *What does this tell us about small samples?*

5. Each squad presents the following to the class:
- Their scenario
- Their computed group means
- Their reasoning: real or random?
- One question they would like to test formally

6. Facilitate a guided discussion:
   - ✔ Why do we need hypothesis testing?
   - ✔ How does it help us make fair decisions?
   - ✔ What role does sample size and variability play?

Strategy 2. **Data Analysis & Visualization**
This strategy will help learners compute group averages, visualize differences, and begin asking whether observed gaps are meaningful or possibly due to chance laying the foundation for hypothesis testing.

1. Prepare a worksheet for each group that includes:
a. Assigned scenario and dataset
b. Space to compute the mean for each group
c. A blank grid for a bar graph or dot plot
d. Reflection prompts:
   - ✔ *Is this enough to say one group is better?*
   - ✔ *Could this be due to chance?*

2. Divide the class into 5 groups (or more if duplicating scenarios) Ask each group to compute for the mean. Walk around to assist with calculations.

	Section A scored higher, but without knowing variability and sample size, we can't be sure this difference is meaningful. b. Mean (AM class) = 25.89 Mean (PM class) = 19.00 *Learners in morning classes looks stronger, but again, we need to test whether this difference is statistically significant.* c. Mean (Eating breakfast) = 39.22 Mean (Not eating breakfast) = 20.67 *A large gap, but we must ask: was the sample size big enough, and could other factors explain it?* d. Mean (Live near the school) = 79% Mean (Commute far) = 63.11% *Proximity seems linked to higher performance, but chance variation could also play a role.* e. Mean (Varsity athlete) = 88.67 Mean (Non-athlete) = 67.56 *Varsity athletes appear to do better, but we must consider whether the sample was small or biased.*	3. Let groups choose between bar graph or dot plot. ○ Bar Graph: *One bar per group showing average score* ○ Dot Plot: *Individual scores plotted for each group* 4. Once graphs are complete, prompt each group to discuss the following: ✔ *What do you notice about the difference?* ✔ *Is this enough to say one group is better?* ✔ *Could this be due to chance?* ✔ *What might affect the reliability of this data?* You may use a data dialogue card with guiding questions. 6. Use the visual comparisons to introduce the idea: *"Even if one group looks better, we need to test if the difference is statistically significant, meaning it's unlikely to be due to chance. That's where hypothesis testing comes in."* Use a visual anchor chart: **Observation → Visualization → Questioning → Hypothesis Testing** **Note:** Although the group averages show noticeable differences, these results alone do not prove that one group is truly better. Without proper statistical testing and larger samples, such differences may simply be due to chance. This example reminds learners that data must be interpreted with caution. Small samples can exaggerate patterns, and random variation can create results that look meaningful but may not reflect reality.
B. Instituting New Knowledge	***B.1. Presenting Examples***	
	Activity B.1 Exploring Key Concepts **Instructions:** Do as indicted. 1. Consider the following real-life situations. a. Learners who commute far have lower attendance.	**Activity B.1 Exploring Key Concepts** This activity will help learners identify and justify the appropriate statistical test (i.e., two-sample t-test, two-sample z-test, or qualitative inquiry) for a variety of real-life claims, by analyzing the nature of the variables, the presence or absence of known population standard deviation, and the measurability of outcomes.

	b. Learners feel more motivated in morning classes. c. STEM learners scored higher on a math quiz than HUMSS learners. d. Varsity athletes have better fitness scores than non-athletes. e. Learners who eat breakfast perform better in first-period exams. f. Learners from Barangay A are more disciplined than those from Barangay B. 2. Identify which real-life claims can be tested using a two-sample t-test, two-sample z-test, or not testable. 3. Answer the following guide questions: ✔ *Is this claim comparing two groups?* ✔ *Is the outcome measurable?* ✔ *Do we know the population standard deviation?* ✔ *Why is it important to know what we can test with data?* ✔ *How does this help us make fair decisions?* ***Answer Key:*** a. Learners who commute far have lower attendance. - Comparing two groups (near vs. far) - Outcome measurable (attendance rate) - Two sample z-test (if large samples and proportions) b. Learners feel more motivated in morning classes. - Comparing two groups (morning vs. afternoon) - Outcome not directly measurable - Not testable (unless the motivation is quantified using a validated scale, then t-test) c. STEM learners scored higher on a math quiz than HUMSS learners. - Comparing two groups (STEM vs. HUMSS learners) - Outcome measurable (quiz scores)	For this activity you may choose any of the following strategies. Strategy 1. **Stat U: Find the Right Test** This strategy will help learners be able to choose the right test (i.e., t-test, z-test, or qualitative method) for different real-life situations by checking if the claim compares two groups, uses measurable data, and has known or unknown standard deviation. 1. Create a set of scenario cards with real-life claims (a – f). Each card contains one statement. 2. Introduce the activity by explaining to the learners: *"Today, you'll become data detectives who decide which claims can be tested using statistics, and which ones need other methods like interviews or observation."* 3. Divide learners to form small groups of 3–5 learners. Give each group a full set of scenario cards and a matching board with three columns: ▪ two-sample t-test ▪ two-sample z-test ▪ not testable 4. Each group discusses and sorts the cards into the correct column. Encourage reasoning: ✔ *Is this claim comparing two groups?* ✔ *Is the outcome measurable?"* ✔ *Do we know the population standard deviation?* 5. Facilitate a whole-class discussion: a. Ask groups to share one card they debated b. Clarify misconceptions (e.g., difference between motivation and test scores) c. Reinforce the logic behind choosing t-test vs. z-test

	- Two sample t-test (population standard deviation σ) is unknown d. Varsity athletes have better fitness scores than non-athletes. - Comparing two groups (varsity athletes vs. non-athletes) - Outcome measurable (fitness scores) - Two sample t-test e. Learners who eat breakfast perform better in first-period exams. - Comparing two groups (eating breakfast vs. skips breakfast) - Outcome measurable (exam scores) - Two sample t-test f. Learners from Barangay A are more disciplined than those from Barangay B. - Comparing two groups (learners from Barangay A vs. learners from Barangay B) - Outcome not measurable - Not testable (discipline is subjective unless measured numerically)	d. Ask the following: ✔ *Why is it important to know what we can test with data?* ✔ *How does this help us make fair decisions?"* Strategy 2. **Visual Mapping Strategy: Test Tracker Wall** This strategy will help learners visually sort real-life claims into the correct statistical category (i.e., t-test, z-test, or not testable) based on the nature of the variables, measurability of outcomes, and availability of population standard deviation. 1. Create a large classroom wall chart or bulletin board with three clearly labeled zones. Zone 1: Two-sample t-test Zone 2: Two-sample z-test Zone 3: Not testable 2. Divide the class to form a group (3 - 5 members). Provide each group a set of real-life scenario cards (a-f). Each card should include space for learners to write their reasoning. 3. Each group discusses their assigned claim using the following guide questions: ✔ *Is this comparing two groups?* ✔ *Is the outcome measurable?* ✔ *Do we know the population standard deviation?* 4. Once the group agrees on the test type. Go to the large classroom wall chart or bulletin board. Post their claim in the correct zone using a colored sticky note or card. Include a brief explanation. 5. Facilitate a whole-class discussion: ✔ *What patterns do we see?* ✔ *Which claims were tricky to classify?* ✔ *How does this help us make fair, data-informed decisions?*

B.2. Discussing the Concept

Activity B.2: Deepening Conceptual Understanding

Instructions:
The following are the important ideas to be understand, listen attentively and participate actively in the discussion.

Hypothesis Testing About the Difference of Two Population Means $\mu_1 - \mu_2$

Case 1: Unpaired Samples

a. Testing Hypotheses About the Difference of Two Population Means $\mu_1 - \mu_2$ when Standard Deviations σ_1 and σ_2 are known

Suppose two independent samples are drawn from normally distributed populations with sample sizes n_1 and n_2, respectively. In order to test the null hypothesis $H_0: \mu_1 - \mu_2 = d_0$ against the chosen alternative hypothesis H_1, the appropriate test statistic to use is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where:

$\bar{x}_1$ = mean of the first group
$\bar{x}_2$ = mean of the second group
σ_1 = population standard deviation of the first group
σ_2 = population standard deviation of the second group
n_1 = number of observations in the first group
n_2 = number of observations in the second group
d_0 = hypothesized difference between two population means

Activity B.2: Deepening Conceptual Understanding

This activity focuses on strengthening learners' conceptual understanding of hypothesis testing for the difference of two population means. Through guided discussion, worked examples, and real-life contexts, learners examine when and how to apply *z*-tests and t-tests for both unpaired (independent) samples and paired samples. The activity emphasizes not only procedural steps but also the reasoning behind choosing the appropriate test based on data conditions and assumptions.

To facilitate the discussion of the following topics you may apply the following strategy:

Small Group Discussion
Part A: Understanding Testing Hypothesis About Mean of Two Populations

Instructions:
1. Divide the class into small groups (4–5 learners per group).

2. Prepare printed or digital copies of the lesson handouts and sample dataset.

3. Distribute handout to all groups and say:
"You learned from the previous lesson the hypothesis testing about the mean of a single population. In this lesson, you will apply the same steps in hypothesis testing but using different test statistics to facilitate the comparisons."

4. Instruct each group to:
• Read the definitions and examples in the handout.
• Highlight key concepts and discuss their insights.
• Answer guiding questions like:
  ✔ *What are the two populations or groups being compared?*
  ✔ *Are the samples independent or paired?*
  ✔ *Is the population standard deviation known or unknown?*

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	H_1	*Reject H_0 if*
Two-tailed test	$\mu_1 - \mu_2 \neq d_0$	$z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
One-tailed test (left-tailed)	$\mu_1 - \mu_2 < d_0$	$z < -z_{\alpha}$
One-tailed test (right-tailed)	$\mu_1 - \mu_2 > d_0$	$z > z_{\alpha}$

Otherwise, fail to reject H_0.

Example:
As part of its commitment to learner well-being and academic engagement, Malawak National High School launched a study comparing average weekly attendance rates between learners in online classes and those in face-to-face classes. The school's guidance office, in partnership with the Supreme Secondary Learner Government (SSLG), collected attendance data over one grading period. They randomly selected 50 learners from each modality and found:

Modality	Mean attendance rate	Population standard deviation	Sample size
Online	88%	4%	50
In-person	91%	5%	50

Is there statistical evidence that online learners have lower attendance rates than in-person learners? Use a 5% level of significance.

Solution:
Step 1. Null hypothesis (H_0): Online and in-person learners have equal attendance rates. (i.e., $\mu_1 - \mu_2 = 0$).

- ✔ *Which test should we use: z-test or two-sample t-test?*
- ✔ *What formula will we use to compute the test statistic?*
- ✔ *What are the sample means, standard deviations, and sizes?*
- ✔ *What formula will we use to compute the test statistic?*
- ✔ *What are the sample means, standard deviations, and sizes?*
- ✔ *Do we reject or fail to reject the null hypothesis?*
- ✔ *Is the difference between the means statistically significant?*

Part B: Computing z-test and t-test (difference of two population means) Using Excel and Jamovi

Instructions

1. Distribute Part 2 of the handout and guide groups to open their laptops.

2. Say:

Ready to analyze your data? Here's how you can use Excel and Jamovi to compute z-tests and t-tests with ease.

3. Instruct groups to:

- Enter or copy the sample dataset into Excel.
- Use (a) Formula function or (b) Data Analysis Toolpack in performing the test.
- Copy and paste the same data into Jamovi.
- Use the *T-test* module to compute the same measures.

4. Reporting and Class Discussion

Alternative hypothesis (H_1): Online learners have lower attendance rates than in-person learners. (i.e., $\mu_1 - \mu_2 < 0$).

Step 2. $\alpha = 0.05$

Step 3. Because population standard deviations are known, the sample sizes are large and drawn independently from normally distributed populations, the appropriate test statistic to use is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step 4. One-tailed test is used since alternative hypothesis is directional (left-tailed). For a one-tailed test at $\alpha = 0.05$:

Reject H_0 if $z < -1.645$

Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x}_1 = 88$ $\bar{x}_2 = 91$ $d_0 = 0$

$\sigma_1 = 4$ $\sigma_2 = 5$

$n_1 = 50$ $n_2 = 50$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(88 - 91) - 0}{\sqrt{\frac{(4)^2}{50} + \frac{(5)^2}{50}}} = -3.31$$

Step 6. Given that computed value of z falls in the critical region, that is, -3.31 < -1.645, then the null hypothesis is rejected.

Step 7. There is sufficient evidence at the 5% level to conclude that online learners have significantly lower attendance rates than their in-person counterparts.

b. Testing Hypotheses About the Difference of Two Population Means $\mu_1 - \mu_2$ when Standard Deviations σ_1 and σ_2 are Unknown and n_1 and n_2 are large

Suppose two independent samples are drawn from non-normal populations with sample sizes n_1 and n_2, respectively. In order to test the null hypothesis H_0: $\mu_1 - \mu_2 = d_0$ against the chosen alternative hypothesis H_1, the appropriate test statistic to use is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

$\bar{x}_1$ = mean of the first group
$\bar{x}_2$ = mean of the second group
s_1 = sample standard deviation of the first group
s_2 = sample standard deviation of the second group
n_1 = number of cases in the first group
n_2 = number of cases in the second group
d_0 = hypothesized difference between two population means

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	H_1	*Reject H_0 if*
Two-tailed test	$\mu_1 - \mu_2 \neq d_0$	$z < -z_{\alpha/2}$ or $z > z_{\alpha/2}$
One-tailed test (left-tailed)	$\mu_1 - \mu_2 < d_0$	$z < -z_\alpha$
One-tailed test (right-tailed)	$\mu_1 - \mu_2 > d_0$	$z > z_\alpha$

Otherwise, fail to reject H_0.

Example:

Two barangay, Barangay Maligaya (a coastal community) and Barangay Masigla (an inland town). partnered with local youth volunteers to study daily water consumption per household. Their goal is to understand how geography might influence water usage and to promote sustainable practices. After a month of data collection, the barangay councils gathered the following summary:

Barangay	Mean Daily Usage	Standard deviation	Sample size
Maligaya	120 liters	15 liters	40
Masigla	110 liters	18 liters	40

Is there a statistically significant difference in average daily water consumption between the two barangays? Use a 5% level of significance.

Solution:

Step 1. Null hypothesis (H_0): There is no significant difference in average daily water consumption between coastal and inland barangays (i.e., $\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): There is a significant difference in average daily water consumption between coastal and inland barangays (i.e., $\mu_1 - \mu_2 \neq 0$).

Step 2. $\alpha = 0.05$

Step 3. Since both sample sizes are large and drawn independently from non-normal populations, the appropriate test statistic to use is

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step 4. Two-tailed test is used because the alternative hypothesis is non-directional. For a two-tailed test at $\alpha = 0.05$:

Reject H_0 if $z < -1.96$ or $z > 1.96$.

Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x}_1 = 120$ $\bar{x}_2 = 110$ $d_0 = 0$
$s_1 = 15$ $s_2 = 18$
$n_1 = 40$ $n_2 = 40$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(120 - 110) - 0}{\sqrt{\frac{(15)^2}{40} + \frac{(18)^2}{40}}} = 2.70$$

Step 6. Given that the computed value of z falls in the critical region, that is, $2.70 > 1.96$, then the null hypothesis is rejected.

Step 7. There is significant evidence at the 5% level to conclude that average daily water consumption differs between the coastal and inland barangays.

c. **Testing Hypotheses About the Difference of Two Population Means μ_1 - μ_2 when Standard Deviations σ_1 and σ_2 are unknown but Assumed Equal**

Suppose two independent samples are drawn from normally distributed populations with sample sizes n_1 and n_2, respectively. In order to test the null hypothesis H_0: $\mu_1 - \mu_2 = d_0$ against the chosen alternative hypothesis H_1, the appropriate test statistic to use is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

with $n_1 + n_2 - 2$ degrees of freedom

where:
$\bar{x}_1$ = mean of the first group
$\bar{x}_2$ = mean of the second group

s_1 = sample standard deviation of the first group
s_2 = sample standard deviation of the second group
n_1 = number of cases in the first group
n_2 = number of cases in the second group
d_0 = hypothesized difference between two population means

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	*H_1*	*Reject H_0 if*
Two-tailed test	$\mu_1 - \mu_2 \neq d_0$	$t < -t_{\alpha/2(n_1+n_2-2)}$ or $t > t_{\alpha/2(n_1+n_2-2)}$
One-tailed test (left-tailed)	$\mu_1 - \mu_2 < d_0$	$t < -t_{\alpha(n_1+n_2-2)}$
One-tailed test (right-tailed)	$\mu_1 - \mu_2 > d_0$	$t > t_{\alpha(n_1+n_2-2)}$

Otherwise, fail to reject H_0.

Example.
At Magalang National High School, two Grade 11 sections, Section Makabansa and Section Makakalikasan, are part of a pilot program exploring different teaching methods in mathematics. Section Makabansa follows a traditional lecture-based approach, with teacher-led discussions and textbook drills. On the other hand, Section Makakalikasan uses an inquiry-based method, where learners explore problems collaboratively, ask questions, and present their own solutions. After the first quarter, both sections take the same summative test. The math department collects the following data:

Section / Method	Mean Score	Sample standard deviation	Sample size
Makabansa (traditional)	78	6.5	20
Makakalikasan (inquiry-based)	81	6	20

Is there statistical evidence that the inquiry-based method leads to higher scores? Use a 5% level of significance, assuming equal population variances.

Solution:

Step 1. Null hypothesis (H_0): There is no difference in mean scores between the two methods (i.e., $\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): The inquiry-based method leads to higher scores. (i.e., $\mu_1 - \mu_2 > 0$).

Step 2. $\alpha = 0.05$

Step 3. Because it is assumed that the unknown variances of the two normal populations are equal, the appropriate test statistic to use is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Step 4. One-tailed test is used since the alternative hypothesis is directional (right-tailed). For a right-tailed test at $\alpha = 0.05$ with degrees of freedom df = 38:

Reject H_0 if $t > 1.686$

Otherwise, fail to reject H_0.

Step 5. Given: $\bar{x}_1 = 78$, $s_1 = 6.5$, $n_1 = 20$; $\bar{x}_2 = 81$, $s_2 = 6$, $n_2 = 20$; $d_0 = 0$

Estimate of the common variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(20-1)(6.5)^2 + (20-1)(6)^2}{20 + 20 - 2} = 39.125$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(78 - 81) - 0}{\sqrt{(39.125)\left(\frac{1}{20} + \frac{1}{20}\right)}} = -1.52$$

Step 6. Given that the value of t does not fall in the critical region, that is, -1.52 < 1.686, then we fail to reject the H_0.

Step 7. There is not enough statistical evidence at the 5% level to conclude that the inquiry-based method leads to higher scores based on this small sample.

Case 2: Paired Samples

Testing Hypotheses About the Difference of Two Population Means μ_1 - μ_2

Suppose paired samples are drawn from two populations, with sample sizes n_1 and n_2, respectively. In order to test the null hypothesis H_0: $\mu_1 - \mu_2 = d_0$ against the chosen alternative hypothesis H_1, the appropriate test statistic to use is

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}}$$

with $n - 1$ degrees of freedom
where:

$\bar{d}$ = mean of differences
d_0 = hypothesized difference between two population means
s_d = standard deviation of differences
n = *sample size*

At the significance level α, the null hypothesis is rejected according to the established decision rules given below.

Type of test	*H_1*	*Reject H_0 if*
Two-tailed test	$\mu_1 - \mu_2 \neq d_0$	$t < -t_{\alpha/2(n-1)}$ or $t > t_{\alpha/2(n-1)}$

One-tailed test (left-tailed)	$\mu_1 - \mu_2 < d_0$	$t < -t_{\alpha(n-1)}$
One-tailed test (right-tailed)	$\mu_1 - \mu_2 > d_0$	$t > t_{\alpha(n-1)}$

Otherwise, fail to reject H_0.

Example:

In a certain municipality, ten Grade 4 learners from Barangay Makabasa Elementary School begin a remedial reading program designed to help them rediscover the joy of reading. These children who are often shy and often overlooked struggle with fluency and confidence. Their teacher, Ma'am Maaruga, believes in their potential and launches a 3-week intervention using storytelling circles, phonics games, and one-on-one coaching. Before the program begins, each child reads a short passage aloud. Their reading fluency is measured in words per minute (wpm). After the sessions, they read a similar passage, and their fluency is measured again. The results are as follows:

Learner	Before (wpm)	After (wpm)
1	45	52
2	50	58
3	42	47
4	48	55
5	46	53
6	44	50
7	49	56
8	47	54
9	43	49
10	46	52

Is there statistical evidence that the remedial reading program improved reading fluency? Use a 0.05 level of significance.

Solution:

Step 1. Null hypothesis (H_0): There is no difference in reading fluency before and after the program (i.e., $\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): Reading fluency improved after the program. (i.e., $\mu_1 - \mu_2 > 0$).

Step 2. $\alpha = 0.05$

Step 3. Since the samples are related, the appropriate test statistic to use is

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}}$$

Step 4. One-tailed test is used since the alternative hypothesis is directional (right-tailed). For a right-tailed test at $\alpha = 0.05$ with degrees of freedom df = 9:

Reject H_0 if $t > 1.833$

Otherwise, fail to reject H_0.

Step 5. The difference between reading fluency before and after the remedial program are computed and summarized.

Learner	Before (wpm)	After (wpm)	d	d^2
1	45	52	7	49
2	50	58	8	64
3	42	47	5	25
4	48	55	7	49
5	46	53	7	49
6	44	50	6	36
7	49	56	7	49
8	47	54	7	49
9	43	49	6	36
10	46	52	6	36
		Total	66	442

$$\bar{d} = \frac{\sum d}{n} = \frac{66}{10} = 6.6$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{442 - \frac{(66)^2}{10}}{10-1}} = 0.843$$

Results showed a mean of $\bar{d} = 6.6$ and a standard deviation of $s_d = 0.843$. Substituting all available information in the test statistic

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}} = \frac{6.6 - 0}{\frac{0.843}{\sqrt{10}}} = 24.76$$

Step 6. Given that the computed value of t falls in the critical region, that is, 24.76 > 1.833, then the null hypothesis is rejected.

Step 7. There is strong statistical evidence that the remedial reading program improved reading fluency.

Computations for t-test and z-test using Excel and Jamovi

Computation Process for MS Excel

I. Using Data Analysis Tool

a. Navigate to the Data Analysis tool:

- ✔ Click the Data tab and select Data Analysis.

b. Choose the test:

- ✔ For a t-test, select t-Test: Paired Two Sample for Means (or other appropriate t-test options).
- ✔ For a z-test, select Z-Test: Two Sample for Means.

c. Input the data:

- ✔ Specify the ranges for your data in *Variable 1 Range* and *Variable 2 Range*.

- ✔ If your data includes a header row, check the Labels box.

d. Enter the hypothesized mean:

- ✔ In Hypothesized Mean Difference, enter a value (usually 0) for a two-sample test to test if the means are equal.
- ✔ For a z-test, you will also need to provide the *Variable 1 Variance* and *Variable 2 Variance*.

e. Set the output location.

- ✔ Choose an *Output Range* for the results.

f. Run the test.

- ✔ Click OK to generate the output table, which includes p-values to help you determine statistical significance.

II. Using T.Test function

If all you are interested in is the p-value, a quick way to calculate this is by entering the following syntax directly into a cell: =T.TEST(array1,array2,tails,type)

Step 1: Input the data sets you want to test.

Step 2: Select a spreadsheet cell and type "=T.TEST(" here.

Step 3: Input the data arrays, the tail distribution value, and test type value.

For array1, type the first data range. For array2, type the second data range.

For the tail distribution value, input 1 for one-tailed test or 2 for two-tailed test.

For the test type value, input 1 for paired test, 2 for two sample equal variance test, or 3 for two sample unequal variance test.

Step 4. Press Enter. The value returned from this formula is the p-value.

Computation Process for Jamovi

a. One sample t-test

The dataset needs one continuous variable (e.g., test grade) and a specific, known value to test against.

Steps:

1. Click Analyses > T-Tests > One Sample T-Test.

2. Move your continuous variable to the Dependent Variables box.

3. Under the "Hypothesis" section, enter your test value in the "Test value" field.

4. To get the sample mean and other summary data, select Descriptives under "Additional Statistics".

5. The result is shown in the right panel.

b. Independent samples t-test

This test compares the means of a continuous variable between two different, independent groups. Thus, the dataset needs a continuous dependent variable and a grouping variable with exactly two levels

Steps:

1. Click Analyses > T-Tests > Independent Samples T-Test.

	2. Move your continuous outcome variable to the Dependent Variables box. 3. Move your two-level grouping variable to the Grouping Variable box. 4. In the options, you can select additional statistics like Descriptives and Effect size. 5. Check for assumptions like Equality of variances (Levene's test) under "Assumption Checks". 6. The result is shown in the right panel. **c. Paired samples t-test** This test compares the means of two continuous variables from the same group of subjects, such as before-and-after measurements. Hence, the dataset needs two continuous variables, with each row representing a single subject or observation. For example, a pretest score column and a posttest score column. Steps: 1. Click Analyses > T-Tests > Paired Samples T-Test. 2. Move your two continuous variables into the Paired Variables box. 3. Under the "Statistics" menu, you can add options for Effect size, Descriptives, and the Mean difference. 4. For Assumption Checks, you can select the Normality test for the difference scores. 5. The result is shown in the right panel. No matter which t-test you run, the output table provides the essential information for interpretation.	

t: The t-statistic, which is the size of the difference relative to the variation in your data.

df: The degrees of freedom, a value related to your sample size.

p: The p-value, which indicates the probability of observing your results if the null hypothesis were true.

If p **< 0.05** (or your chosen alpha level), the result is statistically significant, and you reject the null hypothesis. There is a statistically significant difference between the means.

If p **> 0.05**, the result is not statistically significant, and you fail to reject the null hypothesis.

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Instructions:
Read and solve each problem.

1. As part of a community-based outreach in Barangay Maliwanag, learners are tasked with evaluating battery performance for use in emergency kits. Two brands of battery, Meh Brand and Moh Brand, are being considered for distribution during typhoon season. To help the barangay council make an informed decision, learners test 10 batteries from each brand and record their lifespans (in hours):

Meh Brand:	88	85	89	80	81	72	90	85	84	91
Moh Brand:	80	79	77	82	75	81	77	73	85	83

Assuming that the conditions for inference are satisfied, which statistical test is most appropriate to determine whether Meh Brand batteries have a longer average lifespan than Moh Brand batteries?

2. At Maalalahanin National High School, ten senior high learners are quietly struggling. With deadlines

Activity B.3: Practicing Learned Skills

This activity provides learners with structured opportunities to apply hypothesis-testing procedures in authentic, community-based contexts. Learners analyze real-life situations involving independent samples, paired observations, and known population parameters to determine whether observed differences are statistically significant. The activity reinforces correct test selection, hypothesis formulation, and interpretation of results using both manual computation and conceptual reasoning.

To facilitate this activity the teacher may apply any of the following strategies.

Strategy 1. **Launch Pad: Ready, Rise, Reach**
(Implementing Tiered Problem Sets)
Tiered problem sets allow all learners to engage with the same concept at different levels of complexity. This strategy supports differentiated instruction and helps learners build confidence, deepen understanding, and apply critical thinking.

piling up, expectations rising, and personal worries weighing heavily, they report high levels of academic stress. The school's guidance office, led by Ma'am Mahinahon, launches a peer counseling circle, a safe space where learners can share, listen, and support one another. The program runs for two weeks. Each session begins with a short reflection, a prayer, and a prompt like "What's one thing you're carrying today?" Learners learn to name their emotions, validate each other's experiences, and practice grounding techniques. To assess the impact, Ma'am Mahinahon asks each learner to rate their anxiety level on a scale of 1–10 before and after the program. The results are as follows:

Learner	Before	After
1	8	6
2	7	5
3	9	6
4	6	4
5	8	5
6	7	6
7	9	7
8	6	5
9	8	6
10	7	5

Is there statistical evidence that the peer counseling circle reduced learners' anxiety levels? Use a 5 level of significance.

3. Two barangays, Barangay Malusog and Barangay Masigla launch distinct school feeding programs to address learner nutrition and learning readiness. Both programs run for two months, offering daily meals designed to boost energy, focus, and overall health. Barangay Malusog partners with local farmers to serve fresh vegetable-based meals. Barangay Masigla collaborates with a nutrition NGO to provide fortified rice and protein-rich snacks. After the program, learners from each barangay take a standardized nutrition assessment scored out of 100. Health officers randomly sample 40 learners from each barangay and record the following:

1. Begin with a short review or mini-lesson on the statistical test (e.g., two-sample t-test or z-test). Ensure learners understand:

- ✔ The purpose of the test
- ✔ The formula and steps
- ✔ When to use it

2. Tell learners they will choose or be assigned problems at three levels:

Basic: Straightforward computations with clearly given values
Moderate: Real-life scenarios requiring interpretation
Challenging: Open-ended tasks that ask for justification, critique, or alternative approaches

You may say:
"Today, you'll work on problems that match your current level of confidence. You can move up a tier as you grow more comfortable."

3. Prepare the problem sets.

Tier	Description	Sample Task
Basic	Plug-and-play	Given two groups with means, standard deviations, and sample sizes, compute the test statistic
Moderate	Real-life scenario	Compare sleep hours between early and late sleepers. Interpret the result in context
Challenging	Open-ended	Critique a learner's conclusion: Since the means are different, the review session worked. Is this statistically valid?

Barangay	Mean score	Population standard deviation	Sample size
Malusog	85	5	40
Masigla	82	6	40

Is there statistical evidence that Barangay Malusog's feeding program is more effective? Use a 5% level of significance, assuming known population standard deviations.

Answer Key:

1. Null hypothesis (H_0): There is no difference in the average life span between Meh Brand batteries and Moh Brand batteries. ($\mu_1 - \mu_2 = 0$).

   Alternative hypothesis (H_1): Meh Brand batteries have a longer average lifespan than Moh Brand batteries. ($\mu_1 - \mu_2 > 0$).

   Because it is assumed that the unknown variances of the two normal populations are equal and relatively small sample size, the appropriate test statistic to use is unpaired or independent sample t-test.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Given: $\bar{x}_1 = 84.5$ $\quad \bar{x}_2 = 79.2$ $\quad d_0 = 0$
$s_1 = 5.72$ $\quad s_2 = 3.74$
$n_1 = 10$ $\quad n_2 = 10$

Estimate of the common variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(10-1)(5.72)^2 + (10-1)(3.74)^2}{10 + 10 - 2} = 23.353$$

4. Let learners choose their tier or assign based on readiness. Encourage movement between tiers as confidence grows. Use peer coaching or small group discussions to support struggling learners.

5. After solving, guide learners to reflect using prompts like:
   - ✔ *What did you learn from the activity?*
   - ✔ *What made the problem challenging or meaningful?*
   - ✔ *How would you explain your solution to someone in a lower tier?*

Strategy 2. **The Learning in Tandem: Coach and Catch** *(Implementing Peer Coaching Pairs)*

Peer Coaching Pairs promote collaborative learning, deepen understanding, and build communication skills. One learner explains the steps of a problem, while the other listens actively, checks logic, and asks clarifying questions.

1. Choose a problem that involves multi-step reasoning such as computing a t-statistic, interpreting results, or writing hypotheses.

2. Pair learners intentionally:
   - Mix confidence levels (one explainer, one checker)
   - Rotate partners to build community
   - Consider language comfort, learning styles, or leadership roles
3. Assign roles
   - ✔ **Explainer**: Walks through each step aloud
   - ✔ **Checker:** Listens, asks questions, and verifies logic-

4. Walk around and listen to pairs. You may offer prompts if learners get stuck. Encourage respectful questioning and curiosity.

5. After 1–2 problems, switch roles:
   - ✔ The checker becomes the explainer

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(84.5 - 79.2) - 0}{\sqrt{(23.353)\left(\frac{1}{10} + \frac{1}{10}\right)}} = 2.45$$

For a right-tailed test at 0.05 significance level with degrees of freedom df = 18, use t-critical = 1.734.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

There is enough statistical evidence at the 5% level to conclude that the Meh Brand batteries have longer life span on average than Moh Brand batteries.

2. H_0 There is no difference in learners' anxiety levels before and after the program (i.e., $\mu_1 - \mu_2 = 0$).

H_1: Peer counseling circle reduced learners' anxiety level. (i.e., $\mu_1 - \mu_2 < 0$).

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}} = \frac{-2 - 0}{\frac{0.67}{\sqrt{10}}} = -9.44$$

For a left-tailed test at 0.05 significance level with degrees of freedom df = 9, use t-critical = -1.833.

Reject H_0 if t-computed < t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 0.05 significance level, there is sufficient evidence to conclude that peer counseling circle reduced learners' anxiety level.

✔ The explainer becomes the checker

This builds empathy and reinforces both skills.

6. Reflect together. Encourage gentle correction and affirming feedback. Remind learners: "We learn best when we teach and when we listen."

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3. Null hypothesis (H_0): There is no difference between Barangay Malusog and Barangay Masigla in their nutrition assessment score. ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): The nutrition assessment score of Barangay Malusog is higher than of Barangay Masigla. ($\mu_1 - \mu_2 > 0$).

Given: $\bar{x}_1 = 85$ $\bar{x}_2 = 82$ $d_0 = 0$
$\sigma_1 = 5$ $\sigma_2 = 6$
$n_1 = 40$ $n_2 = 40$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(85 - 82) - 0}{\sqrt{\frac{(5)^2}{40} + \frac{(6)^2}{40}}} = 2.43$$

For a right-tailed test at 0.05 significance level, use z-critical = 1.645.

Reject H_0 if z-computed > z-critical. Since the computed z-value is greater than the z-critical, the null hypothesis is rejected.

At the 5% significance level, there is statistical evidence that the feeding program of Barangay Malusog is more effective than of Barangay Masigla as measured by higher nutrition assessment scores.

</td><td></td></tr>
<tr><td rowspan="2">

C. Demonstrating Knowledge and Skills

</td><td colspan="2">

C.1. Finding Practical Application

</td></tr>
<tr><td>

Activity C.1: Making Real-World Connections

Instructions:
Read and analyze each scenario and perform the indicated task.

Scenario: At Maaral National High School, the senior high department is evaluating the impact of different learning modalities on learner performance. During the

</td><td>

Activity C.1: Making Real-World Connections

This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output.

</td></tr>
</table>

pandemic, learners were assigned to either modular learning (printed modules delivered weekly) or online learning (via Google Classroom and Zoom). Now that face-to-face classes have resumed, the school wants to understand how these modalities affected academic outcomes. The Math department selects a random sample of learners from each group and compares their final exam scores in General Mathematics.

Modality	Mean score	Population standard deviation	Sample size
Online learning	82	6	40
Modular learning	78	5	40

Question: Is there statistical evidence that online learning led to higher academic performance? Use a 5% level of significance, assuming known population standard deviations.

Smarter Together: Does Group Work Boost Math Scores?

Scenario: Two Grade 11 sections at Sipnayan National High School prepare for a quarterly math quiz. The school's math coordinator, Ma'am Masagana, wants to explore how different study strategies affect performance. Section Maramihan reviews through group work, solving problems collaboratively and explaining concepts to one another. While Section Mapag-isa reviews through individual study, using self-paced modules and personal notes. After the quiz, Ma'am Masagana collects the following data:

Section / Method	Mean score	Sample standard deviation	Sample size
Maramihan (group work)	84	5.5	15
Mapag-isa (individual study)	80	6	15

Answer Key:

Learning Across Lanes: Where Do You Learn Best?

Null hypothesis (H_0): There is no difference in the between learners in modular learning and online learning in their final exam scores in General Mathematics. ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): The final exam score of learners in online learning is higher than learners in modular learning. ($\mu_1 - \mu_2 > 0$).

Given: $\bar{x}_1 = 82$ $\quad \bar{x}_2 = 78$ $\quad d_0 = 0$
$\sigma_1 = 6$ $\quad \sigma_2 = 5$
$n_1 = 40$ $\quad n_2 = 40$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(82 - 78) - 0}{\sqrt{\frac{(6)^2}{40} + \frac{(5)^2}{40}}} = 3.24$$

For a right-tailed test at 0.05 significance level, use z-critical = 1.645.

Reject H_0 if z-computed > z-critical. Since the computed z-value is greater than the z-critical, the null hypothesis is rejected.

At the 5% significance level, there is statistical evidence that online learning led to higher academic performance compared to modular learning as measured by the final exam score in General Mathematics.

Smarter Together: Does Group Work Boost Math Scores?

Null hypothesis (H_0): There is no difference in the average score in Mathematics between Maramihan (group work) and Mapag-isa (individual study). ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): Collaborative learning (group work) leads to significantly higher math scores compared to individual study. ($\mu_1 - \mu_2 > 0$).

<table>
<tr><td></td><td>Question: Is there statistical evidence that collaborative learning leads to significantly higher math scores? Use a 5% level of significance, assuming equal population variances.</td><td>
Given: $\bar{x}_1 = 84$ $\bar{x}_2 = 80$ $d_0 = 0$
$s_1 = 5.5$ $s_2 = 6$
$n_1 = 15$ $n_2 = 15$

Estimate of the common variance:
$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$= \frac{(15-1)(5.5)^2 + (15-1)(6)^2}{15+15-2} = 33.125$$
$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(84-80) - 0}{\sqrt{(33.125)\left(\frac{1}{15} + \frac{1}{15}\right)}} = 1.90$$
For a right-tailed test at 0.05 significance level with degrees of freedom df = 28, use t-critical = 1.701.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 5% significance level, there is statistical evidence that collaborative learning (group work) leads to significantly higher math scores compared to individual study.</td></tr>
<tr><td></td><td colspan="2">C.2. Making Generalization</td></tr>
<tr><td></td><td>Activity C.2: Wrapping up the Lesson

Instructions:
Do as directed.3H (HEAD, HEART, HAND) Reflection
Dear Learners,

Now that you've explored what the data tells us, it's time to reflect not just on the numbers, but on what they mean to you. We'll use the HEAD, HEART, HAND format</td><td>Activity C.2: Wrapping up the Lesson

For the summary, you may facilitate the provided activity.

This activity provides learners with a structured opportunity to reflect on their learning experience by integrating cognitive understanding, personal insight, and intended action. Using the 3H Reflection (Head, Heart, Hand) framework, learners are guided to think about what they learned from data analysis, connect statistical findings to their own feelings and experiences, and identify</td></tr>
</table>

	to guide your reflection. This helps you think deeply, feel honestly, and act wisely. **HEAD – What I Learned** Write 2–3 sentences about what you understood from the activity. You can talk about the following: • *What the data showed* • *What the test result meant* • *What surprised or challenged you* **HEART – What I Felt or Realized** Write 2–3 sentences about your feelings or realizations. You can reflect on the following: • *How the result connects to your experience* • *What it made you think about* • *Any personal insight or emotion* **HAND – What I'll Do or Suggest** Write 2–3 sentences about what action you'll take or recommend. You can share the following: • *What you'll try next time* • *Advice for classmates or teachers* • *A change you'd like to se*	meaningful actions or recommendations. The activity reinforces that statistics is not only about numbers, but also about informed thinking, empathy, and responsible decision-making.
	C.3. Evaluating Learning	
	Activity C.3: Assessing Learning Outcomes **Instructions:** Solve the following problems as required**.** 1. A Grade 11 class conducts a mini-research project on screen time. They ask a random sample of learners to report how many hours they spend on screens each day including phones, tablets, and computers. The results are as follows: Boys: Average screen time = 4.2 hours Standard deviation = 0.6	**Activity C.3: Assessing Learning Outcomes** This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation. ***Answer Key:***

Sample size = 25

Girls: Average screen time = 3.8 hours
Standard deviation = 0.5
Sample size = 25

Is there a significant difference in screen time between boys and girls? Use a 5% level of significance.

2. Two Grade 11 sections take the same math quiz. One section attended a review session the day before, while the other did not. After the quiz, the math teacher, Ma'am Sipnayan, collects the following data:

Group	Mean score	Standard deviation	Sample size
With review	18	2	42
Without review	16.5	2.5	42

Did the review session help learners score significantly higher? Use α = 0.05.

3. At a certain school, a group of Grade 11 learners volunteered to join a peer mentoring program aimed at improving study habits and academic performance. Each learner recorded their weekly study hours and scores before joining the program and again after four weeks of participation. The pretest and posttest scores of the learners before and after the peer mentoring program are as follows:

Learner	Pretest Score	Posttest Score
1	12	15
2	14	15
3	15	25
4	13	23
5	13	18
6	16	17
7	17	20
8	19	21
9	15	20

1. Null hypothesis (H_0): There is no difference in screen time between boys and girls. ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): There is a significant difference in screen time between boys and girls. ($\mu_1 - \mu_2 \neq 0$).

Given: $\bar{x}_1 = 4.2$ $\bar{x}_2 = 3.8$ $d_0 = 0$
$s_1 = 0.6$ $s_2 = 0.5$
$n_1 = 25$ $n_2 = 25$

Estimate of the common variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(25-1)(0.6)^2 + (25-1)(0.5)^2}{25 + 25 - 2} = 0.305$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(4.2 - 3.8) - 0}{\sqrt{(0.305)\left(\frac{1}{25} + \frac{1}{25}\right)}} = 2.56$$

For a two-tailed test at 0.05 significance level with degrees of freedom df = 48, use t-critical = ± 2.011.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 5% significance level, there is statistical evidence that there is a significant difference in screen time between boys and girls.

2. Null hypothesis (H_0): The review session did not lead to higher scores. ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): The review session led to significantly higher scores. ($\mu_1 - \mu_2 > 0$).

Given: $\bar{x}_1 = 18$ $\bar{x}_2 = 16.5$ $d_0 = 0$
$s_1 = 2$ $s_2 = 2.5$

10	17	22

Did the peer mentoring program lead to a significant improvement in the scores? Use 0.05 level of significance.

Guide Questions:

1. How do you decide whether to use a z-test or a t-test when comparing two groups?

2. What does a statistically significant result really tell us about the difference between two groups?

3. Why does it matter to know if your data is paired or unpaired when interpreting the results of a hypothesis test?

$n_1 = 42 \qquad n_2 = 42$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(18 - 16.5) - 0}{\sqrt{\frac{(2)^2}{42} + \frac{(2.5)^2}{42}}} = 3.04$$

For a right-tailed test at 0.05 significance level, use z-critical = 1.645.

Reject H_0 if z-computed > z-critical. Since the computed z-value is greater than the z-critical, the null hypothesis is rejected.

At the 5% significance level, there is sufficient evidence to conclude that the review session effectively increased the math quiz scores.

3. Null hypothesis (H_0): Peer mentoring program did not lead to a significant improvement in the scores. ($\mu_1 - \mu_2 = 0$).

   Alternative hypothesis (H_1): Peer mentoring program led to a significant improvement in the scores. ($\mu_1 - \mu_2 > 0$).

The difference between pretest and posttest scores before and after the peer mentoring program are computed and summarized.

Learner	Pretest	Posttest	d	d^2
1	12	15	3	9
2	14	15	1	1
3	15	25	10	100
4	13	23	10	100
5	13	18	5	25
6	16	17	1	1
7	17	20	3	9
8	19	21	2	4
9	15	20	5	25
10	17	22	5	25
		Total	45	299

$$\bar{d} = \frac{\sum d}{n} = \frac{45}{10} = 4.5$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{299 - \frac{(45)^2}{10}}{10-1}} = 3.274$$

Results showed a mean of $\bar{d}$ = 4.5 and a standard deviation of s_d = 3.274. Substituting all available information in the test statistic

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}} = \frac{4.5 - 0}{\frac{3.274}{\sqrt{10}}} = 4.35$$

For a right-tailed test at 0.05 significance level with degrees of freedom df = 9, use t-critical = 1.833.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 5% significance level, there is sufficient evidence to conclude that peer mentoring program led to a significant improvement in the scores.

C.4. Additional Activities

Activity C.4: Extending and Reinforcing Learning

Instructions:
Solve the following problems as required.

1. At a certain junior high school, two groups of 20 learners were surveyed about their daily screen time depending on the mobile phone brand they use. The group using Brand A reported an average screen time of 5 hours per day with a standard deviation of 1.0, while the group using Brand B reported an average screen time of 3.5 hours per day with a standard deviation of 0.8. At the 0.05 level of significance, is the average screen time of Brand A users significantly higher than that of Brand B users?

Activity C.4: Extending and Reinforcing Learning

This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

Answer Key:

1. Null hypothesis (H_0): There is no difference in average screen time between Brand A and Brand B users. (μ_1 - μ_2 = 0).

   Alternative hypothesis (H_1): The average screen time of Brand A users is significantly higher than that if Brand B users. (μ_1 - μ_2 > 0).

2. At Dimakatulog National High School, learners are grouped based on their sleep routines. Early sleepers go to bed before 10 PM and avoid screens an hour before sleeping. On the other hand, late sleepers stay up past midnight and often use phones or gadgets before bed. To explore whether sleep habits affect total sleep hours, the class collects data from a random sample of learners:

Group	Mean Sleep hours	Standard deviation	Sample size
Early sleepers	7.2	0.6	60
Late sleepers	6.5	0.7	60

Question: Is there a significant difference in sleep hours between early and late sleepers? Use α = 0.05.

3. At Mabuhay High School, 15 Grade 11 learners participate in a group math activity where they solve problems together and explain their reasoning to peers. Before and after the activity, each learner rates their confidence in math on a scale from 1 to 10.
The teacher collects the following data:

- Before the activity: Learners report their initial confidence levels.

- After the activity: Learners rate their confidence again.

The results are as follows:

Learner	Before the activity	After the activity
1	6	7
2	5	8
3	4	7
4	5	7
5	6	6
6	4	9
7	3	8
8	4	7
9	5	8
10	4	7
11	5	6

Given: $\bar{x}_1 = 5.0$ $\quad \bar{x}_2 = 3.5$ $\quad d_0 = 0$
$s_1 = 1.0$ $\quad s_2 = 0.8$
$n_1 = 20$ $\quad n_2 = 20$

Estimate of the common variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(20-1)(1.0)^2 + (20-1)(0.8)^2}{20 + 20 - 2} = 0.82$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(5 - 3.5) - 0}{\sqrt{(0.82)\left(\frac{1}{20} + \frac{1}{20}\right)}} = 5.24$$

For a right-tailed test at 0.05 significance level with degrees of freedom df = 38, use t-critical = 1.686.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 5% significance level, there is statistical evidence that the average screen time of Brand A users is significantly higher than that if Brand B users.

2. Null hypothesis (H_0): There is no difference in sleep hours between early and late sleepers. ($\mu_1 - \mu_2 = 0$).

Alternative hypothesis (H_1): There is a difference in sleep hours between early and late sleepers. ($\mu_1 - \mu_2 \neq 0$).

Given: $\bar{x}_1 = 7.2$ $\quad \bar{x}_2 = 6.5$ $\quad d_0 = 0$
$s_1 = 0.6$ $\quad s_2 = 0.7$
$n_1 = 60$ $\quad n_2 = 60$

12	6	7
13	5	7
14	4	8
15	3	7

Question: Did the group activity significantly increase learners' confidence in math? Use α = 0.05.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(7.2 - 6.5) - 0}{\sqrt{\frac{(0.6)^2}{60} + \frac{(0.7)^2}{60}}} = 5.88$$

For a two-tailed test at 0.05 significance level, use z-critical = ±1.645.

Reject H_0 if *z*-computed > *z*-critical. Since the computed z-value is greater than the z-critical, the null hypothesis is rejected.

At the 5% significance level, there is sufficient evidence to conclude that there is a difference in sleep hours between early and late sleepers.

3. Null hypothesis (H_0): Group activity did not lead to a significant improvement in the learners' confidence in math. (μ_1 - μ_2 = 0).

   Alternative hypothesis (H_1): Group activity led to a significant improvement in the learners' confidence in math. (μ_1 - μ_2 > 0).

The difference between learners' confidence in math before and after the activity are computed and summarized.

Learner	Before	After	*d*	d^2
1	6	7	1	1
2	5	8	3	9
3	4	7	3	9
4	5	7	2	4
5	6	6	0	0
6	4	9	5	25
7	3	8	5	25
8	4	7	3	9
9	5	8	3	9
10	4	7	3	9
11	5	6	1	1
12	6	7	1	1
13	5	7	2	4
14	4	8	4	16

15	3	7	4	16
		Total	40	138

$$\bar{d} = \frac{\sum d}{n} = \frac{40}{15} = 2.67$$

$$s_d = \sqrt{\frac{\sum d^2 - \frac{(\sum d)^2}{n}}{n-1}} = \sqrt{\frac{138 - \frac{(40)^2}{15}}{15-1}} = 1.50$$

Results showed a mean of $\bar{d} = 2.67$ and a standard deviation of $s_d = 1.50$. Substituting all available information in the test statistic

$$t = \frac{\bar{d} - d_0}{\frac{s_d}{\sqrt{n}}} = \frac{2.67 - 0}{\frac{1.50}{\sqrt{15}}} = 6.89$$

For a right-tailed test at 0.05 significance level with degrees of freedom df = 14, use t-critical = 1.7613.

Reject H_0 if t-computed > t-critical. Since the computed t-value is greater than the t-critical, the null hypothesis is rejected.

At the 5% significance level, there is sufficient evidence to conclude that group activity led to a significant improvement in the learners' confidence in math.

III. CONTENT	**Lesson 11.5: Correlation** *Recommended time allotment: 2 hours*	
IV. OBJECTIVES	By the end of the lesson, the learners… 1. construct a scatter plot and draw the line of best-fit for a scatter plot 2. compute for the Pearson r correlation and interpret the result 3. determine the degree of relationship between two variables using Pearson r correlation 4. draw conclusions from the results of hypothesis testing on relationship using Pearson r correlation 5. solve real-life problems involving simple linear regression analysis	
V. PROCEDURES	**ACTIVITIES**	**ANNOTATION**

<table>
<tr><td rowspan="2">A. Activating Prior Knowledge</td><td colspan="2">A.1. Eliciting Prior Knowledge</td></tr>
<tr><td>Activity A.1: Leveling Learner Readiness

Instructions:
Perform the indicated tasks.

1. Read the following scenario:
a. As part of a school-wide wellness initiative, the MAPEH department at Malakas National High School decided to measure the average height of Grade 11 learners. The goal was to understand growth patterns and promote health awareness among adolescents. Teachers collected height data from all Grade 11 sections during homeroom period.

b. In Barangay Masigasig, a group of senior high school learners noticed that many of their classmates were feeling tired during morning classes. Curious about the cause, they decided to investigate whether screen time before bed might be affecting their sleep duration.

c. A group of Grade 11 Academic Track learner-researchers from Mapagmahal National High School noticed that some learners consistently perform well while others struggle. To understand why, the group asked learners to reflect on how many hours they study each week, how often they attend class, whether they eat breakfast before school. The group then compared these responses to their quarterly grades to see which habits might be linked to better performance.

d. As part of a homeroom activity in the first week of the school year, Ms. Mahinahon, the Grade 11 class adviser at Maligaya National High School, decided to create a profile of her learners. She collected each learner's birthdate and calculated their age as of October 31. Her goal was to understand the age distribution in her class, especially since some learners had shifted strands or returned after a gap year.

e. A group of Mathematics teachers at Masining National High School wanted to explore how learners' screen habits might relate to their academic performance. With the growing popularity of streaming platforms like Netflix, they decided to investigate the relationship</td><td>Activity A.1: Leveling Learner Readiness

This activity is designed to assess and activate learners' prior knowledge about types of datasets based on the number of variables involved. Through real-life, school- and community-based scenarios, learners analyze data situations and distinguish among univariate, bivariate, and multivariate datasets. The activity encourages collaborative discussion, critical thinking, and conceptual understanding before formal instruction on statistical analysis.

For this activity you may choose any of the following strategies.

Strategy 1. Clue Card Challenge
(Learners will classify real-life data scenarios as Univariate (U), Bivariate (B), or Multivariate (M) by analyzing the number and nature of variables involved).

1. Create 5 clue cards describing scenario involving data.
2. Divide the class into small teams (3–5 members each). Provide each team with a set of clue cards and a classification sheet.
3. The group read each clue card and discuss:
a. How many variables are present?
b. What is being measured or compared?
4. The group labels each card as U, B, or M on their sheet. For each classification, the groups write a brief explanation.
5. After each group posts their answers, everyone walks around the room to look at the other groups' work.

Strategy 2. Station Walk
1. Prepares five (5) different stations, each focusing on one of the scenarios presented.
2. The class is divided into small groups of at least 5 members, and each group is assigned to a different starting station.
3. Learners will read the scenario and discuss how many variables are involved. Classify the data type into U</td></tr>
</table>

<table>
<tr><td></td><td>between the number of hours Grade 11 learners spent watching Netflix shows per week, and their First Semester grades in General Mathematics

2. With your group, compare the different scenario and identify which of them will be under univariate, bivariate, and multivariate.

3. Answer the following questions together:
a. What makes a dataset univariate? What are we measuring, and how many variables are involved?
b. What changes when we look at two variables instead of one?
c. What makes a dataset multivariate? How many variables must be present, and how are they connected?

4. Take turns sharing your group's answers. Be ready to explain:
a. What patterns did you notice in how variables were used?
b. How does the number of variables affect the kind of analysis we can do?
c. Why is it important to classify data correctly before analyzing it?</td><td>(univariate), B (bivariate), or M (Multivariate). Write a short explanation of their answer.
4. After 3-5 minutes, signal the groups to rotate clockwise. Repeat the task at the new station.
5. After visiting all the stations, gather the class. Ask groups to share one scenario they found interesting or tricky. Ask the following questions:
- What made each scenario U, B, or M?
- How does the number of variables affect the kind of analysis we can do?

Note: The teacher facilitates by rotating around the room, assisting and guiding learners as needed.

Answer Key:
a. Univariate
b. Bivariate
c. Multivariate
d. Bivariate
e. Bivsriate

3.a. Univariate dataset is a dataset that involves one variable. We focus on the distribution, central tendency (mean, median, mode), and spread (range, variance, standard deviation) of that single variable.

3.b. Analysis shifts from describing one variable to exploring how two variables are connected. We now look at the relationship between the two variables.

3.c. Multivariate dataset is a dataset that involves three or more variables. We measure relationships among multiple variables simultaneously. Analysis becomes more complex, focusing on how several variables interact together.</td></tr>
<tr><td></td><td colspan="2">A.2. Establishing the Purpose of the Lesson</td></tr>
<tr><td></td><td>Activity A. 2: Appreciating Lesson Relevance

Instructions:
Do as indicated.
1. Look at the following variable pairs on the board.
a. Study hours vs quiz scores</td><td>Activity A. 2: Appreciating Lesson Relevance

This activity encourages learners to connect statistical concepts to their own experiences by exploring how different variables in everyday life may relate to one another. Learners make predictions about the relationship between pairs of variables, share their ideas with peers,</td></tr>
</table>

	b. Attendance vs. performance c. Screen time vs. focus d. Household chores vs. study hours e. Parent involvement vs. academic motivation f. Internet access at home vs. research output g. Commute time vs. punctuality h. Time spent on hobbies vs. academic stress i. Peer support vs. willingness to try difficult task j. Noise level at home vs. ability to study k. Social media use vs. homework completion l. Review time vs. confidence in answering math problems m. Frequency of asking questions vs. test improvement n. Participation in class vs. oral recitation scores o. Weekend study hours vs. Monday quiz scores 2. Choose one pair that interests you. Pick something you've experienced, wondered about, or want to explore. 3. Write your prediction. Use a sticky note or card to complete one of these sentences: "I think more ___ leads to more ___." "I'm not sure if ___ affects ___." "I think these two are not related." 4. Post your prediction on the wall. Place it under the matching variable pair so others can read it. 5. Walk around and read your classmates' ideas. Notice which predictions are similar, different, or surprising. 6. Talk with your group. Share your thoughts: -Do we agree or disagree? -How could we test this using data?	and reflect on how data can be used to confirm or challenge these predictions. The activity fosters curiosity, critical thinking, and appreciation of statistics as a tool for understanding real-life phenomena. For this activity you may choose any of the following strategies. Strategy 1. **Prediction Wall: Think Before You Measure** (The strategy will help learners reflect on possible relationships between two variables using real-life contexts, preparing them to understand correlation and how Pearson's r measures it.) 1. Prepare a large wall space or bulletin board labeled Prediction Wall. 2. Post the different real-life variable pairs reflecting learner life on the board. 3. Give each learner a card or note. Display sentence starters: "I think more ___ leads to more ___." "I'm not sure if ___ affects ___." "I think these two are not related." 4. Write a sample prediction (e.g. "I think more sleep leads to more energy.") and post it under the matching variable pair. 5. Learners choose a variable pair and write their prediction. They will post their note under the correct heading. 6. Learners walk around and read others' predictions. Encourage them to notice the similarities and differences, surprising ideas, and questions they'd like to test. 7. In groups, ask guiding questions: - What made you think those variables might be connected? - How can we move from guessing to measuring? Strategy 2. **Predict and Pair** (This strategy will help learners think critically about how two variables might be related, and to practice forming hypotheses before measuring relationships using Pearson's r.)

		1. Create cards with real-life variables relevant to learners' lives. 2. Give each learner one variable card. Make sure there's a mix of academic, personal, and behavioral variables. 3. Learners walk around the room and find a partner whose variable might be related to theirs. 4. Together, they complete a prediction sentence: "We think more ___ leads to more ___." "We're not sure if ___ affects ___." "We think ___ and ___ are not related." 5. Each pair writes their prediction on a sticky note or mini poster. Post it on the board or wall under a heading like Our Prediction Pairs. 6. Invite a few pairs to share their reasoning. Ask the following: - Why do you think these variables are connected? - How could we test this using data? Strategy 3. **Variable Match Stations** (This strategy will help learners recognize and reflect on possible relationships between variables through matching, discussion, and prediction preparing them to understand correlation and Pearson's r.) 1. Create a set of cards with single variable the reflect the lives of the learners. 2. Arrange tables or areas labeled with Station T, E, A, M. At each station, place a small pile of variable cards and a matching mat (a sheet with two blank spaces labeled Variable 1 and Variable 2). 3. Learners rotate through stations in pairs or small groups. At each station, they will choose two variables they think might be related. Place them on the mat. Write a prediction using sentence starters: "We think more ___ leads to more ___." "We're not sure if ___ affects ___." "We think ___ and ___ are not related." 4.After visiting all stations, learners walk around to read others' matches and predictions. Encourage respectful discussion: - Do we agree?

		- What data would help us test this? *Tips for the Teacher:* 1. Model a pairing first: “I have ‘studying hours’ and I will pair them with ‘quiz scores’ because I think more studying hours leads to more quiz scores.” 2. Encourage learners to think critically and respectfully about each other’s ideas.
B. Instituting New Knowledge	***B.1. Presenting Examples***	
	Activity B.1: Exploring Key Concepts **Instructions**: Follow the task as directed. 1. Read and understand the following scenarios and answer the questions provided. **Scenario 1:** In the Grade 11 class of Mr. Maagap, learners were asked to track their sleep hours for a week. Those who consistently slept 7–8 hours reported feeling more alert and energetic during morning subjects. Meanwhile, learners who slept less than 5 hours often felt tired and struggled to participate in early discussions. **Questions:** 1. Do you think more sleep leads to more energy, less energy, or no change? 2. Would this relationship be strong, moderate, or weak? 3. If we plotted sleep hours and energy levels on a graph, what would the pattern look like? **Scenario 2:** During a study habits survey, learners shared how much time they spent on social media each day. Many who spent over 4 hours scrolling through apps like TikTok and Facebook said they found it harder to concentrate during study time. In contrast, those who	**Activity B.1: Exploring Key Concepts** This activity helps learners investigate and interpret relationships between two variables using real-life classroom and lifestyle scenarios. By analyzing patterns in sleep, study habits, social media use, and nutrition, learners develop a deeper understanding of how variables can be positively, negatively, or not at all related, and how the strength of a relationship can vary. The activity emphasizes critical thinking, reasoning with evidence, and connecting statistics to everyday experiences. For this activity you may choose any of the following strategies. Strategy 1. **From Station to Solution** (The strategy will help learners explore and compare different real-life scenarios involving statistical concepts by engaging in collaborative, movement-based learning. Through discussion and reflection at each station, learners deepen their understanding of how context determines the most appropriate statistical tool, and how relationships between variables can be interpreted meaningfully.) 1. Before class, set up three stations around the room. Each station features a different scenario involving two variables: Station Red: Sleep hours vs energy level Station Blue: Social media time vs study focus Station Yellow: Rice intake vs quiz scores

limited their screen time to under 2 hours reported better focus and fewer distractions.

Questions:

1. As social media time increases, what happens to focus?
2. Would you describe this relationship as positive, negative, or neutral?
3. Do you think the connection is strong, weak, or somewhere in between?

Scenario 3: In a health and nutrition module, learners recorded their rice intake and quiz scores over two weeks. Despite variations in how much rice they ate, there was no clear pattern linking rice consumption to quiz performance. Some high scorers ate a lot; others ate less.

Questions:

1. Do you think rice intake affects quiz performance? Why or why not?
2. Would you say these two variables are connected in any clear way?
3. If we plotted rice intake on the x-axis and quiz scores on the y-axis, what would the scatterplot look like? Would the dots form a pattern, or would they be scattered randomly?

2. At each station, include the scenario, guiding questions, and space for learners to write or discuss their group response.
3. Divide the class into small groups (around 5 learners each) and assign each group a starting station.
4. At each station, learners read the scenario. Talk about it as a group and predict the direction (positive, negative, or none) and strength (strong, moderate, weak, or none) of the relationship. Write their prediction and reasoning.
5. After visiting all three stations, bring the class together for a whole-group discussion.
6. Guide a class sharing session where each group explains:
- What they predicted
- Why they think the relationship is strong, weak, or nonexistent

Strategy 2. **r You Sure? Perfect Match**

(This strategy will help learners understand the strength and direction of relationships between variables by matching real-life examples, visual patterns, and numerical values.)

1. Create three sets of cards:

a. Variable Pair Scenario Cards

b. Scatterplot Cards
Each card shows a scatterplot with a distinct pattern:
- Tight upward trend
- Loose downward trend
- Scattered points
- Tight downward trend
- Loose upward trend

c. SD (Strength and Direction) Cards
- Strong positive
- Weak negative
- No correlation
- Strong negative
- Weak positive

2. Divide the class into small groups (5 learners). Each group gets a mixed set of cards.
3. Learners work together to match each scenario card with the most appropriate scatterplot with the strength and direction.
4. After matching, groups present one set to the class. You guide the discussion:
- What clues helped you match these?
- Could the same scatterplot fit a different scenario?

B.2. Discussing the Concept

Activity B.2: Deepening Conceptual Understanding

Instructions:
The following are the important ideas to be understand, listen attentively and participate actively in the discussion.

Scatter Plot

A **scatter plot** is a graphical representation used to illustrate the relationship between two quantitative variables. It depicts the distribution, degree, and pattern of association within bivariate data by plotting values on a Cartesian coordinate system. Conventionally, the independent or explanatory variable is assigned to the x-axis, while the dependent or response variable is positioned along the y-axis. By examining the plotted points, one can visually assess the presence, direction, and strength of any potential relationship between the two variables.

Example: Engr. Mathiyaga, a Mathematics teacher, sought to investigate the relationship between the number of hours devoted to studying Mathematics and the performance of Grade 11 learners in their summative assessment. To examine this relationship, data were collected on learners' study hours and corresponding assessment scores.

Hours of studying Mathematics	2	4	6	8	10	11

Activity B.2: Deepening Conceptual Understanding

This activity deepens learners' understanding of bivariate data analysis, focusing on scatter plots, correlation, and simple linear regression. Through real-life examples, learners explore how two quantitative variables are related, how to measure the strength and direction of linear relationship**s** using the Pearson Product Moment Correlation Coefficient (r), and how to predict outcomes using the regression line. The activity encourages learners to connect statistical concepts to practical scenarios, such as studying habits, parental involvement, advertising effectiveness, and health practices, promoting critical thinking and data-driven decision-making**.**

To facilitate the discussion of the following topics you may apply the following strategies:

Small Group Discussion

Part A: Understanding Correlational Analysis using Pearson r

Instructions:
1. Divide the class into small groups (4–5 learners per group).

2. Prepare printed or digital copies of the lesson handouts and sample dataset.

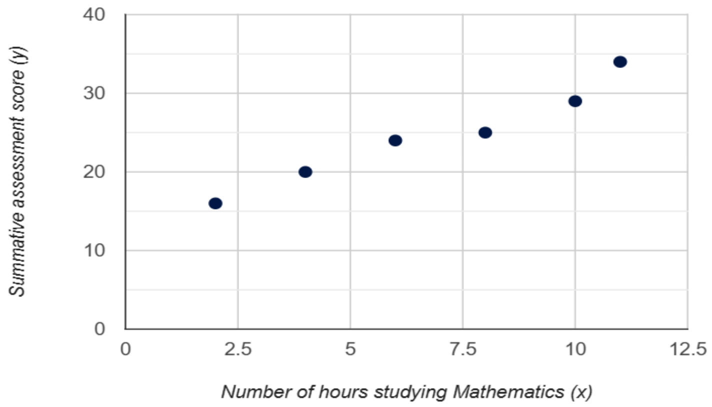
3. Distribute handout to all groups and say:

Summative assessment score	16	20	24	25	29	34

Solution: In the given example, the variables taken into consideration are the number of hours spent studying Mathematics and the summative assessment results among Grade 11 learners. Here, the independent variable is the number of hours spent in studying Mathematics (x) while the dependent variable is the summative assessment score (y).

The following coordinates are used in constructing the scatter plot:

(2, 16), (4, 20), (6, 24), (8, 25), (10, 29), (11, 34)



As seen on the xy-coordinate plane, the points appear to follow a straight line pointing upward to the right. This means that the bivariate data are linearly connected to some degree and that the variables have a positive relationship. It represents a positive trend that as the number of hours spent studying Mathematics rises, the summative assessment scores rise as well.

Example: Dr. Maaruga, a Social Science teacher, conducted a study to examine the relationship between parental involvement and the anxiety levels of Junior High School learners in a particular school. Parental involvement was operationalized as the number of hours parents and their children spent together, while learner

"After exploring different test of means and difference, each group will now explore the concept of measures of relationships (i.e. Pearson product moment correlation coefficient). You will also look at real-world scenarios and interpret the results."

4. Instruct each group to:

- Read the definitions and examples in the handout.
- Highlight key concepts and discuss their insights.
- Answer guiding questions like:
  - What does the sign of r tell you about the direction of the relationship?
  - How strong is the relationship based on the value of r?
  - What does your result tell you about the relationship between the variables?
  - Can this result be trusted based on the sample size and significance test?
  - How might this correlation be useful in real-life decision-making?

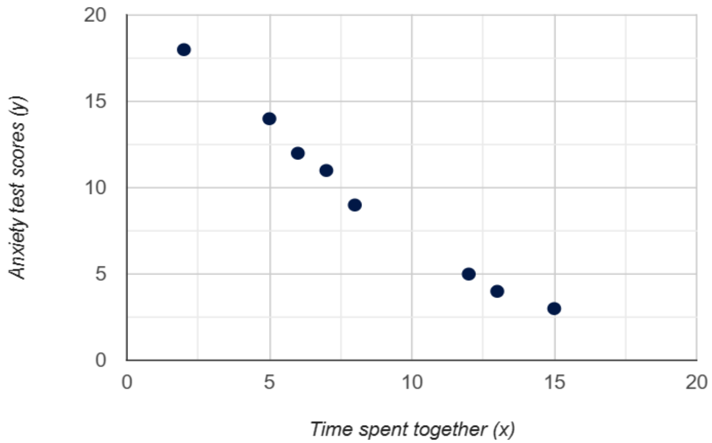
anxiety was measured using standardized scores from the Generalized Anxiety Disorder (GAD) assessment tool. The data collected consisted of paired values representing hours of parental engagement and corresponding anxiety test scores of the learners.

Time spent together (x)	2	6	7	8	5	15	12	13
Anxiety test scores (y)	18	12	11	9	14	3	5	4

Solution: In this given situation, the bivariate data are time spent by the parents and their children together and anxiety test scores among learners. Here, the time spent by the parents and their children together affects the anxiety test scores among learners. Therefore, the independent variable is the *time spent together* (x). On the other hand, the dependent variable is the *anxiety test scores* (y).

In constructing the scatter plot, the following coordinates are used:

(2, 18), (6, 12), (7, 11), (8, 9), (5, 14), (15, 3), (12, 5), (13, 4)



The points appear to be linked by a straight line pointing downward to the right. A negative correlation is represented by the scatter plot. It represents a negative pattern, because as the amount of time parents and their children spend together increases, there is a decrease in the anxiety test scores.

Pearson Product Moment Correlation Coefficient

The *Pearson Product Moment Correlation Coefficient*, denoted by *r*, calculates the degree of relationship or strength of the linear relationship between bivariate data measured on an interval or ratio scale. The formula for *r* is given below.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where:

n = number of pairs of sample data

Σx = sum of all x-values

Σy = sum of all y-values

Σx^2 = sum of all squared x-values

Σxy= sum of all the product of x- and y-values

$(\Sigma x)^2$ = square of the sum of all x-values

$(\Sigma y)^2$ = square of the sum of all y-values

Σy^2 = sum of all squared y-values

Positive relationship
• When one variable has high scores, the other variable also tends to have high scores.

Negative relationship
• When one variable has high scores, the other variable tends to have low scores, and vice versa.

Notes:
• Pearson r can be calculated provided that the distributions of the variables are unimodal and exhibit relative symmetry. Importantly, the assumption of normality is not strictly required, as the measure remains valid under conditions where the data are reasonably balanced and not heavily skewed.

• You may use the following categorization:

r	**Indication**
Between ± 0.80 to ± 1.00	High correlation
Between ± 0.60 to ± 0.79	Moderately high correlation
Between ± 0.40 to ± 0.59	Moderate correlation
Between ± 0.20 to ± 0.39	Low Correlation

Between ± 0.01 to ± 0.19   Negligible correlation

• Always take special care in interpreting the results of correlation:

- ✓ The correlation coefficient (r) serves as a statistical measure that captures only the strength of linear relationships between two variables.

- ✓ It is important to recognize that a high correlation observed within a sample does not necessarily reflect a true relationship in the broader population, as sampling variability may produce misleading results.

- ✓ Moreover, even when a strong correlation is present, it should not be interpreted as evidence of causation. A high value of r merely indicates association between variables X and Y, without implying that changes in one directly cause changes in the other.

Example: Determine the strength of the linear relationship between the number of hours studying Mathematics and summative assessment scores among learners. The data are as follows:

Hours of studying Mathematics (x)	2	4	6	8	10	11
Summative assessment score (y)	16	20	24	25	29	34

Solution. To compute for *r*, we need to find for **Σx, Σx², Σy, Σy², Σxy**. It is easier to arrange the data in a five-column table and then add all of the entries to each column.

X	***Y***	***x²***	***y²***	***Xy***
2	16	4	256	32
4	20	16	400	80
6	24	36	576	144
8	25	64	625	200

10	29	100	841	290
11	34	121	1156	374
Σx = 41	**Σy = 148**	**Σx² = 341**	**Σy² =3854**	**Σxy = 1120**

Using the computational formula for r, the total of each column from the table and n = 6, we have

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$r = \frac{6(1120) - (41)(148)}{\sqrt{[6(341) - (41)^2][6(3854) - (148)^2]}}$$

$$r = \frac{7392 - 6552}{\sqrt{[2046 - 1681][23124 - 21904]}}$$

$$r = \frac{652}{\sqrt{(365)(1220)}}$$

$$r = \frac{652}{\sqrt{445300}}$$

$$r = 0.98$$

The linear correlation coefficient of 0.98 indicates that there is a clear positive linear relationship between the number of hours learners spend studying Mathematics and their summative assessment scores. That is, the more time you spend studying Mathematics, the higher your summative assessment score is likely to be.

Testing the Significance of Correlation Coefficient

To evaluate the statistical significance of the linear association between two quantitative variables, X and Y, a hypothesis test for the population correlation coefficient ρ is conducted. The null hypothesis states that ρ = 0, indicating the absence of a linear relationship, while the alternative hypothesis asserts that ρ ≠ 0, suggesting the presence of a linear association. The test employs the following statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

where r represents the sample correlation coefficient and n denotes the sample size. The resulting t-value is compared against the critical value from the t-distribution with $n - 2$ degrees of freedom. If the computed statistic exceeds the critical threshold, the null hypothesis is rejected, thereby providing evidence of a statistically significant linear relationship between the variables.

At a level of significance α, reject H_0 according to the decision rules given below.

Type of test	*H_1*	*Reject H_0 if*
Two-tailed test	$\rho \neq 0$	$t < -t_{\alpha/2(n-2)}$ or $t > t_{\alpha/2(n-2)}$
One-tailed test (left-tailed)	$\rho < 0$	$t < -t_{\alpha(n-2)}$
One-tailed test (right-tailed)	$\rho > 0$	$t > t_{\alpha(n-2)}$

Otherwise, fail to reject H_0.

Example:
To test the significance of the linear relationship between the number of hours studying Mathematics and summative assessment scores among learners using 0.05 level of significance yields the following:

Step 1. H_0: There is no significant linear relationship between the number of hours studying Mathematics and summative assessment scores among learners.
H_1: There is a significant linear relationship between the number of hours studying Mathematics and summative assessment scores among learners.

Step 2. $\alpha = 0.05$

Step 3. The appropriate test statistic to use is

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

with n – 2 degrees of freedom.

Step 4. Reject H_0 if
$t < -t_{0.025,4} = -2.7764$ or $t > t_{0.025,4} = 2.7764$

Step 5. Given: r = 0.98
n = 6

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{(0.98)\sqrt{6-2}}{\sqrt{1-(0.98)^2}} = 9.18$$

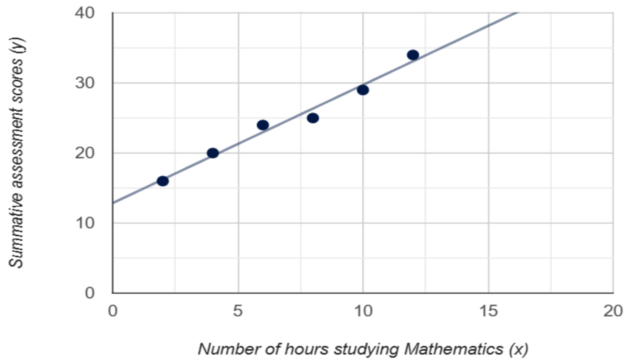
Step 6. Given that the value of t falls in the critical region, that is 9.18 > 2.7764, then reject H_0.

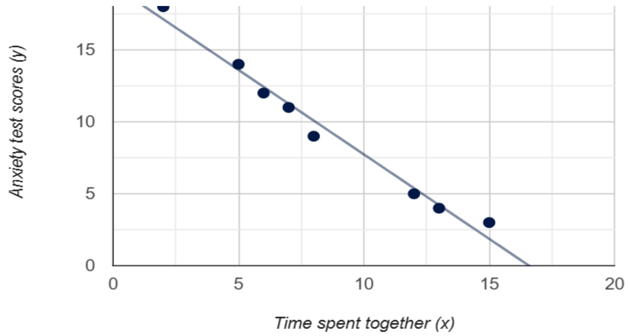
Step 7. At α = 0.05 level of significance, there is sufficient evidence to conclude that there is a significant linear relationship between the number of hours studying Mathematics and summative assessment scores among learners.

Simple Linear Regression

If the correlation coefficient is relevant and meaningful (i.e. significant), the equation of the regression line (line of best fit of the data) that includes the slope and y-intercept must be determined.

The regression line is the line of "best fit" that can be determined from two data points on the scatter diagram. The need for a line of best fit arises from the fact that the values of dependent variable (y) are expected from the values of independent variable (x). Thus, the closest the points are to the line, the better the fit and prediction will be. The following are some illustrations of the line of best fit of bivariate data in the previous examples.





The Regression Line

The regression line is used to predict or estimate the expected value of dependent variable y corresponding to given value of the independent variable x. Thus, the best fit or regression line is expressed as

$$y' = a + bx$$

where

y' = predicted value of the dependent variable y
a = y-intercept of the regression line
b = slope of the regression line
x = value of the independent variable

slope of the regression line:

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

y-intercept of the regression line:

$$a = \frac{(\sum y)(\sum x^2) - (\sum y)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

where:

n = number of pairs of sample data

Σx = sum of all x-values

Σy = sum of all y-values

Σxy= sum of all the product of x- and y-values

$(\Sigma x)^2$ = square of the sum of all x-values

Σx^2 = sum of all squared x-values

Example. Determine and interpret the slope and y-intercept and the regression equation between the number of hours studying Mathematics and summative assessment scores among learners. The data are as follows:

Hours of studying Mathematics (x)	2	4	6	8	10	11
Summative assessment score (y)	16	20	24	25	29	34

Predict the summative assessment score if a learner spends 9 hours studying Mathematics.

Solution. To find for slope *b*, y-intercept *a,* and the regression equation, we need to find for **Σx, Σx², Σy, Σxy**. It is easier to arrange the data in a four-column table and then add all of the entries to each column.

x	*y*	x^2	*xy*
2	16	4	32
4	20	16	80
6	24	36	144
8	25	64	200
10	29	100	290

11	34	121	374
Σx = 41	**Σy = 148**	**Σx² = 341**	**Σxy = 1120**

Using the computational formula for *a* and *b*, the total of each column from the table and n = 6, we have

a. y-intercept of the regression line:

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$
$$a = \frac{(148)(341) - (41)(1120)}{6(341) - (41)^2}$$
$$a = \frac{50468 - 45920}{2046 - 1681} = \frac{4548}{365} = 12.46$$

(A y-intercept of 12.46 indicates that if the number of hours spent studying Mathematics is 0, we expect a summative assessment score of 12.46)

b. slope of the regression line:

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$
$$b = \frac{6(1120) - (41)(148)}{6(341) - (41)^2}$$
$$b = \frac{6720 - 6068}{2046 - 1681} = \frac{652}{365} = 1.786$$

(A slope of 1.786 indicates that an increase in the number of hours spent in studying Mathematics by 1 hour, we expect the summative assessment score to increase by 1.786)

c. Using the slope and y-intercept of the regression line, the equation of the regression line is

$$y' = a + bx$$
$$y' = 12.46 + 1.786x$$

d. Using the equation of the regression line, the summative assessment score of a learner if he/she spent 9 hours studying Mathematics is given as follows:

$$y' = 12.46 + 1.786x$$
$$y' = 12.46 + 1.786(9)$$
$$y' = 12.46 + 16.074$$
$$y' = 28.534 \cong 29$$

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Instructions:
Read and solve each problem.

1. Suppose Masagana Pahunan, a store owner in a growing municipality, wants to evaluate whether advertising expenses are truly effective in boosting monthly sales revenue. She assumes that higher spending on advertising campaigns (posters, radio ads, and social media promotions) leads to higher sales, but she wants statistical evidence to support this belief. To test her assumption, she records data for twelve consecutive months:

Month	Advertising Expenses *(in thousands)*	Sales Revenue *(in thousands)*
January	15	30
February	13	25
March	10	23
April	16	27
May	15	30
June	20	60
July	16	35
August	18	38
September	12	24
October	25	70
November	25	70
December	30	80

Task:

Activity B.3: Practicing Learned Skills

This activity allows learners to apply the concepts of bivariate data analysis, correlation, and hypothesis testing to real-life scenarios. Learners practice calculating the Pearson Product Moment Correlation Coefficient (r), interpreting its meaning, and testing its statistical significance. The activity reinforces the connection between data analysis and practical decision-making in business, health, and education contexts.

To facilitate this activity the teacher may apply any of the following strategies.

Strategy 1. **Individual Work: My Correlation Quest**
Instructions:
1. Provide each learner with a worksheet that includes the following:
   - real-life scenarios with variable pairs
   - a section for calculating Pearson's r
   - a section for testing its significance
2. Learners work independently to estimate or calculate Pearson's r and test the significance.
3. Compare with critical values to decide whether the correlation is statistically significant.
4. After time is up, guide a whole-class review. Discuss common misconceptions (e.g., confusing strength with significance).

Strategy 2. **r You with Me?**
Pair Activity on Pearson r
Instructions:
1. Prepare the needed materials.
2. Each learner will have his/her partner and hand them over the following:

a. Calculate for the Pearson product moment correlation coefficient and interpret the result.

b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.

2. According to the United Nations Children's Fund (UNICEF, May 2020), hand washing with soap and water on a regular basis is important for disease prevention. To prevent the spread of infectious diseases, homes, schools, and healthcare facilities must have adequate water, sanitation, and hygiene services. In certain parts of the world, low levels of coverage of these basic services represent significant disparities between and within countries, making these populations vulnerable to the pandemic.

Suppose Mayumi Kalusugan, a health worker in a municipality, aims to investigate whether regular handwashing with soap and water reduces the risk of illness. To do this, she gathered data from five barangays, recording the proportion of residents who wash their hands after using the toilet alongside the percentage of individuals who fall sick.

Barangay	Percentage of residents with frequent hand washing after using the toilet (%)	Percentage of people who get sick (%)
A	86	10
B	67	23
C	54	30
D	73	11
E	45	25
F	32	27
G	76	11

Task:
a. Calculate for the Pearson product moment correlation coefficient and interpret the result.

- a worksheet involving a real-life scenario
- a small dataset for each scenario
- a formula sheet for Pearson's r and the t-test
- a calculator or spreadsheet (optional)

3. Learners begin by reading each scenario and discussing what kind of relationship they expect (positive, negative, none), how strong they think it might be, what Pearson's r value they might predict.
4. Using the provided data, the pairs will compute Pearson's r using the formula or spreadsheet and test the significance.
5. Each pair writes the computed r value, whether the relationship is significant and what the result means in context.
6. Facilitate a whole-class discussion.

Strategy 3. **Finding Relationship Collaboration**

Instructions:
1. Divide the class into three groups. Assign each group with one scenario involving correlational analysis. Ensure each group has access to available technology tools to compute and visualize data.
2. Each group inputs the data in the available technology tools to compute for the measure of relationship (i.e., Pearson r).
3. Each group analyzes and interprets the results. Discuss insights within the group before presenting the findings.
4. Each group presents their output to the class. Encourage questions and feedback from the class.
5. Teacher facilitates a synthesis on when Pearson product moment correlation coefficient is best used and how to interpret the result. Summarize key takeaways for practical application of measure of relationship (i.e., Pearson r).

Answer Key:

1. a. r = 0.97 (high correlation)
   b. t = 11.93 (significant)

2. a. r = -0.84 (high correlation)
   b. t = -3.48 (significant)

3. a. r = 0.98 (high correlation)
   b. t = 14.69 (significant)

b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.

3. The Department of Education's call for quality and liberating education, reinforced by the K-to-12 Basic Education Program, is consistent with the United Nations' Sustainable Development Goal 4, which promotes equitable access to lifelong learning. Responding to this mandate, Matatag National High School (Senior High School) integrates values of time management and academic excellence into its teaching practices. To evaluate how these values translate into learner outcomes, teacher-researchers at the school designed a study to determine whether there is a correlation between the amount of time spent studying and performance in General Mathematics summative assessments among Grade 11 learners. The data are given below.

Learner	*Amount of time spent on studying*	*Summative assessment score in General Mathematics*
1	8	35
2	7	32
3	9	41
4	7	33
5	6	31
6	5	29
7	2	18
8	3	18
9	8	37
10	6	30

Task:
a. Calculate for the Pearson product moment correlation coefficient and interpret the result.

b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.

C. Demonstrating Knowledge and Skills	*C.1. Finding Practical Application*	
	Activity C.1: Making Real-World Connections **Instructions**: Read and analyze the problem and answer the questions that follow**.** **Problem:** A certain schools division initiated and implemented the PANGKALIKASAN 101 program in support to the National Greening Program of the government. Under this program, it supports the tree planting activity in schools and in the community. This is in line with guidelines of the Department of Education to integrate the *gulayan sa paaralan*, ecological solid waste management and tree growing and caring as key components to attain the goals of DepEd on poverty reduction, food security, biodiversity conservation and climate change mitigation and adaptation. With the progression of climate change, expanding urban greenery by at least 10% could help reduce the anticipated rise in local temperatures in the coming decades (Gill et al., 2007). Suppose a research team from a certain institution aiming to explore the connection between tree density and the average ocean temperature during the years 2016 to 2024. The planned dataset is presented below. Year / Number of trees (in billions) / Mean temperature of oceans (^{o}C) 2016 / 11 / 24 2017 / 9 / 24 2018 / 8 / 25 2019 / 7 / 26 2020 / 6 / 28 2021 / 7 / 27 2022 / 6 / 29 2023 / 5 / 29 2024 / 5 / 30 Answer the following questions:	**Activity C.1: Making Real-World Connections** This activity engages learners in applying statistical concepts to real-world environmental issue**s**. Using actual concerns like tree density and ocean temperature, learners calculate the Pearson Product Moment Correlation Coefficient (r), interpret its direction, strength, and significance, and connect statistical findings to practical implications for climate change and environmental planning. This activity is designed to be completed as group work. Ensure that all learners actively participate and collaborate effectively. This task is an opportunity for them to deepen their understanding and refine their skills, as it will help prepare them for the upcoming performance task. Encourage them to aim for thorough and high-quality work. Provide feedback on their output. ***Answer Key:*** 1. r = -0.93 2. negative 3. The sign of the correlation coefficient r indicates the direction of the linear relationship between two variables. 4. very strong (high correlation) 5. Yes, the correlation is statistically significant at 0.05. 6. Statistical significance tells you that the relationship is consistent and trustworthy (reliable) rather than accidental (due to random chance). 7. a. Correlation does not imply causation. b. A high correlation in a sample does not guarantee that the same relationship exists in the population. c. The correlation coefficient r measures the strength and direction of linear relationships between two variables.

<table>
<tr><td rowspan="3"></td><td>1. Calculate the Pearson product moment correlation coefficient r.
2. Is the value of r positive, negative, or close to zero?
3. What does the sign of r tell you about the direction of the relationship?
4. How strong is the relationship?
5. Is the correlation statistically significant at $\alpha = 0.05$?
6. What does this significance tell you about the reliability of the relationship?
7. What cautions should we take when interpreting correlations?</td><td></td></tr>
<tr><td colspan="2">Activity C.2: Making Generalization</td></tr>
<tr><td>Activity C.2: Wrapping up the Lesson

Instructions:
Do as indicated.

1. Each learner completes three prompts on their sheet:
a. What did I learn?
(e.g., “I learned how to interpret scatterplots and make generalizations.”)
b. What went right?
(e.g., “I worked well with my group and explained our findings clearly.”)
c. What was difficult?
(e.g., “I got confused with the difference between correlation and causation.”

2. Learners pair up and share one insight from each category. They listen actively and affirm each other’s experiences.

3. As a class, gather common themes:
a. What did many of us learn?
b. What helped us succeed?
c. What challenges do we want to work on together?

Visual Reflection Wall
1. Use a bulletin board or a large sheet of Manila paper. Make sure it’s accessible and visible to all learners.</td><td>Activity C.2: Wrapping up the Lesson

For the summary, you may select and facilitate one of the provided activities.

This activity will help learners reflect on their learning journey by identifying key takeaways, successes, and challenges, while supporting shared growth.</td></tr>
</table>

2. Divide into three sections. Label each section clearly with the following:
   a. *What I learned*
   b. *What went well*
   c. *What challenged me*
3. Provide sticky notes, index cards, or cut-out shapes. Let learners choose colors to represent each category (e.g., yellow for learning, green for success, red for challenges).
4. Learners write or draw their responses on sticky notes.
5. Learners post their notes in the appropriate zones. You can read a few aloud (with permission) or invite volunteers to share.

C.3. Evaluating Learning

Activity C.3: Assessing Learning Outcomes

Instructions:
Solve the following problems as required. Show a complete solution to get complete points.

1. Suppose ten (10) Senior High School learners from a local school were randomly chosen, and their final grades in General Mathematics and General Science were recorded. The table reflects their academic performance across these two core subjects.

Gen. Math (x)	90	95	87	94	77	80	82	90	87	82
Gen. Science (y)	91	96	86	91	81	84	82	88	92	86

a. Calculate and interpret the Pearson product moment correlation coefficient.
b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.
c. Calculate and interpret the slope and y-intercept of the regression line.
d. Find the equation of the regression line. What is the expected grade of a learner in General Science if a learner receives a grade of 88 in General Mathematics?

Activity C.3: Assessing Learning Outcomes

Each learner must individually complete this activity. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.

Answer Key:
1. a. r = 0.89 (high correlation)

   b. t = 5.45; t-critical = 2.306
   Since t-computed > t-critical, reject Ho.

   c. slope: a = 26.77
   y-intercept: b = 0.71

   d. equation of the regression line: y = 26.77 + 0.71x

   If x = 88 (grade in General Mathematics), then the expected grade of a learner in General Science will be 89.

2. a. r = - 0.96 (high correlation)

   b. t = - 7.76; t-critical = 2.571

2. Suppose a group of academic track learners launched a study to investigate whether repeated absences in online classes in a certain subject have a measurable impact on learners' semestral grades. Their goal is to uncover patterns that might inform smarter academic interventions. The sample data collected are given below:

Habitual non-presence (x)	2	3	5	6	7	9	10
Semestral grade (y)	96	94	94	90	86	84	79

a. Determine and interpret the Pearson product moment correlation coefficient.
b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.
c. Determine and interpret the slope and y-intercept of the line of best fit.
d. Find the equation of the regression line. Predict the semestral grade of a learner in a certain subject if the habitual non-presence is 8 days.

3. A group of curious and enterprising senior high school learners teamed up to sell a reinvented version of polvoron during Entrepreneurship Week. Treating their venture like a mini business experiment, they deliberately changed the price of their product each day for five consecutive days. Their goal is to explore how daily price adjustments might influence their total sales and earnings, turning their sweet treat into a hands-on lesson in pricing strategy. The data obtained are as follows:

Price per pack (x)	15	25	35	22	30
Number of packs sold (y)	45	35	25	28	30

a. Calculate and interpret the Pearson product moment correlation coefficient.
b. Test the significance of the correlation coefficient to determine if there is a significant linear relationship between the variables considered.
c. Compute and interpret the slope and y-intercept of the line of best fit.

Since the absolute value of t-computed is greater than t-critical, reject Ho.

c. slope: a = 101.23
y-intercept: b = - 2.04

d. equation of the regression line:
y = 101.23 – 2.04x

If the habitual non-presence is 8 days, then the expected semestral grade of a learner in a certain subject will be 85.

3. a. r = - 0.83 (high correlation)

b. t = - 2.61; t-critical = 3.182
Since the absolute value of t-computed is less than the t-critical, fail to reject Ho.

c. slope: a = 54.30
y-intercept: b = - 0.85

d. equation of the regression line:
y = 54.30 – 0.85x

If the price of polvoron is 10 pesos per pack, then the number of packs of polvoron to be sold will be 46.

d. Determine the equation of the regression line. Estimate the number of packs of polvoron to be sold if the price of polvoron is 10 pesos per pack.

Guide Questions:
1. How does the value of Pearson product moment correlation coefficient influence the reliability of predictions made using the regression line?

2. In what ways do correlation and regression complement each other when analyzing relationships between two variables?

3. Why is it important to test the significance of the correlation coefficient before drawing conclusions about the relationship between variables?

C.4. Additional Activities

Activity C.4: Extending and Reinforcing Learning

Instructions:
Read and analyze the scenario and perform the indicated tasks.

Scenario:
The Department of Education emphasizes the importance of equipping learners with 21st century skills to prepare them for higher education, employment, and lifelong learning. These skills (i.e., problem solving, information literacy, and critical thinking) are central to the Basic Education Exit Assessment (BEEA) administered to Grade 12 learners. The BEEA serves as a culminating evaluation of learners' readiness to meet the challenges of the modern world, ensuring that graduates of the K-to-12 program possess competencies aligned with both national priorities and global standards.

Suppose a group of teacher-researchers seeks to examine how these 21st century skills are reflected in learners' academic performance. Specifically, they focus

Activity C.4: Extending and Reinforcing Learning

This additional exercise may be given as an assignment or homework. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

Answer Key:

a.
i. Science & Mathematics: r = 0.142 (negligible correlation)

Almost no linear relationship. Learners who score higher in Science do not consistently score higher in Math.

ii. Science & English: r = 0.303 (low correlation)

Learners who do better in Science tend to do slightly better in English, but the link is weak.

iii. Mathematics & English: r = 0.362 (low correlation) on STEM strand learners, whose mastery of Science, Mathematics, and English is considered vital for success in both tertiary education and professional fields. The assumed data are given below.

Learner	Science	Mathematics	English
1	21	20	17
2	16	16	15
3	11	12	12
4	22	11	13
5	23	14	14
6	21	15	15
7	12	17	13
8	16	13	17
9	18	14	21
10	19	11	15
11	20	15	15
12	21	17	20
13	20	21	19
14	19	17	14
15	19	19	15

a. Determine the Pearson product moment correlation coefficient between the performances in the following areas and interpret the result:

i. Science and Mathematics
ii. Science and English
iii. Mathematics and English

b. Test the significance to determine if there is a significant relationship between the variables considered.

Learners who do better in Mathematics tend to do slightly better in English, but the link is weak.

b.
i. Science & Mathematics: t = 0.52, t-critical = 2.160

Since the computed t-value is less than the critical value, fail to reject the null hypothesis.

There is no statistically significant correlation between Science and Mathematics scores among the 15 Grade 12 STEM learners. While the correlation coefficient was weakly positive, the evidence is insufficient to conclude a meaningful relationship.

ii. Science & English: t = 1.14, t-critical = 2.160

Since the computed t-value is less than the critical value, fail to reject the null hypothesis.

There is no statistically significant correlation between Science and English scores among the 15 Grade 12 STEM learners. Learners who perform better in Science tend to also perform slightly better in English; the evidence is not strong enough to confirm a meaningful relationship in this sample of 15 learners.

iii. Mathematics & English: t = 1.40, t-critical = 2.160

Since the computed t-value is less than the critical value, fail to reject the null hypothesis.

There is no statistically significant correlation between Mathematics and English scores among the 15 Grade 12 STEM learners. Learners who perform better in Mathematics tend to also perform slightly better in English; the evidence is not strong enough to confirm a meaningful relationship in this sample of 15 learners.

VI. SUMMATIVE ASSESSMENTS

Please refer to the attachments for the summative assessment and performance task.

Answer Key:

I. Multiple Choice

1. D	6. B	11. D	16. C	21. B
2. A	7. D	12. A	17. B	22. D
3. B	8. B	13. D	18. A	23. C
4. D	9. D	14. A	19. A	24. C
5. C	10. C	15. B	20. C	25. B

II. Problem Solving

1. a. r = 0.898

An r value of 0.898 represents a strong positive linear relationship. As grades in General Mathematics increase, grades in General Physics also tend to increase.

b. t = 5.757 ; t-critical = 2.306

Since, t-value > t-critical, reject the null hypothesis.

The correlation is significant at the 0.05 level, meaning there is sufficient evidence to conclude that a relationship exists between General Mathematics and General Physics.

2. z = 1.897, z-critical = 1.645

Since the computed z-value is greater than the critical value, reject the null hypothesis.

At a 0.05 level of significance, there is sufficient statistical evidence to conclude that Ms. Marespeto's class performed significantly better than the national average.

3. t = 3.723, t-critical = 2.132

Since the computed t-value is greater than the critical value, reject the null hypothesis.

At the 0.05 level of significance, studying with background music leads to significantly higher quiz scores for the Musika NHS learners.

4. t = 2.539, t-critical = 2.145

Since the computed t-value is greater than the critical value, reject the null hypothesis.

There is sufficient evidence to conclude that there is a statistically significant difference in quiz scores between learners who reviewed until midnight and those who slept early.

	5. t = 5.40, t-critical = 1.97 Since the computed t-value is greater than the critical value, reject the null hypothesis. There is a statistically significant difference in emotional well-being between learners who eat dinner with their families and those who eat alone at the 0.05 level. Learners who regularly eat dinner with family report higher emotional well-being scores.
VII. REFLECTION	Reflection is most effective when done regularly. Make it a habit to reflect after each lesson or weekly to track your progress and growth over time. The reflection questions are designed to help you thoughtfully evaluate your teaching, identify strengths, recognize challenges, and plan improvements to enhance learner learning. The questions provided below may help in reflecting to your teaching performance. **1. What aspects of your teaching went particularly well?** • Which part of the lesson engaged learners the most? • What teaching strategies or methods contributed to successful learning? • How did learners respond to your instruction (e.g., participation, performance, behavior)? • What evidence shows that learning objectives were achieved? • What personal teaching strengths were evident in this session? **2. What challenges did you encounter, and how did you address them?** • What difficulties arose during lesson delivery or classroom management? • Were there moments when learners struggled to understand a concept? Why? • How did you adjust your strategies or pacing to respond to these challenges? • What support or resources would have helped you handle the situation more effectively? • What will you do differently next time to overcome similar obstacles? **3 What will you do differently next time to improve learner learning?** • Which teaching strategies or approaches would you modify or try in the future? • How will you better address diverse learning needs in your classroom? • Are there new resources, activities, or technologies you plan to incorporate? • What steps can you take to enhance learner engagement or participation? • How will you assess whether these changes improve learning outcomes?

Summative Assessment

Instruction:
I. Choose the best answer that corresponds to the question.

1. Which of the following is an example of a null hypothesis?
   A. Studying affects performance positively.
   B. Learners who study less will score lower.
   C. Learners who study more will score higher.
   D. There is no difference in scores between learners who study more and those who study less.

2. Which statement best describes the relationship between H_0 and H_1?
   A. H_0 assumes no effect, H_1 assumes an effect exists
   B. H_0 is optional, H_1 is required
   C. Both predict the same outcome
   D. H_0 is always accepted, H_1 is always rejected

3. Which of the following is a correct alternative hypothesis?
   A. Blood pressure is random.
   B. The new drug lowers blood pressure.
   C. Blood pressure is unaffected by the new drug.
   D. The new drug has no effect on blood pressure.

4. Which sentence illustrates understanding of H_0 and H_1?
   A. H_1 is weaker than H_0.
   B. H_0 and H_1 are identical statements.
   C. H_0 always predicts a positive result.
   D. H_0 predicts no effect, H_1 predicts the effect of a variable.

5. Which scenario demonstrates a directional alternative hypothesis?
   A. Blood pressure is random
   B. A new drug affects blood pressure
   C. A new drug lowers blood pressure
   D. No difference in blood pressure between groups

6. A study finds no significant difference in test scores after a new method. What should the researcher conclude?
   A. Reject H_0
   B. Fail to reject H_0
   C. Study is invalid
   D. H_1 is proven true

7. Which of the following is a Type I error?
   A. Rejecting a false null hypothesis
   B. Accepting a true null hypothesis
   C. Accepting a false null hypothesis
   D. Rejecting a true null hypothesis

8. Which hypothesis is usually tested directly in hypothesis testing?
   A. Sample hypothesis
   B. Null hypothesis (H_0)
   C. Research hypothesis
   D. Alternative hypothesis (H_1)

9. Why do we set a significance level (α) before conducting a hypothesis test?
   A. To calculate the p-value
   B. To decide the sample size
   C. To increase the power of the test
   D. To control the chance of making a Type I error

10. How does increasing sample size affect Type I and Type II errors?
    A. Has no effect

B. Increases both errors
C. Decreases both errors
D. Decreases Type I, increases Type II

11. At an airport, the security system is designed with α = 0.10. Which of the following is true?
A. The system ignores minor threats
B. The system allows fewer false alarms
C. The system reduces Type II errors to zero
D. The system tolerates more false alarms to catch more threats

12. If a drug trial uses α = 0.05 and obtains a p-value of 0.03, what is the correct conclusion?
A. Reject H_0; drug effect is significant
B. Accept H_0; no effect
C. Increase α and test again
D. The result is inconclusive

13. Why might a lower α be chosen for testing a cancer drug but a higher α for marketing a new product?
A. Drug tests have bigger samples.
B. Both cases need low α due to risk.
C. Marketing can tolerate more Type II errors.
D. The drug affects health, so false positives (Type I) are more dangerous, while in marketing, missing a chance (Type II) is costlier.

14. Which choice best shows critical thinking about α and errors?
A. Choosing α depends on context, weighing consequences of errors
B. Errors don't affect real decisions
C. Setting α to 0.05 is always the best practice
D. Lower α always better because errors must be avoided

15. What does setting a significance level α = 0.05 mean?
A. Accepting a 5% chance of Type II error
B. Allowing a 5% chance of Type I error
C. Rejecting the null hypothesis 5% of the time
D. Accepting the alternative hypothesis 5% of the time

16. Ms. Masigasig, a Math teacher at Manalig High School, wants to test if her learners' average score in a 50-item math quiz is significantly different from the national average of 35. She does not know the population standard deviation. Which test should she use?
A. One-sample z-test
B. Paired t-test
C. One-sample t-test
D. Two-sample t-test

17. Which situation best illustrates the use of a one-sample z-test in a Filipino school setting?
A. Comparing the average scores of two sections in Araling Panlipunan
B. Testing if the average height of Grade 10 learners in Parisukat NHS differs from the national average, assuming known population standard deviation
C. Evaluating the improvement in scores before and after a remedial class
D. Checking if the average number of absences differs between male and female learners

18. Engr. Roel was curious whether there is a linear association between Grade 11 learners' grades in the First Semester (i.e. First Quarter and Second Quarter Grades) and if he can predict a learner's Second Quarter grade based on their First Quarter grade. The data are as follows:

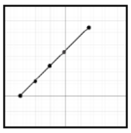
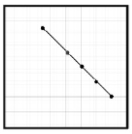
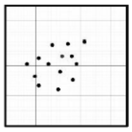
First Quarter (x)	80	89	89	90	98	87	85
Second Quarter (y)	87	92	90	91	97	88	84

If r = 0.87, what can be concluded about x- and y-values?

A. An increase in the first quarter grades yields an increase in the second quarter grades.
B. A decrease in the first quarter grades yields an increase in the second quarter grades.
C. An increase in the first quarter grades yields a decrease in the second quarter grades.
D. A decrease in the first quarter grades yields a decrease in the second quarter grades.

19. If the computed t-value is 2.015 and the critical value is 1.833, which of the following statements could be true?
A. It lies in the rejection region; Ho must be rejected.
B. It lies in the rejection region; we failed to reject Ho.
C. It lies in the non-rejection region; Ho must be rejected.
D. It lies in the non-rejection region; we failed to reject Ho.

20. Which of the following graphs describes a perfect negative correlation?

A. 
B. 
C. 
D. 

21. When is it appropriate to use a z-test instead of a t-test?
A. When comparing paired observations
B. When population standard deviations are known and sample sizes are large
C. When the sample sizes are unequal and small
D. When testing for correlation between two variables

22. Mr. Mataraki, a school principal at Makabayan NHS, wants to compare the average math scores of Grade 10 learners from two different public schools in a certain province. The population standard deviations are unknown. Which test should be used?
A. Two-sample z-test
B. Paired t-test
C. One-sample t-test
D. Two-sample t-test

23. In a one-sample z-test, which of the following is typically known?
A. Sample mean
B. Population mean
C. Population standard deviation
D. All of the above

24. Which of the following pairs of variables is most appropriate for Pearson r analysis?
A. Gender and preferred learning style
B. Grade level and section
C. Number of absences and final exam score
D. Favorite subject and average grade

25. A Filipino teacher finds a strong negative correlation between learners' screen time and their reading comprehension scores. What should she conclude?
A. Screen time causes poor reading skills
B. Learners who spend more time on screens tend to have lower reading scores
C. There is no relationship between the variables
D. Reading comprehension improves with more screen time

II. Analyze and solve the problems below.

1. The following table shows the semestral grades in General Mathematics and General Physics obtained from 10 Senior High School learners selected at random from a certain school.

General Mathematics (x)	89	94	86	95	78	79	80	91	88	81
General Physics (y)	90	95	85	92	82	83	80	89	93	85

a. Calculate and interpret the Pearson product moment correlation coefficient.
b. Test the significance of the correlation coefficient. Use 0.05 level of significance.

2. A certain organization claims that the average score of Grade 10 learners in the National Achievement Test (NAT) for Mathematics is 75. Ms. Marespeto, a teacher from Magalang National High School believes her class performed better than the national average due to her values-integrated teaching approach. She randomly selects 40 learners and finds that their mean score is 78, with a standard deviation of 10. Using a significance level of 0.05, test whether the class performed significantly better than the national average.

3. Learners at Musika NHS tried studying with background music for one week, then without music the next. Their adviser wants to know if music helps improve focus, as measured by quiz scores. The following data were recorded:

Learner	With Music	Without Music
Masinop	45	35
Masipag	43	32
Matibay	39	36
Matapang	45	35
Mapagbigay	41	39

Using a 0.05 level of significance, determine if studying with music leads to significantly higher scores.

4. At Maalalahanin National High School, learners were preparing for a major math quiz. Two groups adopted different strategies based on their beliefs and habits. Group A chose to review until midnight, believing that last-minute cramming would help them retain formulas and problem-solving techniques. On the other hand, Group B decided to sleep early, trusting that rest and prior preparation would lead to better focus and recall. Their adviser, Mr. Alapaap, who often reminds learners that *"ang tagumpay ay bunga ng disiplina at tiwala sa sarili,"* wondered whether the choice of strategy had a measurable impact on quiz performance. The data are given below:

Group A	Group B
45	43
49	34
43	41
35	33
45	25
47	37
34	14
35	23

Is there a statistically significant difference in quiz scores between learners who reviewed until midnight and those who slept early? Use 0.05 level of significance.

5. At a certain school, teachers noticed that learners who regularly eat dinner with their families often seem more emotionally grounded, more cheerful during homeroom, more focused in class, and more open during group reflections. To explore this observation, the Guidance Office, in partnership with the Values Education Department, conducted a survey during Mental Health Awareness Week. Learners rated their emotional well-being on a scale of 1 to 10, with 10 being "very emotionally well" and 1 being "very emotionally distressed." The learners were grouped based on their dinner habits. Under Group A are learners who eat dinner with family at least 5 times a week. On the other hand, Group B are learners who usually eat alone due to work schedules, household setup, or personal preference. Below are the recorded data.

Group	Sample size	Mean	Standard deviation
A (Dinner with family)	100	8.5	2
B (Eat alone)	105	6.8	2.5

Is there a statistically significant difference in emotional well-being between learners who eat dinner with family and those who eat alone? Use 0.05 level of significance.

Performance Task

Task Title: "Study Smart: A Statistical Investigation of Study Habits and Academic Performance"

Overview: You are a data analyst tasked with investigating whether there is a significant relationship between learners' study habits (e.g., hours of study per week) and their academic performance (e.g., grades in Mathematics). This inquiry is supported by various research studies including the study of Gumapac et al. (2025), which found a positive correlation between consistent study routines and improved math achievement among Filipino secondary learners. Similarly, Hermano and Gaudicos (2025) examined the predictive relationship between study habits and mathematical achievement among STEM learners, highlighting how learning styles (visual, auditory, tactile) and study preferences influence performance. Also, Zubair et al. (2024) revealed that learners who studied more than six hours per week consistently outperformed those with fewer study hours. Your task is to collect data, analyze using appropriate statistical tools, and draw conclusions based on your findings and reflect on whether your results support or verify the claims.

References: DOI: https://doi.org/10.54660/.IJMRGE.2024.5.3.720-728
Findings: Significant relationship between study habits and math achievement; private unaided school learners performed better. (Zubair, T., et. al., 2024)
DOI: 10.47857/Irjms.2025.V06i03.04913
Findings: Auditory learners benefited most from study habits like time on task and test-taking skills. (Hermano, R. H., Gaudicos, AG. P., 2025)
Gumapac, JC. S., Yulo, J.C., Nanud, MN. D. and Baluyos, J. R. (2025). Learners` Basic Math Skills and Study Habits in Relation to their Performance in Mathematics. *International Journal of Science & Engineering Development Research, 10(1)*:a551-a564, January-2025. Retrieved http://www.ijrti.org/papers/IJRTI2501066.pdf

Instructions:
Data Collection:

A. Survey at least 20 learners about their (e.g. weekly study hours) and their latest Mathematics test scores.
B. Organize the data in a spreadsheet.

Formulate Hypotheses:

A. Create a research question (e.g., “Is there a significant relationship between study hours and math performance?”).
B. Formulate the null and alternative hypotheses.

Determine Statistical Parameters:

A. Choose an appropriate level of significance (e.g., 0.05).
B. Identify possible Type I and Type II errors in your context.

Statistical Testing:

A. Use statistical software (e.g., Excel, GeoGebra, or any calculator) to perform:
   1. z-test (known and unknown standard deviation)
   2. One-sample t-test
   3. Independent sample t-test (e.g., comparing two groups: those who study <5 hours vs. >5 hours)
   4. Paired sample t-test (e.g., pre-test and post-test scores if applicable)

Correlation and Regression:

A. Create a scatter plot (e.g. study hours vs. test scores).
B. Draw the line of best fit.
C. Calculate Pearson’s r and interpret the strength and direction of the relationship.

Conclusion:

A. Interpret the results of your hypothesis tests and correlation analysis.
B. Discuss whether your data supports your hypothesis.
C. Reflect on the implications of your findings for learners and educators.

Output Requirements:

1. A written report (3–5 pages) including:
    a. Introduction and research question
    b. Data table
    c. Hypotheses and statistical analysis
    d. Graphs and charts
    e. Interpretation and conclusion
2. A short presentation (5–7 minutes) summarizing your findings.

Group Assigned task:

1. Hours of Study per Day/Week
2. Number of Study Sessions per Week
3. Duration of Each Study Session (in minutes or hours)
4. Sleep Hours per Night
5. Number of Subjects Studied
6. Test-Taking Skills (e.g., preparation skills)
7. Number of Distractions (e.g., phone checks per hour)
8. Time Spent on Social Media per Day
9. Attendance Rate (%)
10. Number of Practice Tests Taken

Performance Task Rubric

Criteria	Description	Points
1. Formulation of Hypotheses	Clearly states null and alternative hypotheses relevant to the research question.	10
2. Identification of Statistical Concepts	Accurately illustrates and explains null/alternative hypotheses, level of significance, and types of error.	10
3. Data Collection & Organization	Gather appropriate data, present it clearly in tables/spreadsheets.	10
4. Application of Statistical Tests	Correctly applies z-tests and t-tests using appropriate tools and interprets results.	15
5. Graphical Representation	Constructs a clear scatter plot and accurately draws the line of best fit.	10
6. Correlation Analysis	Calculates Pearson's r and interprets the strength and direction of the relationship.	10
7. Interpretation and Conclusion	Provides a well-reasoned conclusion based on hypothesis testing and correlation results.	15
8. Report Quality	Organized, clear, and complete report with proper formatting, grammar, and visuals.	10
9. Presentation	Engaging, clear, and concise presentation of findings. Uses visuals effectively.	10
	TOTAL	**100**

Criteria	Excellent (Full Points)	Proficient (80%)	Developing (60%)	Beginning (40%)
1. Hypotheses Formulation	Clear, correct null & alternative hypotheses aligned with the research question.	Minor errors in formulation.	Vague or partially correct hypotheses.	Incorrect or missing hypotheses.
2. Statistical Concepts	Accurately illustrates & explains all 4 concepts (null/alt, significance level, errors).	1–2 minor inaccuracies.	Some concepts are missing or unclear.	Most concepts are missing or incorrect.
3. Data Collection & Organization	Complete, relevant data; well-organized tables/spreadsheets.	Mostly complete; minor issues in organization.	Incomplete or disorganized data.	Data missing or irrelevant.
4. Application of Statistical Tests	All tests applied correctly with accurate interpretation.	Most tests correct minor interpretation errors.	Some tests are misapplied or misinterpreted.	Tests misapplied or not attempted.
5. Graphical Representation	Accurate scatter plot with labels & correct line of best fit.	Minor issues in plot or line.	Plot present but unclear or inaccurate.	Plot missing or incorrect.
6. Correlation Analysis	Pearson’s r correctly calculated & interpreted.	Minor errors in calculation or interpretation.	Significant errors in calculation or interpretation.	Missing or incorrect analysis.
7. Interpretation & Conclusion	Insightful, evidence-based conclusions; deep understanding.	Reasonable conclusions; minor gaps.	Weak or unclear conclusions.	Conclusions unsupported or incorrect.
8. Report Quality	Well-organized, clear, error-free, with visuals.	Mostly clear; minor errors.	Some organization or grammar issues.	Disorganized or unclear.
9. Presentation	Clear, engaging, confident, with effective visuals.	Clear with minor issues.	Lacks clarity or engagement.	Unclear or incomplete.